

Nonparametric Estimation of Voters' Ideological Preferences

Nina Nikiforova*

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Abstract

This paper develops a method to recover voters' preferences from aggregate election returns under spatial voting with multiple dimensions. The method exploits the convex partitions of the policy space induced by candidate positions, and requires only that a voter's utility decrease in the distance between the candidate's position and the voter's ideal point. I apply it to bring new insights into the rise of populist parties, studying both the electorate of the radical-right Alternative for Germany (AfD) and the scope for establishment parties to win it back. I use municipal-level Bundestag results from 2013 to 2025 in a policy space spanned by a left-right and an anti-elite dimension. The results indicate that over time the AfD electorate shifted from stronger anti-elite attachment towards ideological alignment with the party's radical-right platform. Counterfactual simulations for the 2025 election suggest that establishment parties could capture a third of the AfD electorate by hardening their immigration policy alone.

*Bocconi University. E-mail: nina.nikiforova@unibocconi.it. I am grateful to Guido Tabellini for his constant support and guidance, and to Jerome Adda, Christian Cox, Catherine De Vries, Francesco Decarolis, Thomas Le Barbanchon, Nicolas Longuet-Marx, Massimo Morelli, Eduard Osipov, Carlo Schwarz, Cailin Slattery, Piero Stanig, Pietro Tebaldi, Edoardo Teso and Francesco Trebbi for their helpful comments and suggestions.

1 Introduction

Politics nowadays is more than just a conflict along the redistributive dimension. Politicians build their electoral campaigns around cultural and social issues, confrontation with the elites, demands for institutional reforms and climate policy among other topics. Voters, too, tend to base their electoral choices at least partly on issues unrelated to the economic policy dimension (Gethin, Martínez-Toledano and Piketty, 2022; Danieli et al., 2023; Gennaioli and Tabellini, 2025; Longuet-Marx, 2025; Noy and Rao, 2025; Acemoglu, Egorov and Sonin, 2026). Understanding voters’ preferences in such multidimensional environments is essential for a better understanding of voters’ policy demands, and for crafting appropriate responses.

How can one credibly learn about voters’ preferences? Doing so requires two ingredients: reliable data on voting choices, and a plausible model of voting behavior. The first ingredient points to aggregate electoral data: unlike surveys, they do not suffer from misreporting or sampling bias (Bursztyrn and Jensen, 2017; Tyler et al., 2026). The second points to the canonical frameworks of voting choice, of which the spatial voting model is the most common (Downs, 1957).

This paper contributes to the literature by developing a novel approach to estimate voters’ preferences using aggregate voting data within a multidimensional spatial voting model. I apply this approach to provide new insights into the rise of populism in Germany. While the literature has established many causal drivers of populist voting, it remains divided on the underlying dimension of this rise: is populist voting grounded in a rejection of the elites and of how representative democracy currently functions, or in the ideological Left-Right positions of populist candidates? (Guriev and Papaioannou, 2022)

I recover the preferences of voters of the radical-right populist party Alternative for Germany (AfD) over 2013–2025, in a voting model with Left–Right and Anti-Elite dimensions. The results show that the party’s earlier support was grounded in anti-elite sentiment, whereas its more recent support is rooted in the party’s radical-right stance. I further show that these patterns differ between East and West Germany and across demographic groups. Counterfactual simulations suggest that in the most recent election, establishment parties could have won about one third of the AfD electorate by moving moderately to the right on the immigration policy alone.

In the first part of the paper, I develop an approach to recovering voters’ preferences. I adopt a spatial voting model in which voter i ’s valuation of candidate j depends only on the weighted Euclidean distance between the voter’s ideal point (bliss point) X_i and the candidate’s position Y_j in the policy space. Weights represented by a diagonal weight matrix W capture the importance voters attach to each dimension.

I formalize a nonparametric approach to recovering voters’ preferences (the common dimension weights W and the distribution of ideal points X_i) from aggregate election returns. The approach relies solely on the geometry of choice implied by the multidimensional spatial voting model and imposes no distributional assumptions. It is broadly applicable across electoral settings in which voters face different choice environments (candidates’ positions Y_j).

Two observations are key. First, the weight matrix W determines how the policy space is partitioned in each choice environment: each candidate is assigned the region of ideal points for which they are the closest option. When similar voters face different choice environments, a given W can rationalize only some joint voting patterns and not others. Weights are therefore set-identified: any W that cannot rationalize the joint voting patterns observed in the data is ruled out, and the identified set consists of those that can.

Second, there are generally infinitely many ideal-point distributions consistent with the data. This fundamental underidentification can be addressed by adding structure—either by strengthening assumptions on the choice model or by imposing structure on unobserved heterogeneity.¹ An alternative approach is to answer the questions of interest without recovering a single best-fitting distribution of ideal points at the cost of obtaining bounds rather than point estimates. This paper adopts the latter approach, building on [Tebaldi, Torgovitsky and Yang \(2023\)](#), who develop a nonparametric revealed-preference framework to estimate sharp bounds for the quantities of interest about demand for health insurance plans in the California marketplace.

Specifically, I focus on bounding *target parameters*: interpretable functionals of the underlying distribution of preferences (W, X_i). A target parameter could be, for example, the share of voters whose ideal points lie in a given area of the policy space, or the share of voters who would abandon a party if it shifted its position by a given amount.

In the second part of the paper, I apply the framework to Germany, where the electoral landscape has undergone a pronounced transformation over the past two decades, with the traditional establishment parties losing ground to the populists. The central empirical object in this paper is the preferences (W, X_i) of the electoral base of the very successful populist party Alternative for Germany (AfD), which by the mid 2026 leads national polls with 6 p.p. margin across the country.² This party combines radical-right positions with an explicitly anti-elite appeal, which motivates the choice of the two dimensions used in the paper – Left-Right and Anti-Elite dimensions.³

¹This is the approach taken, among others, by [Merlo and De Paula \(2017\)](#); [Iaryczower, Montero and Kim \(2024\)](#).

²Politico, National Parliament Voting Intentions – [Link](#). Accessed: 6th July 2026.

³These dimensions also connect to how the political science literature conceptualizes populism. A body of work distinguishes the role of *thin* and *thick* ideologies in populist support, that is, anti-elite appeals and substantive ideological stances. See [Mudde and Kaltwasser \(2017\)](#); [Vries and Hobolt \(2020\)](#) for a discussion, and [Guriev and Papaioannou \(2022\)](#) for an overview of the political economy literature.

The analysis leverages rich within-region (state) variation in political supply. In each election year (2013, 2017, 2021, 2025), voters in each of Germany’s 299 electoral districts face a district-specific set of party-affiliated candidates. I estimate candidate positions on two dimensions (Left–Right and Anti-Elite), using the German Longitudinal Election Candidate Study and a complementary candidate survey distributed by a nonpartisan organization *Wen Wählen* (*Whom to Vote For?*). Positions are constructed from multiple survey items per dimension (e.g., “Government should stay out of the economy” and “Immigrants should adapt to German culture” for Left–Right; “Laws reflect citizens’ interests” for Anti-Elite), standardized, and aggregated into indices that are consistent throughout the study period.

On the voters’ side, the unit of analysis is an age–gender–education group within a state and year (e.g., female, aged 18–24, higher education, Saxony, election year 2021). I combine municipality-level party-list vote shares for the six main parties (AfD, CDU, FDP, Greens, Left, SPD) with municipality-level joint demographic data for 24 age–gender–education bins. Together, these data allow the model to relate observed vote shares to latent group-specific preferences.

Two institutional features are important for implementation. First, Bundestag elections are held under a mixed electoral system: voters cast one ballot for a district candidate and one for a party list. I focus on party-list vote shares and treat them as sincere among the six major parties, while using district candidates’ positions as proxies for locally perceived party positions.

Second, the identifying variation comes from differences in candidate positions across electoral districts within a state and year. The identifying assumption is that within-state variation is captured by the observed age–gender–education composition of municipalities and is unrelated to unobserved voter heterogeneity across districts. I view this assumption as plausible because, in practice, single-member district candidates are approved by the state party leadership, which is typically more concerned with advancing preferred candidates to the Bundestag than with responding to specific local voter demands. I test the robustness of my results to this assumption in three ways: first, by focusing on very small municipalities (fewer than 5,000 inhabitants), which are least likely to be targeted, as they are least decisive for candidates’ electoral success; second, by restricting the analysis to non-contender AfD candidates, who have weaker incentives to cater to the local electorate; and third, by allowing for preference heterogeneity between electoral districts in which AfD candidates are more or less conservative—an instrument similar in spirit to the one used in [Tebaldi, Torgovitsky and Yang \(2023\)](#).

I start by estimating the weight of the Anti-Elite dimension relative to the Left–Right dimension among AfD voters. Since obtaining an informative estimate of the weights requires substantial variation in choice environments (a point discussed in the Appendix simulations, Section [B](#)), I estimate

a single weight common to all demographic groups and states in a given election year, separately for Eastern and Western Germany. The results suggest that in Western Germany the Left–Right dimension carries roughly three to five times the weight of the Anti-Elite dimension throughout the period. The same does not hold in Eastern Germany, where the Anti-Elite dimension carries a substantial weight in every year and is estimated to outweigh the Left–Right dimension in 2013 and 2025.

Next, I quantify the distribution of voters’ bliss points in the two-dimensional policy space. Within the AfD’s attraction area, I partition voters by whether their ideal points are distinctively Anti-Elite, distinctively Right, or both, using a fixed cutoff in each dimension. This delivers the following result: AfD support is increasingly grounded in the party’s conservative Left–Right positions rather than in its anti-elite appeal. Specifically, the share of voters with distinctively Right ideal points rises steadily over the period, and by 2025 its lower bound exceeds the upper bound of the Anti-Elite share. The Anti-Elite and Both regions display no comparable trend, despite accounting for larger shares than the Right region in 2013.

I then recover the *motivations* behind the AfD vote, in an exercise that combines the estimated weights with the underlying preference distribution. Suppose the AfD split into two parties: one keeping its radical-right economic position but moving towards the establishment, the other keeping its anti-establishment appeal but moderating its Left–Right position. What share of voters would each attract? The exercise assigns each AfD voter to whichever dimension—anti-elite or right—their ideal point leans towards, relative to the AfD’s own position. The share of voters drawn primarily by the party’s right position rises over time, reaching at least 53% in 2021 and 2025; because the exercise is symmetric, at most 47% are anti-elite-motivated by then.

The results are heterogeneous across Western and Eastern Germany, with the latter being relatively more anti-elite, as well as across demographic groups. Among Western voters without higher education, the party’s right-wing positions became the dominant motive only in later years; the same holds for voters under 35.

I next run two counterfactual repositioning exercises. In the first, I explore the sensitivity of the AfD electorate to shifts in the AfD’s own position towards more pro-elite or more moderate Left–Right stances. This analysis combines the estimated voter preferences with the positions of the other parties. The results suggest that up to 2025 voters were relatively insensitive to small shifts in party positions. This already indicates that competition between the AfD and other parties used to be relatively weak, with the parties largely competing for different groups of voters. The picture changes in 2025, when even a small moderation in the AfD’s Left–Right position would lead to substantial electoral losses.

These results point to intensified competition for voters between the AfD and the establishment

parties. In the second exercise, I ask how establishment parties should compete with populist challengers. To do so, I focus on the Christian Democratic Union (CDU, the largest establishment party to the right of center) and estimate how much of the AfD electorate it could capture by shifting towards the AfD—that is, towards more conservative and more anti-elite positions. The results suggest that a small repositioning (0.25 estimated units) on the Left–Right dimension would capture up to 10% of the AfD electorate before 2025. In 2025, however, the same shift would allow the CDU to capture about one third of the AfD electorate. Crucially, hardening the party’s anti-elite positions is not nearly as effective in any year.

Because the candidate position scores are constructed from candidate questionnaires, I can translate this 0.25-unit hardening on the Left–Right dimension into concrete survey answers. Hardening the CDU candidates’ answers on the three immigration questions by one point on a 1–5 scale produces a shift of 0.252 units along the Left–Right dimension in 2025. Thus, by hardening its stance on immigration alone—leaving its other economic and cultural positions unchanged—the CDU could win up to one third of the AfD electorate in 2025.

I conclude by discussing to what extent the results above are driven by parties’ repositioning rather than by demand-side changes (which could include genuine preference evolution as well as compositional shifts). To answer this, I compute the Shapley values (Shapley, 1953) for the position channel in the counterfactual exercise estimating the motivations behind the AfD vote, for each pair of consecutive years. The results suggest that parties’ repositioning does not explain the increase in Right motivations over 2013–2017 and 2017–2021 (and, if anything, decreases the share of Right-motivated voters), but does contribute between 1 and 9 percentage points to the increase over 2021–2025. Overall, this suggests that the evolution of voters’ preferences documented in this paper is not an artifact of supply-side repositioning.

Related Literature. This paper is related to several strands of literature. First, to a growing body of research that estimates structural models of voting in elections (Coate and Conlin, 2004; Ujhelyi, Chatterjee and Szabó, 2021; Iaryczower, Montero and Kim, 2024; Kawai and Sunada, 2025; Berry, Cox and Haile, 2025; Longuet-Marx, 2025). This paper differs from these in that its underlying framework is a multi-dimensional spatial voting model, in which individuals vote for the candidate closest to them in the weighted policy space.⁴

The paper is closely related to Degan and Merlo (2009), who first formalize the partitioning of the policy space under the multidimensional spatial voting model using Voronoi cells, and to Merlo

⁴Of these papers, only Iaryczower, Montero and Kim (2024) estimates a spatial voting model, and only in one dimension. As noted in Merlo and De Paula (2017), estimation of a single-dimensional model is considerably easier, since the problem of separately identifying *weights* from bliss points does not arise.

and De Paula (2017), who provide nonparametric identification and estimation of voter preferences under such a partition. Conceptually, this paper is closest to the latter. Merlo and De Paula (2017) obtain identification under a large-support assumption on candidates’ positions and estimate the model with a sieve minimum-distance estimator. This approach fixes the flexibility of the bliss-point distribution and recovers that distribution and the weights jointly, so *both* can be biased if the imposed distribution is too inflexible relative to the true one.⁵ This paper instead uses Voronoi cells and their superimposition, requiring neither a full-support assumption on candidates’ positions for identification nor estimation of the exact voter preference distribution. It separates the problem into recovering (i) the weights and (ii) the bliss-point distribution. This comes at the cost of potentially recovering not a single best-fitting weight and bliss-point distribution, but sets for each.

Second, this paper is related to a literature that employs a range of methodologies to recover voters’ preferences, including the use of survey responses on political attitudes and self-placement, which are then mapped into ideological positions (Fiorina and Abrams, 2008; Draca and Schwarz, 2024; Algan et al., 2017; Gethin, Martínez-Toledano and Piketty, 2022; Hangartner et al., 2019; Desmet, Ortuno-Ortín and Wacziarg, 2024). Such methods face two common shortcomings: the views voters hold do not necessarily translate into voting choices, and some views may be stigmatized or socially desirable, and thus not reported truthfully. Survey-based stated preferences can therefore differ substantially from the revealed preferences inferred from electoral outcomes, and the extent of this divergence can only be assessed by estimating preferences from actual election data.

Third, this paper contributes to the large and growing literature on the causes of the rise and success of populist parties and leaders in the Western world. An extensive body of work examines the various drivers of populist voting, including economic factors such as trade and the 2008–2009 Global Financial Crisis (Autor et al., 2020; Colantone and Stanig, 2018*b,a*; Algan et al., 2017; Guiso et al., 2021; Dippel et al., 2022), immigration (Dustmann, Vasiljeva and Piil Damm, 2019; Alesina and Tabellini, 2024), and cultural factors (Margalit, 2019; Algan et al., 2018; Gennaioli and Tabellini, 2025), among others. While these drivers can be mapped onto distinct ideological dimensions (see, e.g., Rodrik, 2021, for a clear formalization of this logic and a meta-analysis of mechanisms related to globalization shocks), they do not reveal the relative importance of each, or their interplay. This paper addresses this gap by estimating at large voters’ ideal points and dimension weights in a multidimensional spatial model, and tracing the evolution of preferences across demographic groups.

In this respect, the paper is closely related to Danieli et al. (2023), who recover the preferences of voters of the radical-right populist parties in Europe. Unlike the present paper, however, that study

⁵The flexibility of the estimator is limited by the amount of data available, so the problem is especially acute in small samples. I provide a Monte Carlo simulation of weight estimation using the approach of this paper and that of Merlo and De Paula (2017) in Appendix Section B.

relies on survey-based self-reported vote choices and issue preferences; moreover, while its conceptual framework can be cast as a bliss-point model, it cannot separate a strengthening association between individual characteristics and ideal points on a given dimension from a change in the weight on that dimension.

Lastly, this paper contributes to the debate on the relative importance of ideological and anti-elite dimensions in populist voting. A large part of the literature emphasizes the role of host ideologies in shaping populist support (Acemoglu, Egorov and Sonin, 2013; Mudde, 2013; Norris and Inglehart, 2019; Guenther and Nunnari, 2025), whereas another strand adopts an *ideational* approach and highlights the role of populist attitudes distinct from Left–Right ideology (Mudde, 2004; Akkerman, Mudde and Zaslove, 2014; Van Kessel, 2015). Several studies have sought to compare the importance of these two factors directly using survey and experimental data (Silva, Fuks and Tamaki, 2022; Neuner and Wratil, 2022; Neuner et al., 2025). This paper shows that populist voting can take different forms, with different factors prevailing across regions and over time.

The rest of the paper is structured as follows. Section 2 presents the nonparametric model of choice, and Section 3 describes the scope of the approach. Section 4 describes the institutional context and the data. Section 5 covers identification and estimation. Section 6 presents the main results. Section 7 concludes.

2 Nonparametric Model of Party Choice

This section develops a nonparametric model of choice in a voting environment and outlines its identification. It draws on Merlo and De Paula (2017), which formalizes the logic of Voronoi partitions in the estimation of voter preferences, and on Tebaldi, Torgovitsky and Yang (2023), which develops a nonparametric approach for discrete choice models. Relative to the former, the framework differs in two key respects: identification does not rest on the assumption of a large support for candidate positions, and it does not aim to simultaneously recover a single distribution of voter preferences together with weights, but does it separately. Relative to the latter, it shifts attention from a product-valuation model in a price space to a pure characteristics model in a policy space, in which the main object of interest is individuals’ preferences⁶, and extends that model by estimating the relative importance of the different dimensions.

The section proceeds in three steps. Subsection 2.1 introduces the primitives, the identifying assumptions, and the target parameters (functionals of bliss point distributions). Subsection 2.2 shows that for known weights the estimation problem for target parameters is equivalent to a finite-dimensional linear problem. Subsection 2.3 establishes the set identification of weights, and combines

⁶See Berry and Pakes (2007) for an extensive discussion of the differences between these models.

the two steps into a single algorithm.

2.1 Setup

Individual i chooses a candidate⁷ j from a set $\mathcal{J} = \{1, \dots, J\}$. Each choice environment is characterized by a vector of potentially endogenous candidate positions $Y_i = (Y_{i1}, \dots, Y_{iJ})$, which I model as discretely distributed. An individual may face one or more such environments at the same time. In an electoral setting, the ballot provides a natural representation of this set. Individual i has a bliss point position X_i known to themselves, but latent from the perspective of the researcher.

I adopt a spatial voting model (Downs, 1957; Hinich and Munger, 1996) and assume that valuations are a function of distance between the individual's own bliss point and the positions of the candidates:

$$V_{ij} = V(X_i, Y_{ij}) = U(d(X_i, Y_{ij})),$$

where $d(X_i, Y_{ij})$ denotes the Euclidean distance between the two points in space, and $U(\cdot)$ is a strictly decreasing function. Note that this functional form does not presuppose any specific cardinal loss function, and is invariant to strictly monotone transformations of the (weighted) Euclidean distance.⁸

For simplicity of exposition, in this and the next subsection I consider the unweighted Euclidean distance ($d(X_i, Y_{ij}) = \sqrt{(X_i - Y_{ij})'(X_i - Y_{ij})}$). Subsection 2.3 extends the analysis to the weighted Euclidean distance and discusses separate identification of the weights.

The policy space $\mathcal{R}$ is assumed to be nonempty and compact. For the purposes of this paper I focus on the two-dimensional policy space, $\mathcal{R} \subset \mathbb{R}^2$, and, hence, $X_i, Y_{ij} \in \mathcal{R}$. An individual i 's choice is:

$$\text{Choice}_i = \text{Choice}(X_i, Y_i) = \arg \max_{j \in \mathcal{J}} V_{ij} \quad (1)$$

The primitive object of interest is the conditional distribution of bliss points X_i , given candidate positions Y_i , market (region) M_i , and demographic covariates D_i . I assume throughout that this conditional distribution is absolutely continuous, so that, for each realization $Y_i = y$, $M_i = m$, and $D_i = d$, it admits a density $f(x | y, m, d)$; since the set of bliss points indifferent between any two candidates has Lebesgue measure zero, ties between choices then occur with probability zero. Let $\mathcal{F}$

⁷ *Candidate* is used here to denote any choice option appearing on the ballot, including parties, individual candidates, or referendum choice options, among others.

⁸ See Kendall, Nannicini and Trebbi (2015); Cruz et al. (2024) for the estimation of the functional form of the loss function. The analysis is not invariant, however, to arbitrary alternative notions of distance. This is because the identification argument relies on the geometry of the regions of the ideological space consisting of points closest to each candidate, i.e. Voronoi cells. Alternative metrics may generate different choice regions. For example, in a multidimensional space, using the sum of absolute coordinate-wise differences between a bliss point and a candidate position corresponds to imposing Manhattan, rather than Euclidean, geometry.

denote the set of all such conditional density functions.

Conditioning on Y_i allows for the possibility that the choice environment faced by an individual is *endogenous*: candidate positions may reflect voter characteristics beyond the observed demographics d and market m , in which case $f(x \mid y, m, d)$ varies with y . I assume there exists an observed variable Z_i (an instrument) such that, for all (z, m, d) and all y in the support of Y_i given $(Z_i, M_i, D_i) = (z, m, d)$, the conditional density of bliss points satisfies

$$f(x \mid y, z, m, d) = f(x \mid z, m, d) \quad \text{for all } x. \quad (2)$$

The leading case is when candidate positions are unrelated to voters' unobserved characteristics conditional on (m, d) alone; the assumption then is $f(x \mid y, m, d) = f(x \mid m, d)$ for all y . To save on notation, I proceed under this special case throughout; when a non-degenerate instrument is required, every argument below applies with z added to the conditioning set (m, d) , and objects conditional on (m, d) alone are recovered by averaging over the observed distribution of the instrument, $f(x \mid m, d) = \mathbb{E}[f(x \mid Z_i, m, d) \mid M_i = m, D_i = d]$. The assumption has identifying content only if the instrument is *relevant*, i.e., if candidate positions vary with the instrument. Section 3 discusses institutional settings, and identification strategies from the literature, under which the Assumption 2 is plausible.

Common quantities of interest can be written as functionals of f . For instance, one could be interested in estimating a share of individuals with given demographics with bliss points that are left-leaning on a given dimension, $\underline{\mathcal{X}} = \{x \in \mathcal{R} : x_1 \leq \underline{x}_1\}$ for a known threshold $\underline{x}_1$:

$$\theta(m, d; f) = \int_{\underline{\mathcal{X}}} f(x \mid m, d) dx$$

Denote values of interest similar to the one just above as *target parameters*. Other quantities of interest could characterize support for political alternatives or the consequences of institutional rules – for instance, the share of candidate j 's supporters with bliss points in a given region, or counterfactual vote shares under alternative positions or rules. The goal is to infer the values of $\theta(m, d; f)$ that are consistent with both the observed data and the assumptions imposed.

The first assumption is that the support of f is contained in a known subset of $\mathcal{R}$. In the case of the voting application, this naturally translates into assuming a bounded policy space, preventing unreasonably extreme ideal points:

$$\int_{\bar{\mathcal{X}}} f(x \mid m, d) dx = 1 \quad \text{for all } (m, d), \text{ where } \bar{\mathcal{X}} \text{ is a known subset of } \mathcal{R} \quad (3)$$

In voting contexts, only vote shares aggregated over specific administrative units (e.g. municipalities, precincts) are observed; I refer to these units as *submarkets* of a market and index them by s , with $\mathcal{S}(m)$ denoting the submarkets of market m . For each submarket a researcher observes its demographic composition; denote $q_{s,d}$ to be the share of group with $D_i = d$ in submarket s , and $\sum_d q_{s,d} = 1$. The observed data for the submarket s in market m with candidate positions y are then:

$$s_j(y, s) = \sum_d q_{s,d} \mathbb{P}[\text{Choice}_i = j \mid Y_i = y, M_i = m, D_i = d]$$

Each density of bliss points $f \in \mathcal{F}$ also implies a set of vote shares:

$$s_j(y, s; f) = \sum_d q_{s,d} \int_{\mathcal{X}_j(y)} f(x \mid m, d) dx$$

where $\mathcal{X}_j(y) = \{x \in \bar{\mathcal{X}} : V(x, y_j) > V(x, y_k) \ \forall k \neq j\}$ is the attraction region of candidate j under positions y , with y_j the position of candidate j in configuration y . A density f is consistent with the observed choice shares, if

$$\underbrace{s_j(y, s)}_{\text{observed}} = \underbrace{s_j(y, s; f)}_{\text{predicted}} \quad \text{for all } j \text{ and all pairs } (y, s) \text{ observed in the data} \quad (4)$$

The sharp identified set of bliss-point densities is defined as the subset $\mathcal{F}^*$ of $f \in \mathcal{F}$ that satisfy (2), (3), (4). The main object of interest in the paper throughout is the sharp identified set for the target parameter, constructed as:

$$\Theta^* = \{\theta(f) : f \in \mathcal{F}^*\}$$

2.2 Equivalent finite-dimensional problem

How could one compute Θ^* ? Note that $\mathcal{F}$ is an infinite-dimensional object, which does not allow solving the problem via grid search. [Tebaldi, Torgovitsky and Yang \(2023\)](#) develop an approach for checking whether there exists some $f \in \mathcal{F}^*$ that would generate a given conjectured value $t = \theta(f)$ for the target parameter. Their paper proves that the existence of such an f is equivalent to the existence of a solution to a finite-dimensional system of linear equations.

The finite-dimensional system is constructed by first partitioning the policy space into the smallest number of sets needed to describe choice behavior under all relevant candidate positions. Let $\mathcal{Y}$ denote a finite set of candidate position vectors that contains the observed candidate positions, as well as any additional positions relevant for evaluating the target parameter. The Minimal Relevant

Partition (MRP) is the coarsest partition of the policy space $\mathcal{R}$ with the following property: any two bliss points in the same subset produce the same choice behavior for all $y \in \mathcal{Y}$, while any two bliss points in different subsets produce different choice behavior for at least one $y \in \mathcal{Y}$. Formally, the MRP is the coarsest partition $\mathbb{X}$ of $\mathcal{R}$ such that

$$x, x' \in \mathcal{X} \text{ for some } \mathcal{X} \in \mathbb{X} \iff \text{Choice}(x, y) = \text{Choice}(x', y) \text{ for all } y \in \mathcal{Y}.$$

In other words, whenever two different bliss points x, x' belong to the same set from the MRP, this implies that x, x' yield the same choice behavior across all choice environments in $\mathcal{Y}$. Note two distinctive features of this setup. First, it does not require researchers to normalize any of the options. Intuitively, this is because individuals' valuations of the options are well defined relative to one another (see discussion in [Merlo and De Paula \(2017\)](#) on identification of the spatial voting model). Second, because only relative comparisons between candidates are exploited, the framework does not naturally accommodate the turnout decision, which would require an absolute utility threshold for abstention⁹.

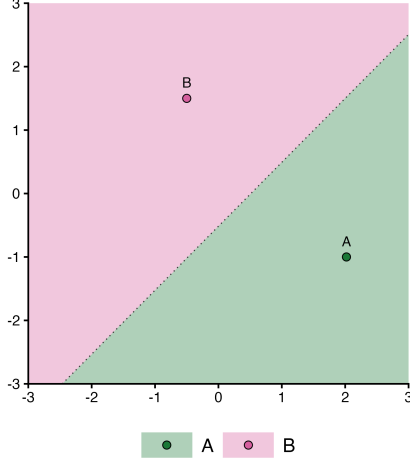
Figures 1(a) and 1(b) illustrate the partitions of the policy space generated by two hypothetical position vectors, y_1 and y_2 , in a setting where individuals face different candidate choice sets across elections. For consistency with the application that follows, the illustration uses a two-dimensional policy space. Figure 1(c) shows the two partitions overlaid.

This mirrors the structure of the application ahead, where voters face different choice environments across geographic areas: individuals in the same market m confront different candidate sets (y_1 and y_2 in Figure 1) depending on the electoral district to which they belong. The identifying variation comes from the differences in choice environments of individuals with plausibly similar preferences (same market m , demographics d).

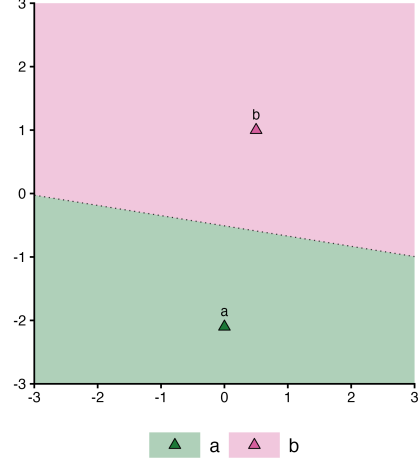
The MRP is obtained by partitioning the continuum of bliss points into sets that generate identical choice behavior for all positions in $\mathcal{Y}$. When the object of interest is behavior under these candidate positions, attention can therefore shift from densities to *mass functions defined over the sets of the MRP*.

To illustrate this, consider the target parameter defined above, share of voters with left-leaning bliss points. Figure 1 (d) graphically represents this quantity as the share of voters whose bliss points lie in area T , where $T = \{x \in \bar{\mathcal{X}} : x \leq (0, 0)\}$. In the MRP representation, evaluating such a target parameter requires refining the partition by the boundary of T , so that T can be represented as a union of MRP cells. In the Figure 1 (d) T consists of three subsets, $\mathcal{X}_1, \mathcal{X}_2, \mathcal{X}_3$. Hence, the target parameter can be written as:

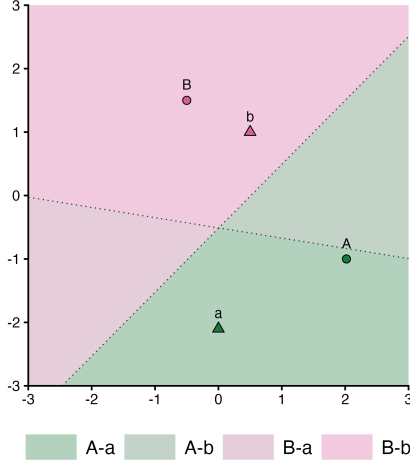
⁹In the application ahead, I account for the differential turnout rates across demographic groups across markets.



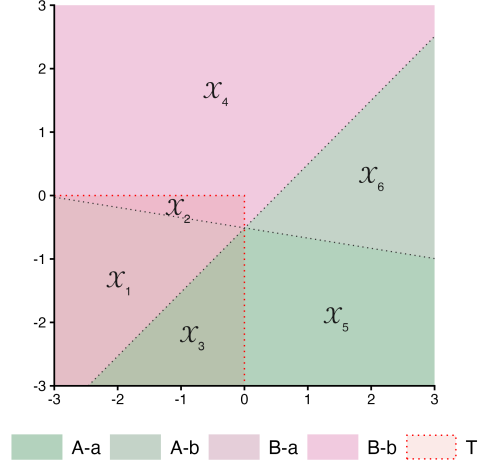
(a) Election 1



(b) Election 2



(c) Overlay



(d) MRP: Overlay with Target Parameter

Notes: The figure illustrates a hypothetical setting with elections involving two candidates in each contest, (A, B) and (a, b) . In the two upper panels, the shaded pink region denotes the attraction area of candidates B and b , the green region – of candidates A and a in Elections 1 and 2, respectively. Panel (c) shows the overlay of the partitions in panels (a) and (b). Panel (d) presents the same overlay together with the region corresponding to a target parameter. $\mathcal{X}_n$ denote unique cells that characterize observed behavior in Elections 1 and 2 or a target parameter.

Figure 1. Example of partitions of the policy space in two elections with two parties ((a), (b)) and their overlay (c) with a target parameter (d).

$$\theta(m, d; f) = \int_{\mathcal{X}_1} f(x | m, d) dx + \int_{\mathcal{X}_2} f(x | m, d) dx + \int_{\mathcal{X}_3} f(x | m, d) dx,$$

which depends not on f per se, but only on the mass that f places on $\mathcal{X}_1, \mathcal{X}_2, \mathcal{X}_3$. Focusing on these masses transforms the intractable infinite-dimensional problem of searching over all $f \in \mathcal{F}$ into a tractable finite-dimensional problem of searching over a finite set of non-negative numbers that sum to unity.

Denote the set of conditional mass functions on $\mathbb{X}$:

$$\Phi = \{\phi \in \mathbb{R}_+^{d_\phi} : \sum_{\mathcal{X} \in \mathbb{X}} \phi(\mathcal{X} | m, d) = 1 \ \forall (m, d) \in \text{supp}(M_i, D_i)\},$$

where d_ϕ is the cardinality of $\mathbb{X} \times \text{supp}(M_i, D_i)$. Denote $\mathbb{X}_j(y)$ the sets in the MRP associated with choosing j under y . Then $\phi \in \Phi$ is consistent with some f that satisfies 4 only if it satisfies:

$$\underbrace{s_j(y, s)}_{\text{observed}} = \underbrace{s_j(y, s; \phi) = \sum_{\mathcal{X} \in \mathbb{X}_j(y)} \sum_d q_{s,d} \phi(\mathcal{X} | m, d)}_{\text{predicted}} \quad \forall j \text{ and observed } (y, s). \quad (5)$$

Similarly, $\phi \in \Phi$ is consistent with an f that satisfies 3 only if it satisfies:

$$\sum_{\mathcal{X} \in \bar{\mathbb{X}}} \phi(\mathcal{X} | m, d) = 1 \text{ for all } (m, d) \quad (6)$$

where $\bar{\mathbb{X}}$ is the subset of cells of $\mathbb{X}$ that are contained in $\bar{\mathcal{X}}$ (the partition can always be refined so that every cell lies either inside or outside $\bar{\mathcal{X}}$).

For any $f \in \mathcal{F}$, define $\bar{\phi}(f) \in \Phi$ as

$$\bar{\phi}(f)(\mathcal{X} | m, d) = \int_{\mathcal{X}} f(x | m, d) dx \quad \text{for all } \mathcal{X} \in \mathbb{X}, (m, d) \in \text{supp}(M_i, D_i).$$

Next, assume that the target parameter is representable in terms of the cell masses: there exists a known function $\bar{\theta} : \Phi \rightarrow \mathbb{R}^{d_\theta}$ such that

$$\theta(f) = \bar{\theta}(\bar{\phi}(f)) \quad \text{for all } f \in \mathcal{F}. \quad (7)$$

This assumption can always be satisfied by choosing $\mathcal{Y}$ (and hence the partition) to include all positions and regions relevant for evaluating the target parameter; in the leading example above,

$$\bar{\theta}(\phi) = \phi(\mathcal{X}_1 \mid m, d) + \phi(\mathcal{X}_2 \mid m, d) + \phi(\mathcal{X}_3 \mid m, d).$$

The main propositions in [Tebaldi, Torgovitsky and Yang \(2023\)](#) can then be adapted to the voting context as follows:

Proposition 1. *Suppose that [7](#) is satisfied. Let $\Phi^* \equiv \{\phi \in \Phi : \phi \text{ satisfies } \text{5, 6}\}$, and for any $t \in \mathbb{R}^{d_\theta}$, let $\Phi^*(t) \equiv \Phi^* \cap \{\phi \in \Phi : \bar{\theta}(\phi) = t\}$. Then $t \in \Theta^*$ iff $\Phi^*(t) \neq \emptyset$.*

The nontrivial direction of Proposition 1 is sufficiency: any $\phi \in \Phi^*(t)$ can be extended to a density $f \in \mathcal{F}^*$ with $\theta(f) = t$ by distributing the mass assigned to each cell arbitrarily (for instance, uniformly) within that cell; see Appendix in [Tebaldi, Torgovitsky and Yang \(2023\)](#) for the formal argument.

Proposition 2. *If $\bar{\theta}$ is continuous on Φ and $d_\theta = 1$, then $\Theta^* = [t_{lb}^*, t_{ub}^*]$, where*

$$t_{lb}^* = \min_{\phi \in \Phi^*} \bar{\theta}(\phi), \quad t_{ub}^* = \max_{\phi \in \Phi^*} \bar{\theta}(\phi).$$

Existence of the extrema and the interval form follow because Φ^* is a compact convex polytope and $\bar{\theta}$ is continuous; when $\bar{\theta}$ is linear in ϕ , as in the leading example, both bounds are computed by linear programming.

2.3 Weighted spatial voting model

Assume individual valuations of choice options can be expressed as:

$$V_{ij} = V(X_i, Y_{ij}; W_i) = U(d_{W_i}(X_i, Y_{ij})),$$

where W_i is a weighting matrix whose diagonal elements govern the relative importance of the dimensions, and whose off-diagonal elements determine the interaction between the two dimensions. For computational and identification purposes, I set the off-diagonal elements to zero:

$$W_i = \begin{pmatrix} w_{i1} & 0 \\ 0 & w_{i2} \end{pmatrix}, \quad w_{i1}, w_{i2} > 0.$$

In the two-dimensional setup the distance is $d_{W_i}(X_i, Y_{ij}) = \sqrt{(X_i - Y_{ij})^\top W_i (X_i - Y_{ij})}$. The case considered before is a special case with $w_{i1} = w_{i2}$. Since choices depend only on the relative weights assigned to the dimensions, not on their absolute levels, I normalize $w_{i1} = 1$ without loss of generality and treat $w_{i2} \in \mathbb{R}_+$ as the relative weight of the second dimension.

Throughout this section I impose that the weighting matrix is common across individuals in a

given market and denote: $W_i = W = \text{diag}(1, w_2)$ for all i . In the application, w_2 is allowed to differ across region–years but is common within them.

This subsection demonstrates that one can separately identify the weight w_2 and the probability mass function of bliss points ϕ and derive the sharp bounds for the target parameters using those.¹⁰

The two observational-equivalence conditions 4, 5 for f and ϕ respectively can now be recast as follows. The predicted vote share of candidate j is:

$$s_j(y, s; f, w) = \sum_d q_{s,d} \int_{\mathcal{X}_{j;w}(y)} f(x \mid m, d) dx,$$

where $\mathcal{X}_{j;w}(y) = \{x \in \bar{\mathcal{X}} : V(x, y_j; w) > V(x, y_k; w) \ \forall k \neq j\}$. And it matches the data if

$$s_j(y, s; f, w) = s_j(y, s) \quad \forall j \text{ and observed } (y, s). \quad (8)$$

Likewise, for the equivalent finite dimensional problem, denote by $\mathbb{X}_w$ the MRP generated by $\mathcal{Y}$ under weight w , and by $\mathbb{X}_{j;w}(y)$ its cells on which j is chosen under y ; then

$$\sum_{\mathcal{X} \in \mathbb{X}_{j;w}(y)} \sum_d q_{s,d} \phi(\mathcal{X} \mid m, d) = s_j(y, s) \quad \forall j \text{ and observed } (y, s) \quad (9)$$

The algorithm for obtaining bounds for the target parameter in the presence of an unknown weight is then the following. First, obtain the identified set of weights over a candidate space $\mathcal{W} \subseteq \mathbb{R}_+$,

$$W^* = \{w \in \mathcal{W} : \exists \phi \in \Phi \text{ such that (9) and (6) are satisfied}\}.$$

Second, for each $w \in W^*$ obtain $\Theta^*(w)$ by applying Propositions 1–2 to the partition $\mathbb{X}_w$. The identified set for the target parameter is then $\Theta^* = \bigcup_{w \in W^*} \Theta^*(w)$. Since each $\Theta^*(w)$ is an interval, Θ^* is contained in

$$\left[\min_{w \in W^*} t_{lb}^*(w) ; \max_{w \in W^*} t_{ub}^*(w) \right],$$

¹⁰The closest argument in the literature is [Merlo and De Paula \(2017\)](#), who prove nonparametric point identification of weights and distributions of bliss points in a spatial voting model. Their analysis relies on a large support assumption for candidate positions. In particular, to establish joint identification of bliss points and weights, they rely on the results of [Ichimura and Thompson \(1998\)](#), who prove nonparametric identification in a binary discrete choice model using the Cramer–Wold device: there exists a two-candidate configuration such that the affine hyperplane separating the two candidates delivers different election outcomes under two distinct preference distributions. Building on these results, they extend them to 1) include multiple candidate positions, and 2) introduce weights. Both 1) and 2) are based on the large support assumption, i.e. that one could sample arbitrary candidate configurations from $\mathbb{R}^2$. This assumption is infeasible in political-economy environments, where the set of candidate positions is typically finite and confined to a bounded subset of $\mathbb{R}^2$. Hence, the setup of this paper accommodates the fact that there are just finitely many observed configurations in a bounded policy space.

and these are the bounds I report. Appendix E describes this estimation procedure in the application.

Note two features of this problem. First, the equations in the finite-dimensional problem (9) are defined by the available partition cells in a minimal relevant partition $\mathbb{X}_{j;w}(y)$, and these cells vary with the weight. Second, given the cells, the distribution of bliss points is matched by fitting $\phi(\mathcal{X} \mid m, d)$ over the available cells. Intuitively, the former is the source of identification for the weights, and the latter the source of identification for the bliss-point distribution: the weight is identified by *which* cells the data require, the mass function by *how much* mass those cells must carry. I now provide intuition for the argument for weight identification.

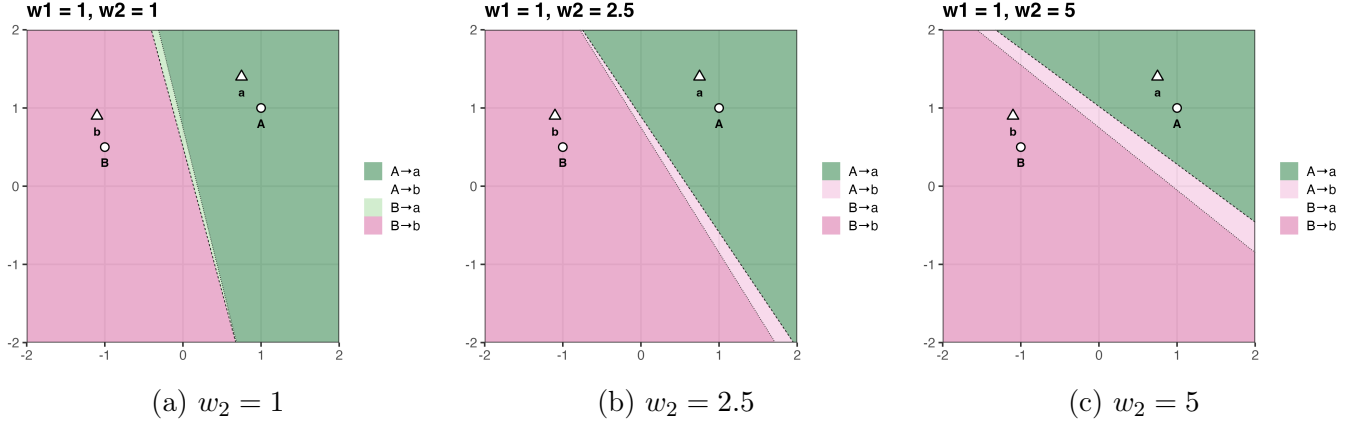
Imagine a researcher observes the results of two elections with two candidates each (candidates A, B and a, b , respectively). The goal is to use these two elections to obtain information about f and W . The observed marginals (vote shares) – s_A, s_B, s_a, s_b – are informative about the joint values $(s_{Aa}, s_{Ab}, s_{Ba}, s_{Bb})$ of interest. Consider the hypothetical scenario from Table 1, in which candidate A and candidate b obtain the majority of votes in respective elections.

	a	b	Total
A	$[0, 0.1]$	$[0.6, 0.7]$	0.7
B	$[0, 0.1]$	$[0.2, 0.3]$	0.3
Total	0.1	0.9	1

Table 1. Marginals (observed) and Fréchet bounds on the joint cell shares

This scenario implies that the cell A – b hosts at least 60% of all voters. Hence, the minimal relevant partition of those elections must contain a cell that would rationalize such voter behavior. The claim is that such a cell might not appear under every configuration of weights, and that the weights yielding the partition that contains this cell are best suited to explain the behavior of voters. Figure 2 illustrates this point and demonstrates that, in this hypothetical example, the weight configuration $w_2 = 1$ is not well suited to explain the data. Moreover, Figure 2 illustrates the nature of the set-identification of weights in this setup: $w_2 = 2.5$ and $w_2 = 5$ effectively yield the same partition of the policy space and cannot be told apart.

The example shows that weights are identified by the available cells in the minimal relevant partition. Appendix A proves that each cell of the minimal relevant partition is non-empty only for weights in an interval, so the weights consistent with the cells the data require form an intersection of such intervals—itsself an interval containing the true weight. Thus the identified set for w_2 is contained in an interval on $\mathbb{R}_+$. The informativeness (length) of this interval is then an empirical question. Lieber, Tebaldi and Torgovitsky (2026) provide a similar argument about the set-identification of relative weights for different characteristics in an additive RUM.



Notes: The figure illustrates a hypothetical setting with elections involving two candidates in each contest, (A, B) and (a, b) . Each panel overlays the two elections in the policy space $\mathbb{R}^2$: triangles mark the positions of candidates A and a , circles mark candidates B and b . The shaded regions denote the attraction areas of the joint outcomes: $A \rightarrow a$ (dark green), $A \rightarrow b$ (light green), $B \rightarrow a$ (light pink), and $B \rightarrow b$ (dark pink), where $X \rightarrow y$ denotes voters who prefer X in Election 1 and y in Election 2. The cells present in a given panel constitute the minimal relevant partition for that weight configuration. Across panels, the dimension weight w_2 increases ($w_1 = 1$ throughout).

Figure 2. Illustration of minimal relevant partitions induced by the same candidate configurations $(A, B), (a, b)$ under different values of w_2 , $w_1 = 1$.

In practice there need not be a unique partition that best explains the data: because the number of feasible cells (over which voter mass is estimated) weakly exceeds the number of observed vote-share marginals, the problem is an under-identified linear system, and several partitions may rationalize the data equally well. Two observations are worth highlighting. First, as Lemma 1 in Appendix A shows, each cell is generated only by weights lying in an interval. Identifying power then comes from Fréchet-type necessity constraints¹¹. Second, additional elections with distinct vote-share configurations weakly increase the number of such constraints. The admissible weights must lie in the intersection of the intervals associated with every cell required to carry positive mass; since the data were generated at the true weight, every cell the data require is feasible there, so this intersection is a non-empty interval containing the truth, and it weakly shrinks as elections accumulate. The size and shape of the identified set are thus an empirical object.

Appendix B reports simulations comparing the approach of this paper—which exploits the fact that different weights generate different cells—with that of Merlo and De Paula (2017), who recover the weights jointly with the distribution of bliss points using a sieve minimum-distance estimator. The simulations suggest that the true weight value belongs to the recovered set under the approach of this paper, whereas the weights recovered jointly with the bliss-point distribution fall outside the identified set. The intuition is that identification of the weights from the cell structure requires no

¹¹For example, if candidate (or coalition) i receives a share of 0.52 in election e and candidate j receives 0.51 in election e' , at least 3% ($0.52 + 0.51 - 1$) of voters must choose the pair (i, j) , so the corresponding cell must be non-empty.

assumption on the distribution of bliss points, so it is not contaminated when that distribution is misspecified; the sieve estimator, by contrast, must approximate the bliss-point density, and any residual misspecification is absorbed into the weight estimate. Appendix B also demonstrates that the identified set for weights is shrinking with a number of elections. Introducing noise in voters’ perceptions of candidate positions leaves this conclusion unchanged: the identified region tightens around the true weight, while the sieve estimates remain outside it.

The identification argument above does not rely on the weight being common across demographic groups within a market, and a similar argument goes through when W varies across demographic groups too. In practice, however, group-specific weights are considerably more demanding on the data. Informative group-level intervals would require either group-level choice data or far richer variation in candidate configurations than is available in the application.

3 Scope of the Approach

This section discusses the settings in which the identifying assumption of the approach introduced in the previous section is plausible, highlights its broad applicability, and considers possible extensions and generalizations of the model, including the incorporation of strategic voting.

The approach offers a method for estimating voters’ preferences within a spatial voting framework using real-world aggregate voting data. Identification rests on comparing the behavior of voters who share the same preference distribution across different choice environments: either directly, $f(x \mid y, m, d) = f(x \mid m, d)$ for all y , or conditional on an instrument z for candidate positions, $f(x \mid y, z, m, d) = f(x \mid z, m, d)$. While such settings may at first seem uncommon, they are in fact pervasive, and closely related identifying assumptions are already used in the political economy literature and beyond. I organize the discussion around three types of settings.

First, consider simultaneous voting, where the same voters face different choice environments because several elections take place at the same time. The main advantage of this setting is that the identifying condition $f(x \mid y, m, d) = f(x \mid y', m, d)$ holds by construction. Its principal limitation, however, is that the number of simultaneous elections is typically small, resulting in a coarse partition of the policy space. Consequently, the inferences that can be drawn using this approach may be less informative.

Among other cases, simultaneous voting is used in the United States, where elections for multiple offices often take place on the same day; in Australia, where voters cast ballots for both chambers of parliament simultaneously; in Italy, where general elections fill both the Chamber of Deputies and the Senate on the same day; or in Switzerland, where voters may face multiple electoral ballots as

well as referendum questions on the same election day.

Second, *ranked-choice voting (RCV)* provides another setting in which the same voters face varying choice environments. A ranking can be read as a sequence of choices from shrinking choice sets: the ballot contains the same candidates throughout, but the relevant choice environment changes across ranking stages, since a candidate who has already been ranked is no longer available in the voter’s subsequent choice set. As a result, lower-ranked choices can be especially informative for identifying voters’ positions in the policy space. As with simultaneous elections, the identifying condition holds by construction.

RCV is used in national elections in Australia, Ireland, Malta, and Sri Lanka, among others, and has been adopted for state and federal elections in Maine and Alaska as well as for local races in several US cities.

Third, consider the case when *different* voters face *different choice environments across geographic areas*. Such variation is generated by design in most electoral systems, since geographically based representation is a standard feature: examples of first-past-the-post systems across districts include U.S. Senate and House of Representatives elections and House of Commons elections in Canada and the United Kingdom; examples of mixed systems, combining elements of first-past-the-post and proportional representation, include legislative elections in Germany, Hungary, Mexico, Japan, and South Korea, among many others.

In this setting, however, the identifying condition no longer holds by construction. The political economy literature posits that politicians’ and voters’ preferences are closely related, whether because of politicians’ opportunistic incentives, intrinsic preference congruence, or voters’ responsiveness to politicians’ cues (Persson and Tabellini, 2002; Osborne and Slivinski, 1996; Callander and Carbajal, 2022). A researcher should therefore justify the comparison of different choice environments. In practice, this can be addressed in several complementary ways: (1) by identifying contexts or subcontexts in which the condition holds directly; (2) by instrumenting candidate positions; or (3) by relaxing the equality condition and assessing the sensitivity of the results to such relaxation. I now discuss each approach with reference to papers that adopt it.

(1) Contexts where the condition holds directly. Longuet-Marx (2025) exploits congressional district borders within states, following the approach by Spenkuch and Toniatti (2018), and compares only contiguous precinct pairs, which lie in different electoral districts but are very similar otherwise. In the terminology of this paper, he posits that $f(x \mid y, m, d) = f(x \mid y', m, d)$ holds within each precinct pair.¹² This implies that the overlay of choice situations should be done only within the

¹²In terms of Assumption 2, this amounts to letting the instrument z index the precinct pair: conditional on the pair, the two precincts share the same preference distribution while facing different candidate sets.

pairs of precincts, which limits the available identifying variation. However, by considering many such pairs, a researcher may still be able to recover more about the underlying primitives of interest than from single elections alone. I illustrate this with a simple example in Appendix C.

Merlo and De Paula (2017) and Danieli et al. (2023) treat national-level party positions across European countries as exogenous to voters’ preferences and estimate voter preferences by exploiting cross-country variation in party positions. While the former is closer to the main setup considered here and relies on aggregate voting data, Danieli et al. (2023) combines survey data with party positions. Nevertheless, the same logic of preference estimation applies in this setting as well.

(2) Instrumenting candidate positions. Iaryczower, Montero and Kim (2024) exploit variation in candidates’ positions across Brazilian municipalities to estimate voters’ preferences. Their empirical strategy relies on using the policy positions of mayoral and state-level candidates in the most recent local electoral cycle as an instrument z for current local party positions – the non-degenerate case of Assumption 2.

(3) Relaxing the equality condition. Tebaldi, Torgovitsky and Yang (2023) suggest a way to relax the equality assumption on preference distributions by allowing for pointwise differences in the density. In the voting context, one adaptation of their idea would be to impose $f(x \mid y, m, d) \in [(1 - \kappa)f(x \mid y', m, d), (1 + \kappa)f(x \mid y', m, d)]$ for all x and all y, y' , where κ is a researcher-chosen parameter governing the degree of difference across preference distributions in different regions. Setting $\kappa = 0$ recovers the baseline condition, and the identified set is weakly increasing in κ , so reporting bounds across several values of κ amounts to a sensitivity analysis with respect to the identifying assumption.

The model can also be extended to incorporate other important features of voting behavior, such as strategic voting. Under strategic voting, some individuals may abandon their most preferred candidate if a vote for that candidate is perceived as *wasted*. The key implication is that such behavior generates different effective candidate choice sets across voters. Those who vote for underdog candidates can plausibly be treated as sincere and therefore as facing the full choice set. By contrast, for voters who support viable contenders, it is unclear whether they are choosing their most preferred candidate overall or only their most preferred candidate among viable alternatives. For these voters, the relevant choice set should therefore be restricted to viable candidates. Although this coarsens the partition and may therefore widen the resulting bounds, it makes them more robust to strategic voting.

I now turn to describing an application of the model to one of the most widespread electoral regimes: the mixed electoral system.

4 Application: Rise of Populism

One of the central puzzles in the political economy literature is the rise of populism, and in particular radical-right populism, in Western democracies. The empirical literature has largely focused on identifying the causal mechanisms underlying the rise of populist politics, examining their effects on voters' preferences, and analyzing the contextual and individual-level factors that shape these preference responses (see [Guriev and Papaioannou \(2022\)](#) for an overview).

While understanding specific mechanisms is essential, it is equally important to develop a broader understanding of the preferences of populist voters. One recent attempt in this direction is [Danieli et al. \(2023\)](#), who study the rise of populist radical-right parties in Europe using survey evidence over time. To my knowledge, this paper is the first to provide revealed-preference evidence on the preferences of populist voters at large. Focusing on revealed-preference data rather than surveys is important because it avoids problems of survey misreporting, in particular the underreporting of populist vote choice, and makes it possible to recover the relative importance of different dimensions of political competition, which is especially difficult to measure using survey data.¹³

4.1 The Context

Over recent decades, the German electoral landscape has undergone important changes, most notably with the rise of the Alternative for Germany (AfD), a populist radical-right party. For the first time in postwar Germany, a party to the right of the Christian Democrats (CDU/CSU) surpassed the 5% threshold and entered the Bundestag. In the most recent federal election, in 2025, the AfD finished second, winning more votes than several parties of the traditional political establishment, including the Social Democratic Party (SPD), one of Germany's two major parties.

Alternative for Germany was established in 2012 to oppose the policies pursued by the German government to fight the Euro crisis. From the beginning, the party was distinctive in its radical-right and anti-establishment stance. In the run-up to the 2013 Bundestag Elections, its main concerns were Euro Crisis and the Greek bailout, as well as the adherence to the rule of law and stronger participatory democracy ([AfD, 2013](#)). In 2015, the party split; many founding members resigned, and the conservative, anti-immigrant wing took over the leadership, which it has held since.

Hence, throughout the study period (2013–2025), the AfD was consistently ranked as the most right-wing and anti-elite among Germany's main parties ([Jolly et al., 2022](#)), and it stood out as the most conservative and anti-elite party with a realistic chance of winning seats in the Bundestag.

¹³This is because voters tend to adjust their reported positions to align with those of their party, particularly when the issue is not especially important to them. Yet, when relative weights are estimated using survey data, such proximity in reported positions may be mistakenly interpreted as evidence that the dimension is more important to voters.

This motivates the choice of the two dimensions of policy space in my application – Left-Right and Pro-Elite – Anti-Elite (I refer to the latter as just *Anti-Elite* throughout).¹⁴ I discuss what I mean by those dimensions below in the data section.

The focus of this application is the estimation of AfD voters’ preferences over time. I now turn to the relevant institutional details and data.

My analysis draws on data from German federal parliamentary (Bundestag) elections. Germany is divided into 299 electoral districts across 16 states (e.g. Bavaria, Saxony). Under Germany’s mixed electoral system, each voter casts two votes: a first vote for a direct candidate in a single-member district, and a second vote for a party list. First votes are likely to be heavily shaped by the strategic incentives of first-past-the-post competition, whereas second votes among viable parties can more plausibly be treated as sincere (more on this below). In what follows, I rely exclusively on second-vote party shares.

My analysis is based on the assumption that *candidates’ positions in single member districts shape perceived party positions*. In other words, I use estimated candidate positions (see below) as proxies of parties’ positions in given districts. This allows me to exploit the variation in candidate positions, while preserving the assumption of the sincere voting for the party list. To document the validity of this assumption, I demonstrate that within federal states (which I treat later as separate *markets*) and within parties individual perceptions of party positions correlate with estimated candidate positions on the Left - Right dimension (Table F1). Moreover, the rich survey data from the time around 2021 Bundestag Elections document that voters extensively interact and know their local candidates¹⁵.

Two additional institutional features are important for the analysis. First, state-level (regional) party elites play a substantial role in candidate selection across electoral districts. Although German election law formally assigns nominations in single-member districts to local (district) party chapters, state party organizations act as de facto gatekeepers within the region. They can influence nominations both (i) by blocking district candidates they consider unsuitable and (ii) by determining the state party list for Bundestag elections. The latter matters because candidates may enter the Bundestag either through district races or through state lists; in practice, around two-thirds of candidates on state lists also run in single-member districts. Candidate selection is moreover often uncontested, as elites tend to preselect candidates in advance — particularly when incumbents are involved (Det-

¹⁴This aligns with the literature in political science that highlights that populist parties build their agenda on two main dimensions: ideological and anti-system (See [Mudde \(2004\)](#); [Guriev and Papaioannou \(2022\)](#) among others), and the adjacent paper [Nikiforova \(2026\)](#) that identifies Left-Right and Anti-Elite as the two central dimensions of voting in Germany in recent years.

¹⁵The phenomenon that candidates’ characteristics affect parties’ perception in mixed electoral systems is referred to as "contamination effect" in the political science literature ([Ferrara, Herron and Nishikawa, 2005](#)).

terbeck, 2016; Schüttemeyer et al., 2024). These considerations both motivate the choice of states as separate markets and inform the identification argument.

Section H.1 provides more details and descriptive evidence on the nomination practices employed by the major parties. In particular, it highlights persistence in candidate nominations and patterns of nomination across single-member districts and state lists.

Second, electoral districts are drawn primarily on the basis of population size. Unlike in the United States, redistricting in Germany is nonpartisan and district boundaries are largely stable. A permanent Constituency Commission reviews population changes each electoral cycle and recommends adjustments to the Bundestag. Constituencies should remain within 15% of the average population, must be redrawn if deviations exceed 25%, and must lie entirely within a single state. In practice, only a small share of constituencies changes in any given electoral period.¹⁶

4.2 Data

In the application that follows a state (N=16) constitutes a market; markets are subdivided into electoral districts (N=299); and each electoral district contains multiple municipalities (N=11,000). Voting and demographic data are observed at the municipality level. Each electoral district features a distinct set of candidates. Figure F1 illustrates this division. I now describe the data used for estimation.

The demographic group taken as a unit of analysis consists of age-gender-education bracket in a given market (state), and distribution of preferences is estimated separately for each group within each state in a given year. An example of such a group is *female age 18-24 with higher education in Saxony in 2021*. Hence, to proceed with the estimation I need the data on age-gender-education composition and voting at the same level. I now turn to describing such data.

Municipal-level data. I collect data on turnout and second vote shares for the six main parties at the level of municipalities^{17,18}: *AfD*, *CDU/CSU*, *FDP*, *Greens*, *Left*, *SPD*. I abstract from coalition considerations and treat those parties to be *viable* in the years studied, which implies that if an individual truly prefers one of the considered parties, then there are no strategic considerations not to vote for this party.¹⁹ In turn, this implies that my estimates elicit true voting preferences among the voting population.

¹⁶See, for example, [Die Bundeswahlleiterin](#) (n.d.) for changes made prior to the 2025 Bundestag elections.

¹⁷The cumulative party vote shares for the studied viable parties are 94%, 95%, 91% in years 2013, 2017 and 2021 respectively. This implies that the analysis excludes only a small group of voters who supported non-viable parties

¹⁸The municipality is the smallest unit of analysis in Germany that can be geo-coded.

¹⁹It also assumes that voters expected these parties to clear the 5% threshold; the exceptions across years and parties are FDP/AfD (2013), Left (2021) at 4.7–4.9%, and FDP (2025) with 4.3%.

I also collect demographic data at a municipal level for each electoral year. I construct municipality-level information on the population composition of eligible voters in 24 age–gender–education brackets, comprising 6 age groups (18–24, 25–34, 35–44, 45–59, 60–69, 70+), two genders, and a binary distinction between individuals with and without higher education.²⁰ These demographics are measured precisely for 2021, drawing on the 2022 Census. To construct demographics for 2013 and 2017, I use changes in the size of each age–gender group at the district level ($N \approx 400$) and project these changes onto municipalities within each district, using the 2021 municipal composition as the baseline.

I restrict the benchmark sample to municipalities with fewer than 10,000 inhabitants.²¹ Cross-market variation in the demographic distribution is what identifies preferences separately by demographic group (Berry and Haile, 2021); smaller units retain more distinctive compositions, widening that variation. I test the sensitivity of my results to this threshold value below.

Candidate positions. A key feature of the considered setting is that voters face distinct party-affiliated candidates across single-member districts. To estimate candidates’ ideological positions, I combine two data sources:

- 1) GLES Candidate Study for the Bundestag elections 2013–2025, administered by the German Society for Electoral Research and Leibniz Institute for the Social Sciences (GLES, 2023),
- 2) Wen Wählen (*Whom to Vote For?*), a nonpartisan platform that collects standardized questionnaires from election candidates.

The data sources include candidates from parties with a high likelihood of clearing the 5% threshold. Consequently, candidates from all parties considered in my analysis were contacted for participation in each election year.

First, using the GLES Candidate Study, I construct a left–right score and an anti-elite score for each candidate from multiple survey items corresponding to each dimension (see Table H1 for the exact wording).²² I treat the left–right dimension as a bundle of economic, cultural and social policies, and construct it from questions spanning economic issues (e.g., whether the government should reduce income differences, or stay out of the economy), cultural issues (e.g., whether same-sex marriage should be banned, or whether women should be free to decide about abortion), and immigration (e.g., whether immigrants should adapt to German culture, or whether immigration

²⁰To do so, I use data on the age–gender composition of the population at the municipality level and complement them with data on the age–gender–education composition, which are available only for larger municipalities (with more than 10,000 inhabitants), municipal associations (*Gemeindeverbände*), or higher-level administrative units (*Kreise*).

²¹Note that this restriction effectively captures almost exclusively rural municipalities, making a separate urban–rural distinction unnecessary in this context. It also naturally excludes the city-states (Berlin, Bremen, and Hamburg) from the analysis.

²²I use the terms *anti-establishment* and *anti-elite* interchangeably.

benefits the German economy). In line with the literature, I treat the anti-elite dimension as the axis capturing antagonism toward elites and the claim that the people (rather than professional politicians or technocrats) should hold authority, as well as general distrust in democratic institutions (Mudde, 2004; Guriev and Papaioannou, 2022).²³ To quantify a candidate’s position on this dimension, I use items such as whether democracy offers citizens enough opportunity to participate, whether there should be nationwide referendums, and whether citizens have lost trust in politicians. See Table H1 for the list of questions used from GLES Candidate Study.

The same questions are used in each election year.²⁴ I standardize each item using z-scores and compute, for each candidate, the average across all available items within a given dimension to obtain a single index.²⁵

The participation in the survey is voluntary, and the response rate among the single member district candidates remains moderate – about 35% across the years.

To improve data coverage, I supplement the GLES Candidate Study with a second source, Wen Wählen (Whom to Vote For?), a nonpartisan platform that collects standardised policy questionnaires from election candidates across all major parties. Because the two surveys use distinct question sets, I bridge them using an elastic net regression. The training sample consists of candidates who responded to both surveys across the 2013, 2017, 2021 and 2025 federal elections; their estimated ideological positions from the GLES Candidate Study serve as the outcome variable, while their responses to the Wen Wählen questionnaire serve as predictors. Hyperparameters are selected by 10-fold cross-validation. I then apply the trained model to impute positions for candidates who appear only in the Wen Wählen dataset. Figure H5 plots the median values of candidate positions by party, dimension, and year; and Appendix H.2 provides more details on the estimation of candidates’ positions.²⁶

Complementing both surveys allows me to recover comparable positions for about 60% of candidates across the years. Parties differ by responding rates with CDU having the lowest response rates across the years, and Greens – the highest (37% and 77% respectively). About 50% of AfD

²³An important part of the AfD’s rhetoric is nationalism. While it could be attributed to either dimension, I treat it as belonging to the Left–Right dimension. As is clear from the imputation coefficients reported in Section H.2, the item most closely related to economic nationalism (“Reintroduce the Deutsche Mark!”) is a strong predictor of a candidate’s Left–Right placement in 2013.

²⁴Several questions aren’t available in selected years.

²⁵I prefer this method for its transparency. The correlation with indices based on Item Response Theory exceeds 0.85 across all years and dimensions.

²⁶External validation of such measures is difficult; one available benchmark is the within-party evolution of positions over time in the Chapel Hill Expert Survey (CHES) (Bakker et al., 2015). The within-party dynamics implied by my estimates are broadly consistent with CHES. In particular, the rightward and pro-elite shift of the CDU between 2021 and 2025, and the moderation of the AfD’s anti-elite positioning over the same period, are both mirrored in the change between the 2019 and 2024 CHES waves. Similarly, the hardening of AfD positions between 2017 and 2021 is consistent with the change between the 2014 and 2019 waves. Note that the CHES waves (2014, 2019, 2024) do not coincide with the German federal elections (2013, 2017, 2021, 2025), so the correspondence is approximate.

candidates are available across the years. Importantly, candidates who respond to the survey and those who do not achieve comparable first-vote shares and have similar winning probabilities across parties. For those candidates who are missing in the data, the median positions across the remaining single member districts are imputed. For the estimation that follows I keep only the electoral districts in which there are no more than three imputed candidates.

I also construct second best party choices using survey data across federal states and across elections. Standard discrete choice models are prone to badly predict substitution behaviour (second choice) of consumers, and second best choices as explicit target moments are known to be very helpful in better fitting substitution patterns (Conlon and Gortmaker, 2023). While stated voting behaviour is biased, I believe that second-best party choices less so. See adjacent paper Nikiforova (2026) for the discussion on this. Figure F4 provides an evolution of second best party choices of those who indicate AfD as their first choice over time at a national level.

5 Estimation & Identification

5.1 Estimation

Section E describes how the empirical restrictions implied by the spatial voting model are translated into a finite-dimensional optimization problem, and how (i) the weights and (ii) the bounds on the target parameters are computed.

Importantly, when computing the target parameters I construct a single linear program for each part of the country (East/West) in a given year, introducing coupling of the preference distribution across states. This allows me not to confine the analysis to individual states, but to explore richer heterogeneity both within and across them, under a weak assumption on preference similarity between demographic groups across states. See Section E for more details.

Section E.3 discusses the fit of the model. The correlation between predicted and observed AfD vote shares exceeds 0.9 for 2017–2025 and is 0.59 in 2013 (Figure E1, E3). Moreover, the correlations of predicted and observed municipal AfD shares with municipalities’ demographic characteristics are almost identical (Figure E2). This suggests that the model fits the data well and can be used for the preference estimation that follows.

5.2 Identification

The approach in this paper exploits variation in candidate positions across electoral districts within a state in a given year to estimate preferences of AfD voters. Identification assumption then posits that *in a given year variation in candidates’ positions across electoral districts within a state*

(market) m is unrelated to districts’ unobserved voter heterogeneity, $f(x | y, m, d) = f(x | y', m, d)$. I first discuss the reasons why this assumption may be viewed as plausible, and then discuss how I test validity of this assumption in later sections.

First, survey evidence suggests that age-gender-education brackets explain large share of variation in individuals’ self-identification among the voters of one party. To demonstrate this I focus on a more restrictive version of the model using the self-perceived Left-Right position from survey, and estimate an OLS *mean-prediction* regression predicting Left-Right position separately for each demographic group across states within parties. The R^2 of such regressions across federal states takes values 0.3 – 0.85 with a mean of 0.519.

Second, as discussed in the data section, candidate nominations are largely controlled by regional party elites and influenced by state-list considerations, which limits local discretion in candidate selection. More generally, the political science literature highlights the relatively small pool of potential candidates and the constraints this places on candidate choice (see, e.g., [Höhne \(2023\)](#)). I provide more information on candidate nomination practices in Section [H.1](#).

I further restrict the sample to municipalities with fewer than 10,000 inhabitants. Because the marginal electoral return to targeting a locality is low when its electorate is small, candidates have little incentive to tailor their platforms to these voters. Observed preferences in small municipalities are therefore less likely to reflect strategic candidate positioning than those in larger, more contested urban centers.

I also present two additional pieces of observational evidence that strengthen this assumption. First, across the municipalities used in the estimation, the estimated candidate positions of a given party are largely uncorrelated with local socio-economic characteristics within a state.²⁷ Note, however, that at the district level positions do correlate with a district’s socio-economic characteristics within party and state, and with the expected signs. For example, in districts with higher unemployment or larger shares of benefit recipients, candidates are more left-leaning. Figure [F5](#) reports the balance of socio-economic and demographic characteristics at the municipal level, and Figure [F6](#) at the district level. Second, Table [F2](#) demonstrates that candidates’ positions are uncorrelated with individuals’ self-reported views in surveys.

I assess the robustness of the identification assumption in three ways: (i) by restricting the analysis to small municipalities (smaller than 5k inh.), (ii) by excluding AfD candidates with stronger incentives to cater to local voters, (iii) by relaxing the exogeneity assumption and separating the districts into conservative and moderate (an approach similar to [Tebaldi, Torgovitsky and Yang](#)

²⁷Correlations remain significant for three variables after controlling for state-by-year and party-by-year fixed effects. Two of these are demographic characteristics, which I therefore account for in the estimation.

(2023)). The first approach follows logic similar to that in [Spenkuch and Toniatti \(2018\)](#), which suggests that politicians are more likely to target the preferences of voters in larger municipalities and cities. I therefore re-estimate the results using only municipalities with 5,000 or 10,000 inhabitants or fewer. The second approach exploits the idea that politicians’ incentives to cater to local voters vary with their electoral prospects, as competitive candidates are more likely to tailor their positions to constituency preferences. The third approach effectively removes the variation used for the estimation of the target parameters, and provides sensitivities to such removal.

6 Results

The main goal of this section is to provide the new insights into the rise of populism using the revealed-preference evidence from the approach developed above. To do so, I first estimate voters’ preferences: the weight on the Anti-Elite dimension relative to the Left–Right dimension, w_{AE} , and the distribution of ideal points in a policy space defined by Left–Right and Anti-Elite dimensions separately for demographic groups d in markets m $f(x \mid m, d)$. I next use those estimates for the counterfactual analysis.

Throughout the section I often present the results separately for eastern and western Germany. There are many reasons for this distinction, such as different histories and present-day institutional environments, as well as different levels of AfD support.²⁸ Crucially, the results from both parts of the country, analyzed separately, could inform our understanding of the rise of populism across countries with different institutional environments, such as those in eastern and western Europe.

6.1 Relative importance of dimensions

I begin by describing how the relative importance of the Left–Right and Anti-Elite dimensions has evolved over time. The weights come first because they determine the shape of the partition of the policy space used to evaluate the target parameters below.

Only relative weights can be recovered, and I normalize the weight on the Left–Right dimension to $w_{LR} = 1$ and search for the best-fitting value of w_{AE} . Precise estimation of weights requires especially rich data, as demonstrated in the simulations in the Appendix Section B. To increase the observational richness of the analysis, I estimate the weights separately for East and West Germany, and I allow ideal-point distributions to vary across states – that is, each demographic group d in each market m (state) has its own bliss point distribution $f(x \mid m, d)$; while I assume the same weight

²⁸At the moment of writing this draft, the AfD tops the polls nationwide with almost 30% support and 6 p.p. winning margin. The within-country heterogeneity is striking. In eastern Saxony-Anhalt, which holds elections in September 2026, the AfD is expected to win an outright majority, whereas in western Rhineland-Palatinate and Baden-Württemberg the party obtained less than 20% in March 2026.

w_{AE} for all groups in a given year and region (East or West).

The constructed grid to search for the best weight includes relative weights across the two dimensions of $(0.2, 0.3, \dots, 0.9, 1)$ and the corresponding inverse ratios.²⁹ The best relative weight is defined then as a weight that yields the smallest error value for the model fit.

Figure 2 presents examples of the partition of the policy space for different weight configurations. It illustrates the intuition behind the weighting: a lower (higher) weight on the anti-elite dimension means that individuals tolerate a larger (smaller) distance along that dimension for a given distance along the left–right dimension between their bliss point and a candidate’s position.

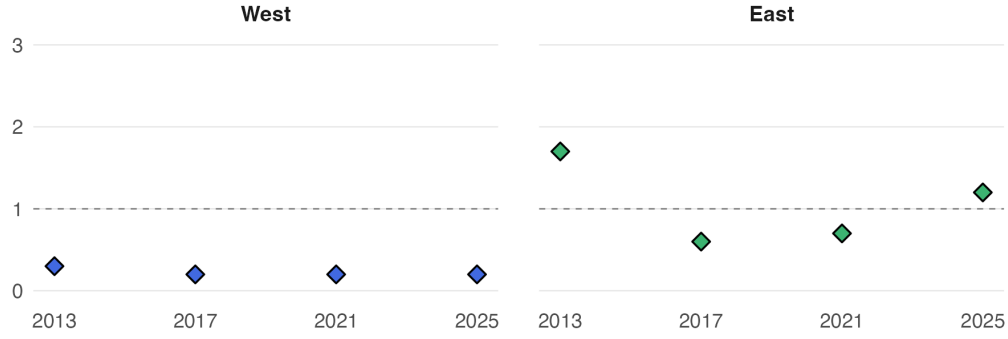
The model’s ability to yield the best fit in the context of changing partitions (i.e. changing relative weights) depends on the number of cells available under each setup, which, in turn, depends on the patterns of co-location of candidate positions on anti-elite and left-right dimensions³⁰. To alleviate this concern, I fix the number of available cells at 500, restricting attention to the AfD attraction region. Specifically, I retain the 500 largest cells and merge all remaining smaller cells into the larger cells with which they share the longest boundary. Additionally, Figure F7 shows that the correlation between left–right and anti-elite positions differs little between Eastern and Western Germany, both within parties (Figure F7a) and across parties (Figure F7b); and Table F4 shows that the standard deviations also do not differ much across years and parts of the country. This suggests that the relative positioning of parties on these two dimensions is unlikely to account for the regional differences reported below and that the results are instead driven by demand-side differences in the relative importance attached to each dimension.

Figure 3 presents the results for the optimal weights. Points denote the point estimates corresponding to the best-fitting weight. Because the data are noisy, the optimization problem always yields a single best-fitting estimate. To reflect the fact that the weights are set identified, I also report intervals that allow for discrepancies of up to 1% in the objective-function error (Figure G1).

The results suggest that the Anti-Elite dimension carried more weight in Eastern Germany than in Western Germany throughout the period. In Western Germany, the Left–Right dimension remains consistently the more important of the two. In Eastern Germany, the Anti-Elite dimension carried a larger weight than the Left–Right dimension in 2013 (although this estimate is imprecise once a

²⁹While it is possible to construct wider bounds, I do not do so for two reasons. First, wider bounds would amplify the measurement error in the estimated candidate positions. Second, the weights are identified only insofar as they meaningfully change the partition of the space, which for each pair of candidates depends on the weight and the ratio of their position differences, $\frac{|x_j^{LR} - x_k^{LR}|}{|x_j^{AE} - x_k^{AE}|}$. More than 80% of the values of this ratio across pairwise comparisons involving AfD candidates lie within $[0.2, 5]$.

³⁰Note that this argument is analogous to a standard result on variance decomposition in linear regression: for a fixed regressor X_1 , the improvement in R^2 from adding a second regressor X_2 is smaller the larger the correlation between the two regressors.



Notes. $w_{LR} = 1$ is fixed throughout, w_{AE} should be interpreted as the relative weight placed on the Anti-Elite dimension. Separate exercises are run for East and West Germany. I allow ideal-point distributions to vary across federal states – that is, each demographic group d in each market m (state) has its own bliss point distribution $f(x | m, d)$; while imposing the same weight w_{AE} for all groups within a given year and region. To run this exercise, the grid of possible weights is constructed. The grid is symmetric around 1 and imply relative weights across the two dimensions of $(0.2, 0.3, \dots, 0.9, 1)$ and the corresponding inverse ratios. See G2 for full results over the grid. Points denote the point estimates corresponding to the best-fitting weight.

Figure 3. Weights for Anti-Elite dimension w_{AE} relative to Left-Right $w_{LR} = 1$ across years and parts of the country

1% difference in the objective-function error is allowed for, see Figure G1) and in 2025. In 2017 and 2021, the Left-Right dimension is relatively more important, though the weight on the Anti-Elite dimension remains sizable.³¹

These results are in line with the differential role played by both dimensions across the two parts of the country. A broad literature documents the saliency with which Eastern Germans perceive themselves as neglected by federal elites, a perception grounded in measurable underrepresentation: as of 2024, Eastern Germans held only 12.1% of positional elite roles across twelve societal sectors, far below their population share of roughly 20% (Bundesministerium für Wirtschaft und Energie, 2025). The Anti-Elite dimension is much more muted in the West, where this region-wide sense of elite underrepresentation is far less pronounced.

Figure G2 plots the objective-function value for different values of the weight on the second dimension.³² Figures G4, G5 demonstrate consistency of estimates across subsamples by plotting the objective-function value on different subsamples of electoral districts across years and parts of the country. I interpret these results as evidence supporting the validity of the exercise.

In the following analysis I use the values for the estimated weights to define the partition of the policy space across the two regions unless stated otherwise.

³¹Note that here and in what follows the evolution of preferences of AfD voters is a combination of at least two factors – (i) genuine within-person evolution, and (ii) compositional change.

³²In the West 2017–2025 the solution 0.2 is the corner value. Figure G3 shows that when the grid is extended below 0.2 (adding 0.1), a weight of 0.2 remains the best fitting weight in 2021 and 2025, but not in 2017. For the reasons discussed above, I nonetheless retain 0.2 as the best-fitting weight for all of these years.

6.2 Distribution of ideal points

Next preference object is the distribution of voters' ideal points $f(x|m, d)$. To provide evidence on the evolution about this object, I focus in this subsection on the location of AfD voters in the weighted policy space.

6.2.1 Voters' ideal points in fixed areas of policy space

The natural question to ask about voters' ideal points is *Where voters' ideal points are located in the policy space?* In particular, I ask this question in relation to the AfD voters in 2013-2025. The AfD median candidate position across the years is about (1.2, 0.8) for Left-Right and Anti-Elite positions respectively and was around this area over the period (Figure H5). To study the bliss points of AfD voters and their evolution I then first partition the attraction area of AfD in three fixed regions: 1) Anti-Elite: Area with distinctively Anti-Elite Bliss points: those to the upper left from (0.5, 0.5); 2) Right: Area with distinctively Right Bliss points: those to the lower right from (0.5, 0.5); 3) Both: Area with both Anti-Elite and Right bliss points: those to the upper right from (0.5, 0.5). Figure 4a provide an illustration for these areas.

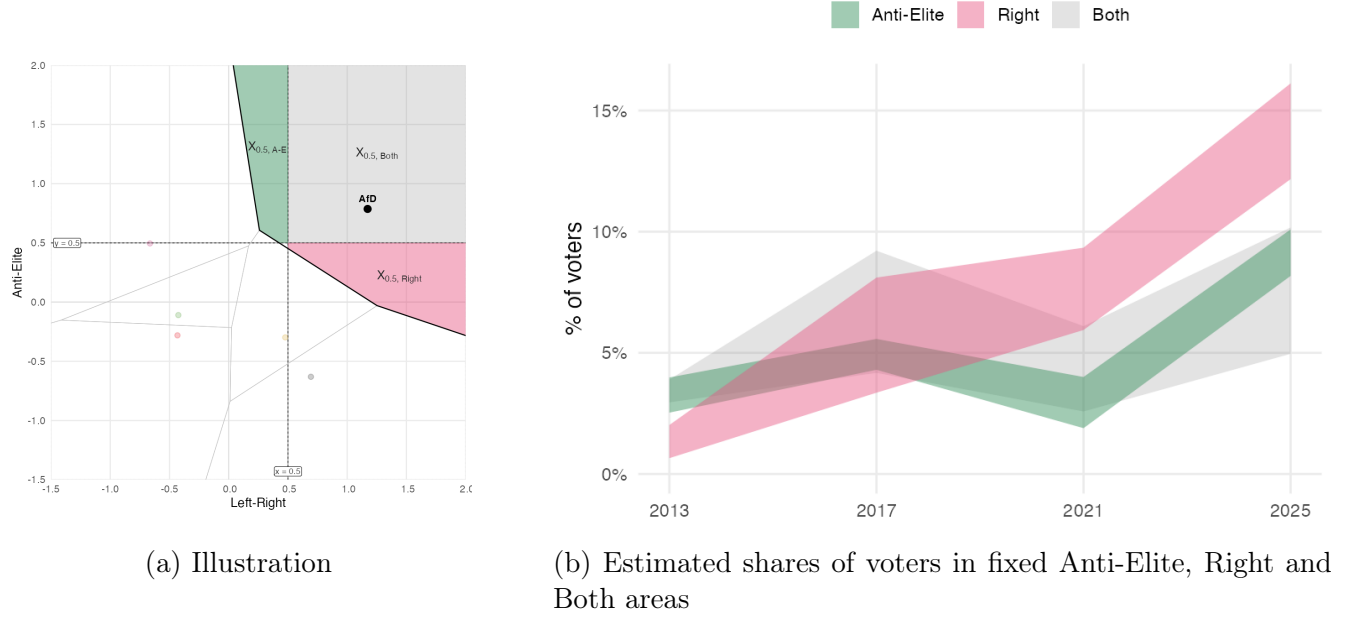
Let $f(x | m, d)$ be the density of ideal points among eligible voters in state m and demographic group d , and $F_{M,D}$ their joint (observed) distribution. The target parameter is the share of AfD voters in each region, e.g. for anti-elite:

$$\theta_{\text{AE-Fixed}} = \iint_{X_{\theta_{\text{AE-Fixed}}}} f(x | m, d) dx dF_{M,D}(m, d)$$

Because $f(x | m, d)$ is only partially identified from the municipal vote shares and demographic composition, I report bounds on $\theta_{\text{AE-Fixed}}$ and other target parameters; note that the bounds reported here and below are sharp (see [Tebaldi, Torgovitsky and Yang \(2023\)](#) for details).³³ Figure 4b plots the results.

The overall increase in target-parameter values over time reflects the rise in AfD support from 2013 to 2025. This increase is heterogeneous across masks: a large and steady increase is observed for the share of AfD voters in the Right fixed mask, which contains up to 15% of all voters by 2025. This is evidence that the AfD's electoral base is shifting towards a more ideologically conservative, right-wing support base rather than a more anti-elite one.

³³Also note that the fixed masks are contingent, so the lower bounds are the most informative: they give the smallest possible share of voters located in the region (since the estimated objects are the *sharp* bounds on the target parameters). The upper bounds, by contrast, may reflect both the degree to which the problem is underidentified and the mass in the cells shared by contiguous fixed masks.



Notes: Panel (a) illustrates party positions at their national median with an isotropic partition of policy space; the areas of AfD voters with distinctively anti-elite bliss points (green, $X_{0.5, A-E}$), Right bliss points (pink, $X_{0.5, Right}$) or Both (gray, $X_{0.5, Both}$). Panel (b) plots bounds on the shares of voters whose bliss points lie in the green, pink and gray regions across years. The results are aggregated across the two parts of the country.

Figure 4. Illustration of a target parameter (a) and results (b)

Figure G8 shows broadly similar patterns across Western and Eastern Germany, with a pronounced rise in the Right group over time. The Anti-Elite group is relatively larger in the East, but by 2025 the Right group overtakes it there as well (its lower bound exceeds the Anti-Elite group's upper bound).

This subsection has provided a first piece of evidence on the distribution of AfD voters' bliss points and its evolution. While instructive and intuitive, it does not pin down voters' motivations, which, beyond voters' locations, would also require explicitly accounting for the weights voters attach to the different dimensions. I now turn to the analysis that resolves this.

6.2.2 Voters ideal points and their motivations for AfD support

The AfD is distinctive in combining an ideologically radical-right stance with an anti-elite appeal, so a ballot cast for it is uninformative about why the voter chose it. Recovering that motivation matters, because it maps to the policy demands the vote expresses, and could help policymakers craft more appropriate responses. The analysis in this section resolves the ambiguity with a counterfactual: suppose the AfD split into two parties, one retaining its radical-right position on Left-Right dimension but moving toward the elite, the other retaining its anti-elite appeal but moderating its right radicalism — what share of AfD voters would each attract? Equivalently, it measures the share

of AfD voters whose ideal point in a weighted policy space leans anti-elite versus right, relative to where the AfD itself stands.

The estimated composition therefore has a behavioral interpretation: *a voter in the anti-elite region is one for whom protest against the elites, rather than Left-Right ideology, is the pivotal reason to vote AfD, and conversely for the right region.*

Let $\bar{Y} = (\bar{Y}_{LR}, \bar{Y}_{AE})$ be the nation-wide median AfD position. Displacing the AfD by δ along each dimension defines two hypothetical successor parties and partitions the AfD win region $X_{AfD}(\bar{Y})$ into an anti-elite region X_{A-E} (voters nearer the party that retains the anti-elite appeal; green) and a Right region X_{Right} (voters nearer the party that retains the same Left-Right position; pink), with $X_{AfD} = X_{A-E} \cup X_{Right}$.³⁴ Let $f(x | m, d)$ be the density of ideal points among eligible voters in state m and demographic group d , and $F_{M,D}$ their joint (observed) distribution. The target parameter is the share of AfD voters in each region:

$$\theta_{AE} = \frac{\iint_{X_{A-E}} f(x | m, d) dx dF_{M,D}(m, d)}{\iint_{X_{AfD}} f(x | m, d) dx dF_{M,D}(m, d)}$$

I now report bounds on $\theta_{AE}, \theta_{Right}$. Note that by construction, $\theta_{Right} = 1 - \theta_{AE}$.

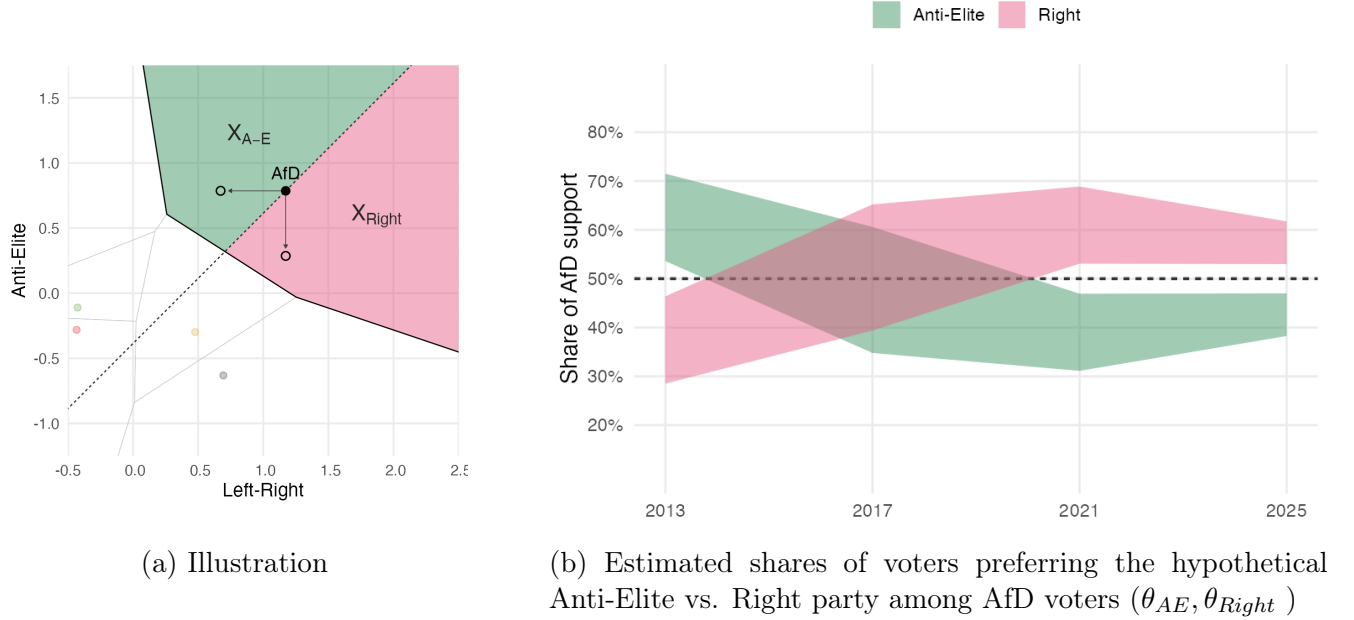
Figure 5b provides the bounds for the target parameters over years. The results indicate that AfD support has changed over time. In earlier years the support of AfD was driven more by anti-elite motivations, while it flipped over time and in 2021 and 2025 at least 53% of AfD support comes from individuals with stronger ideological Right motivations.

These aggregate results mask substantial regional and subgroup heterogeneity. Figure G10 reports the estimates separately for Eastern and Western Germany. The pattern for Western Germany resembles the national-level pattern presented above; in turn, voters in Eastern Germany are more strongly driven by anti-elite attitudes than voters in Western Germany, especially in 2017.³⁵ The patterns are clearer and more precisely identified for Western Germany, reflecting the fact that it is larger in population and offers more electoral districts across which to exploit variation.

I now want to highlight two more subgroup heterogeneity results among Western voters over time, as Western Germany allows for detecting rich heterogeneity patterns in the data. Figure 6

³⁴The size of a shift used for the estimations in the paper that follow is $\delta = 1$. Figure G9 presents the results for the other shifts for θ_{Right} . Results are quantitatively similar for different step values.

³⁵Except in 2013, when anti-elite sentiment dominated in the West. The party's anti-elite appeal then was arguably of a different nature—rooted in opposition to the Eurozone bailouts rather than in the hostility towards the establishment and democratic institutions that came to define the party after 2015.



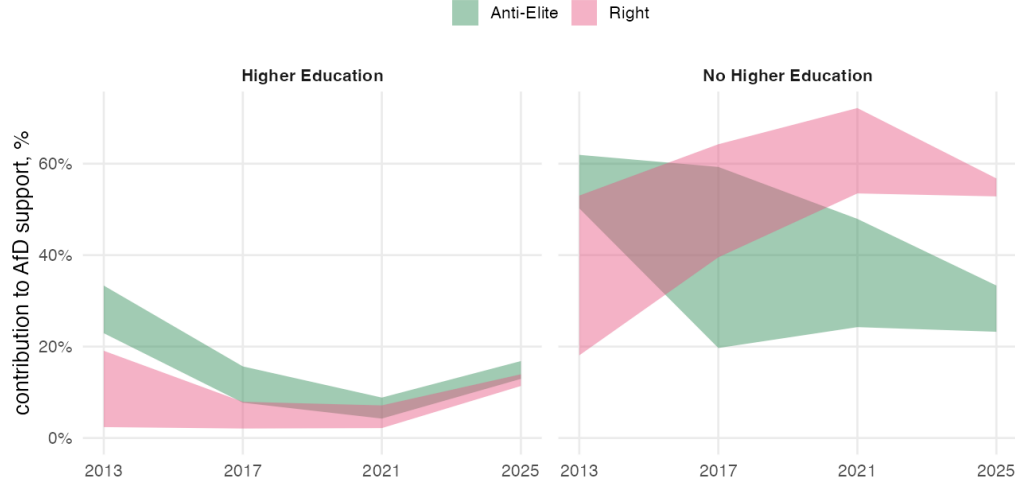
Notes: Panel (a) illustrates party positions at their national median, and hollow circles illustrate the counterfactual AfD shifts. The areas of AfD voters motivated more by anti-elite (green, X_{A-E}) or right (pink, X_{Right}) ideological stance of AfD are plotted. The illustrative weight used is $w_{AE} = 1$, weights used in the estimation are the weights estimated in Subsection 6.1. Panel (b) plots bounds on the shares of voters whose bliss points lie in the green and pink regions across years. The results are aggregated across the two parts of the country.

Figure 5. Illustration of a target parameter (a) and results (b)

demonstrates shares of AfD voters with and without higher education belonging to the Anti-Elite and Right areas of Figure 5a (these are similar target parameters as just above, but taking into account also the relative sizes of the groups at each year); Figure 7 provides the same analysis for young (< 35 y.o.) and old (> 35 y.o.) individuals.

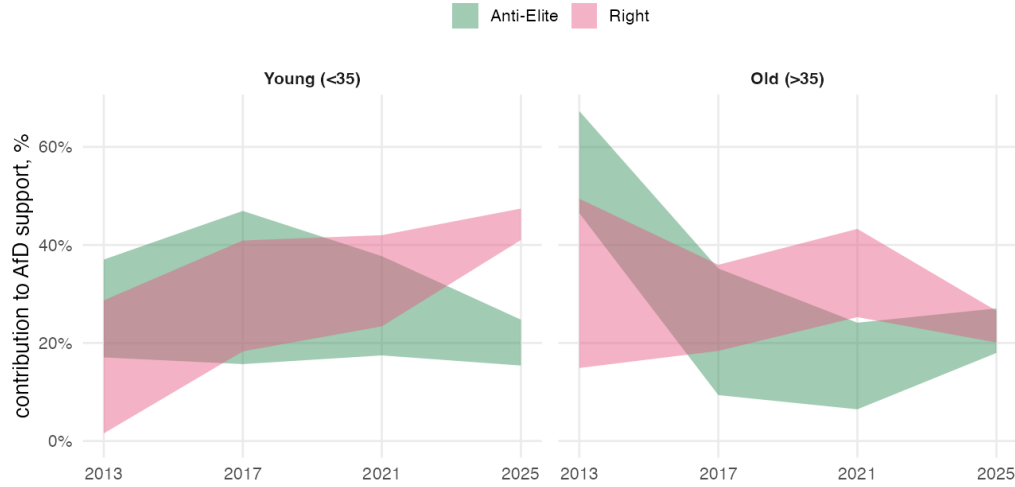
These figures are intended to highlight that individuals in different demographic groups have different rationales for supporting the same party, and that these rationales may evolve over time. In particular, for individuals without higher education the party's radical-right stance came to outweigh its anti-elite appeal only in later years. This could suggest that voters with lower educational attainment hold preferences that are more malleable, or more responsive to party messaging. Another important pattern is the rising share of young AfD voters whose preferences lie in the Right group. The growing support for radical (and in particular radical-right) political movements among young people has been documented in the political science literature, though the motivations behind these choices remain contested (van der Brug, Hobolt and Popa, 2026). The results of this paper suggest that young individuals may increasingly support these parties because of their radical-right ideology rather than their anti-elite stance.

A large literature in political economy relies on stated vote choices and stated issue preferences



Notes. Bounds on the shares of voters whose bliss points lie in the green and pink regions across years are plotted. The results are only for the Western Germany.

Figure 6. Estimated shares of AfD voters in both subspaces of policy space by educational attainment



Notes. Bounds on the shares of voters whose bliss points lie in the green and pink regions on the Figure 5a across years are plotted. The results are only for the Western Germany.

Figure 7. Estimated shares of AfD voters in both subspaces of policy space by age

to estimate voters' preferences and their relationship to electoral behavior (Le Pennec and Pons, 2019; Gethin, Martínez-Toledano and Piketty, 2022), including in the context of support for populist politicians (Guiso et al., 2017; Algan et al., 2017; Danieli et al., 2023; Bartels, 2023). A natural question, then, is how the preferences inferred from survey data differ from those recovered using the revealed-preference approach developed in this paper. Section D presents this analysis. The main takeaway is that the shares of voters located in the Anti-Elite and Right regions display patterns similar to those obtained from survey data. Note that this comparison concerns bliss points only: the weights used in the exercise are those recovered from the revealed-preference framework, since they

cannot be estimated from survey data (see Section D for more details).

Given that second-choice data are not widely available, I re-estimate the analysis in this subsection without using second choices as moments in the estimation (Figure G11). The results remain quantitatively similar, with slightly wider bounds, suggesting that the second-choice moments were not heavily restricting the distribution of voters' bliss points, and that the aggregate data alone provide enough identifying variation for this exercise.

The results presented in this subsection speak to the changing nature of AfD support over time, across regions, and across demographic subgroups. As I regard these as one of the paper's main contributions, I now verify that they are robust to relaxing the selection-on-observables assumption used to identify these patterns.

Robustness. I provide three robustness exercises that relax a selection-on-observables assumption on candidate positions. First, I run the main analysis on municipalities with fewer than 5,000 inhabitants (Figure G12). The idea is that parties are likely to target more populous, more electorally relevant constituencies; small municipalities therefore face a set of candidates pre-selected for larger constituencies. The results remain qualitatively similar in the West and deliver a more stable Anti-Elite lean in the East across 2017–2025.

Second, I restrict the analysis to districts in which AfD candidates were not competitive for victory. Specifically, I exclude every district in which the AfD candidate trailed the winning candidate by no more than 5 (10) percentage points in the Eastern (Western) part of the country. This removes 12 districts in 2017 and 28 in 2021 in the East, and 22 in the West in 2025. The different thresholds reflect the party's greater popularity in the East, where a uniform 10-point threshold would remove up to 80% of all districts in 2021. Moreover, for the East the analysis is restricted to 2017 and 2021: in 2013 there were no serious contenders, while in 2025 there were too many even under the 5-point threshold, as AfD candidates won almost every district.

The idea behind this analysis is that candidates who recognize they have little chance of winning will cater less to the voters of their particular district, while still shaping the party's image through their more intrinsic ideological stances. Figure G13 presents the results separately for Eastern and Western Germany. The results for the presented years remain quantitatively similar.

Third, I exploit an instrument similar in construction to the one in Tebaldi, Torgovitsky and Yang (2023), which directly relaxes the assumption of preference similarities across districts with different candidates. In particular, I allow that the mass of voters of a given demographic group g within a state in each partition cell between conservative and moderate districts could differ by at most a factor of $1 + \kappa$ for some fixed κ . I set $\kappa = 2$, so that masses could be different no more than

three in both directions. Figure G14 presents the results. Relaxing the assumption on preference similarity does not quantitatively change the results, but provides somewhat wider bounds, especially in the East.

The bounds reported above reflect partial identification only and do not account for any statistical uncertainty. Section E.4 quantifies sampling uncertainty by re-estimating the bounds on bootstrapped samples of municipalities.

The analysis so far has focused on the evolution of AfD voters’ preferences. The next subsections turn to counterfactual exercises, examining how changes in the positions of the (i) AfD and (ii) its center-right, pro-elite rival (The Christian Democratic Union, CDU) would affect the AfD vote share.

6.3 What explains the success of AfD in competition with other parties: conservative positions or anti-elite stance?

The party does not exist in isolation, but competes with other parties for voters. Answering the question of which dimension matters more for voters requires assessing the sensitivity of voters’ choices to counterfactual party positions.

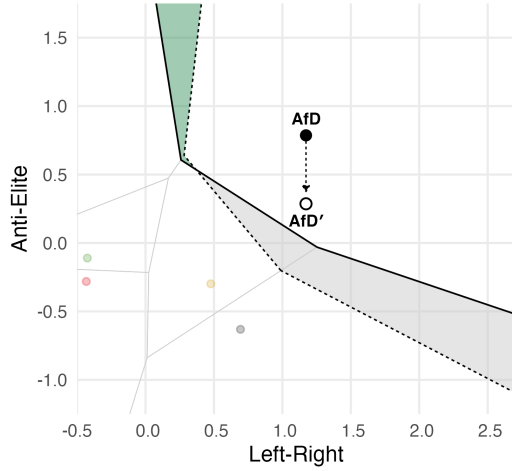
To assess to what extent the AfD’s success is driven by right-wing ideology or by anti-elite sentiment, I combine the estimated bliss-point distributions of AfD voters, the estimated relative weights, and the median positions of all the other parties, and ask: *How sensitive are AfD voters to the party shifting its positions along Left-Right and Anti-Elite dimension?* Put differently, the question is how many of the AfD’s baseline supporters the party would lose if it adopted a more pro-elite or a more left-wing position.

I hold the estimated preferences fixed and shift the AfD’s position relative to its national-level median in each year, $\bar{Y}$, by 0.25/0.5/1 units along each dimension.³⁶ Figure 8 illustrates the shifts along both dimensions under median party positions across years and isotropic weights for clarity.

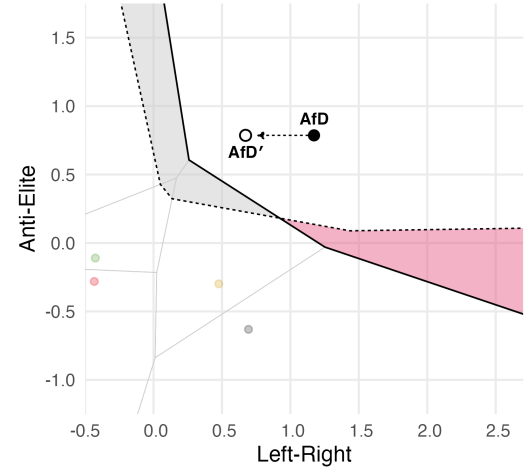
Before presenting the results, I provide a substantive interpretation of what a shift of 0.25, 0.5, or 1 unit actually means for a party. To do so, I trace back the scoring pipeline to determine what such a shift would require in terms of changes in candidates’ stated opinions. Note that the units of the dimensions—and the scoring procedure as a whole—are identical across years, so these steps are comparable over time.

A candidate’s position on each dimension is the average of her standardized answers to the relevant survey items, so moving a single question by one scale point shifts the dimension score by some σ , unique to each question. On the anti-elite dimension, the most influential single question—

³⁶Positions are standardized by construction, so a given shift is comparable across dimensions.



(a) AfD shift towards 0.5 more pro-elite position



(b) AfD shift towards 0.5 more left position

Notes: The panels present an illustration of the unweighted cropped policy space, with partitions determined by median party positions across years. Shaded dots mark the median positions of parties other than the AfD. **AfD** denotes the AfD's median position across years, and **AfD'** denotes the shifted AfD position—either towards the pro-elite position (Panel (a)) or towards a more left-wing position (Panel (b)). Solid lines determine the existing partition, the dashed line – counterfactual partition. The illustrated shift is 0.5 units along each dimension. Since positions are standardized by construction, a 0.5-unit shift is comparable across dimensions. The exercise estimates the share of voters the AfD would lose under the shift, i.e. those located in the green or pink areas in each panel.

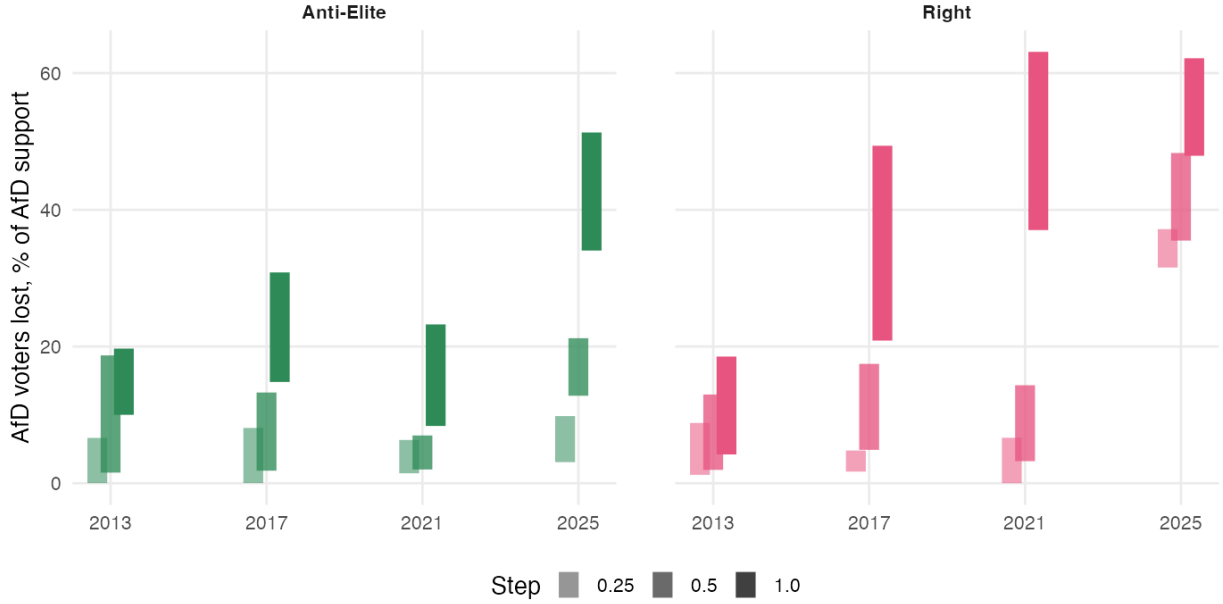
Figure 8. Illustration of counterfactual moderating shifts by AfD by 0.5 units separately on two dimensions

Satisfaction with the way democracy works (a 1–4 item)—moves the score by about 0.18 per scale point across years, and a typical five-point item such as *Legislation reflects citizens' interests* by about 0.14. A 0.25 step therefore corresponds to revising one such question by roughly 1.5–2 scale points (say, from *rather agree* to *strongly disagree*), or two questions by 1 point each. Moving every anti-elite question by one scale point would imply a strong repositioning of the party, and would change the score by about 0.95, so the 0.25/0.5/1 steps amount to softening (or hardening) a corresponding share of all anti-elite answers.

The Left–Right dimension behaves similarly: potent five-point items such as *Immigration is good for the economy* or *Marriage only for heterosexual couples* move the score by about 0.10 per point, and moving all economic items by one point shifts the score by about 0.69. In short, a 0.25 step is a modest, one-to-two-question adjustment; a 0.5 step is a substantial repositioning that must touch many questions at once; and a full 1-unit step amounts to broadly shifting one's stance across an entire dimension. Table F5 provides score sensitivities across questions for 2017.

The target parameter is the share of baseline AfD voters who would abandon the party had

it shifted to **AfD'**.³⁷ Figure 9 plots the results separately for different values of steps and different directions of a shift.



Notes. Each bar is the identified interval for the share of *baseline* AfD voters who would abandon the party if it moderated by one step along the indicated dimension, holding all other parties at their observed positions. In the left panel (**Anti-Elite**) the AfD moves toward the elite along the anti-elite dimension; in the right panel (**Right**) the AfD moderates along the left–right dimension. Within each year, shading denotes the size of the moderation step (0.25/0.5/1.0 in standardized policy-space units; see the discussion just above for its substantive interpretation). National figures aggregate the East and West regional bounds. Partitions are built under the weighted (anisotropic) spatial metric with the estimated relative weight w_{AE} .

Figure 9. Estimated shares of AfD voters loss under counterfactual AfD shifts

The results suggest that under small steps of 0.25 (0.5), the share of voters abandoning the AfD is no more than 10% (20%) across years and dimensions before 2025. Shifts of a full step of 1 yield much larger losses, especially along the Right dimension, reaching up to 60%.

The results also suggest that competition between the AfD and other parties became much tighter by 2025, in the sense that even small AfD shifts (particularly along the Left–Right dimension) would produce large defections of voters to other parties. This was not the case before 2025, indicating that the party had previously existed in relative isolation.

If competition with the establishment parties has indeed intensified, those parties should be able to capture a large mass of AfD voters by moderating their positions only slightly. I now turn to such an analysis.

³⁷Precisely, $\theta_{\text{Loss, A-E}} = \frac{\iint_{X_{\text{Loss, A-E}}} f(x | m, d) dx dF_{M,D}(m, d)}{\iint_{X_{\text{AfD}}} f(x | m, d) dx dF_{M,D}(m, d)}$, $X_{\text{Loss, A-E}} = X_{\text{AfD}} \setminus X_{\text{AfD}'}$

6.4 The response from the parties of the establishment

German establishment parties, which once commanded a large majority of the electorate, are now losing ground to the AfD, which leads the polls nationwide as of mid-2026.³⁸ The development is by no means unique to Germany. Across Western democracies, populist and radical parties have risen sharply over the past two decades, in many cases entering national legislatures and governments. Examples include Italy, France, the Netherlands, Spain, Sweden among others. Beyond Europe, Donald Trump’s rise is widely regarded as part of the same populist wave.

This raises a natural question: how should establishment parties compete with populist challengers? This section examines how effective it would be for them to adopt positions closer to the AfD’s in the two-dimensional space defined by the Left–Right and Anti-Elite dimensions. The natural candidate for such analysis is the Christian Democratic Union (CDU), one of Germany’s two largest parties and the main establishment party on the right of the Left–Right spectrum, on which I focus throughout this section.

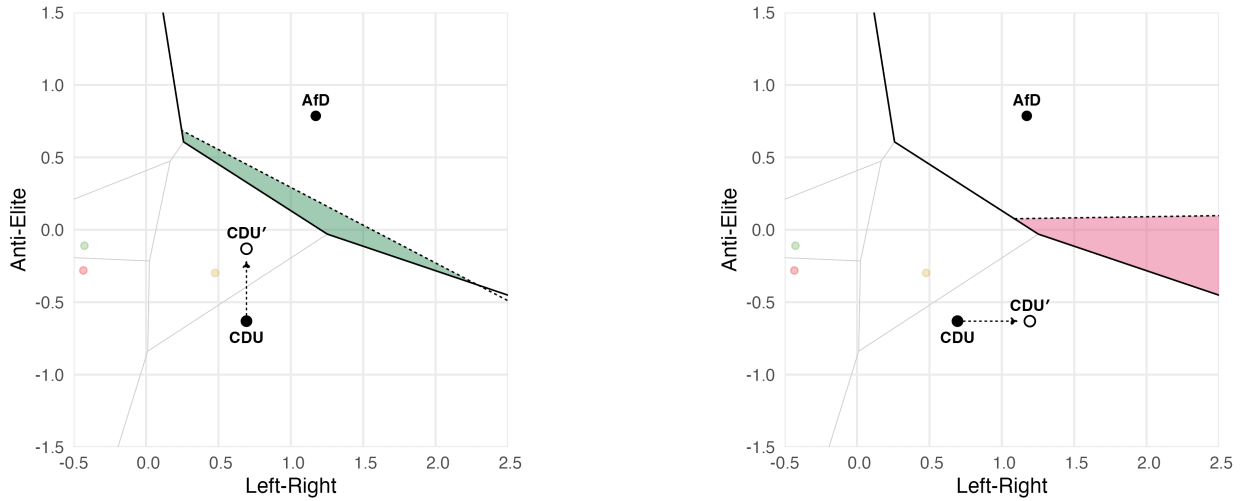
In particular, I conduct an analysis similar to that in the previous subsection, but ask the reverse question: *if the CDU moves a given step towards the AfD along the Left–Right or Anti-Elite dimension, how many AfD voters would it capture?* Figure 10 illustrates these counterfactual scenarios.

Figure 11 plots the results separately for different step sizes and different directions of the shift. The results suggest that even a small shift (0.25) along the Left–Right dimension could allow the CDU to capture up to 10% of AfD support across 2013-2021; shifts of size 0.5 and 1 would yield larger gains. However, as discussed above, such larger steps may be too costly for a party with a relatively stable policy platform over the years, leaving the 0.25 step as the only realistic option.

Note that in 2025 a 0.25 shift in the CDU’s position along the Left–Right dimension could allow it to capture about 35% of the entire AfD electorate. While this exercise is not designed to solve the CDU’s vote-share maximization problem, it is clear that by 2025 35% of AfD voters constituted a large mass of the electorate, and the CDU would be unlikely to lose a comparable mass of voters on the other (left) side of its attraction area.

Putting this result into perspective, it implies that if the CDU had slightly hardened its positions on immigration in the 2025 elections, it could have captured about a third of the entire AfD electorate. Concretely, this could mean shifting from “Agree” to “Rather Agree” on the question “*Immigration is good for the economy*”, from “Agree” to “Strongly Agree” on “*Immigrants should adapt to German*

³⁸As of July 2026, national polling averages place the AfD at roughly 28% and the CDU/CSU at roughly 22%. See Politico’s Poll of Polls, <https://www.politico.eu/europe-poll-of-polls/germany/> (accessed July 2026).



(a) CDU shift towards 0.5 more anti-elite position (b) CDU shift towards 0.5 more right position

Notes: The panels present an illustration of the unweighted cropped policy space, with partitions determined by median party positions across years. Shaded dots mark the median positions of parties other than the AfD and CDU. **CDU** denotes the median CDU position across years, and **CDU'** denotes the shifted CDU position—either towards the anti-elite position (Panel (a)) or towards a more right-wing position (Panel (b)). Solid lines determine the existing partition, the dashed line – counterfactual partition. The illustrated shift is 0.5 units along each dimension. Since positions are standardized by construction, a 0.5-unit shift is comparable across dimensions. The exercise estimates the share of voters the AfD would lose under the shift, i.e. those located in the green or pink areas in each panel.

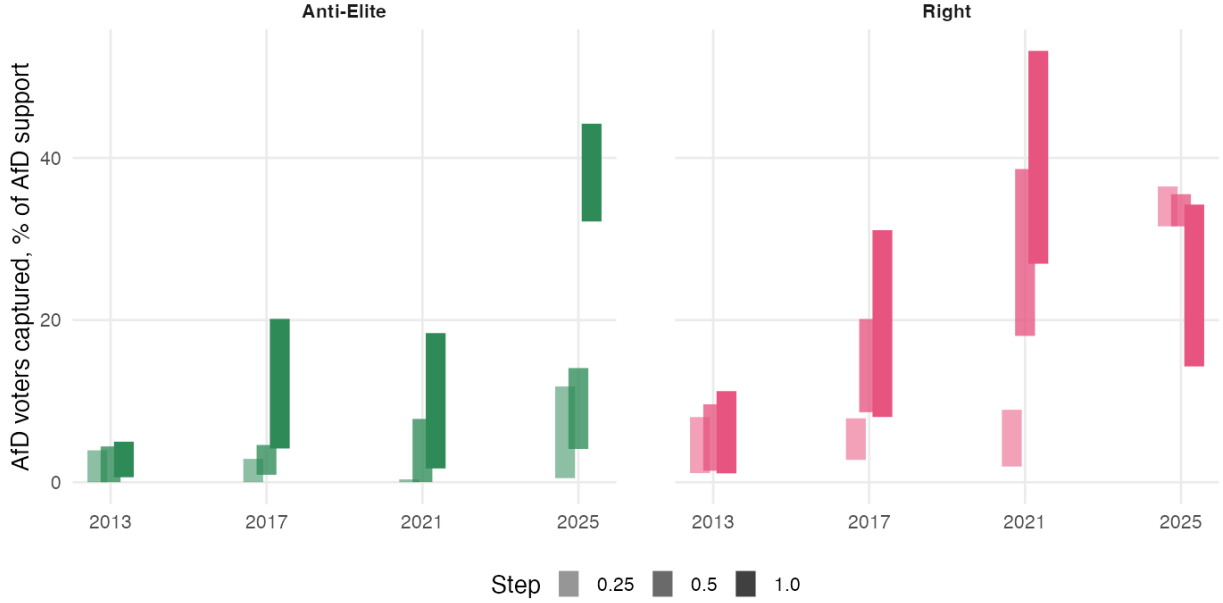
Figure 10. Illustration of counterfactual shifts by CDU by 0.5 units separately on two dimensions

culture”, and from 8 to 10 on “*Self-Position on Immigration, 1–11*”.³⁹

These results are aligned with the experimental findings of [Guenther and Nunnari \(2025\)](#), who find that perceiving the CDU as one unit more liberal on immigration (on a 0–10 scale) raises the intention to vote for the AfD by an average of 5 percentage points – about 25% of the AfD vote. Putting the results on the same scale, my results suggest that the 2 units CDU hardening on immigration on the 0–10 scale would capture about 35% of the AfD electorate. The results are also consistent with the documented defection of voters between parties in 2025: about 1.2 million CDU supporters in 2021 voted for the AfD in 2025, making this the largest such flow between any two parties in the country ([Vehrkamp and Posthofen, 2026](#)). The counterfactual is thus likely driven by the fact that the CDU could win these voters back with a slightly hardened Left–Right position.

This section was intended to provide some intuition for what the best strategy might be for establishment parties competing with populists. Notice that the party’s hardening of its anti-elite positions for realistic step values (0.25, 0.5) does not appear to be as effective in any year—in particular, not even in earlier years, when the AfD electorate was less pronouncedly right-wing. The party selected for this exercise was a large center-right establishment party, which could be affecting

³⁹All illustrative answers here are given relative to the real median answers of a CDU candidate in the 2025 survey. See Table F6 for the values of relative contribution of shifts in each survey question.



Notes. Each bar is the sharp identified interval for the share of *baseline* AfD voters who would abandon the party if CDU moved by one step along the indicated dimension, holding all other parties at their observed positions. In the left panel (**Anti-Elite**) the CDU moves toward the anti-elite positions; in the right panel (**Right**) the AfD moves towards the right positions. Within each year, shading denotes the size of the moderation step (0.25/0.5/1.0 in standardized policy-space units; see the discussion just above for its substantive interpretation). National figures aggregate the East and West regional bounds. Partitions are built under the weighted spatial metric with the estimated relative weight w_{AE} .

Figure 11. Estimated shares of AfD voters loss under counterfactual CDU shifts

the results. Nonetheless, the finding that competition with populists could be effective along the Left–Right dimension at least for this party is a reassuring one, since competing on anti-elite rhetoric could, among other things, contribute to the erosion of democratic institutions.

6.5 The Role of the Supply Side (and other factors)

To what extent the results presented above are driven by the supply side repositioning versus the demand side changes (genuine evolution of preferences, or compositional changes)? To answer this question I compute Shapley values for the target parameter from the Section 6.2.2.⁴⁰ Let $\theta_{Right}(Pref_t, Comp_t, Pos_t)$ denote the share of AfD voters whose ideal points fall in the *Right* region (Figure 5a), given voter preferences $Pref_t$ across demographic groups, the demographic composition of AfD voters $Comp_t$, and national median party positions Pos_t . I decompose the change in this

⁴⁰See [Shapley \(1953\)](#) for the original reference, or [Guriev et al. \(Forthcoming\)](#); [Longuet-Marx \(2025\)](#) for recent applications in political economy.

share between t and t' as follows:

$$\begin{aligned} & \theta_{Right}(Pref_{t'}, Comp_{t'}, Pos_{t'}) - \theta_{Right}(Pref_t, Comp_t, Pos_t) \\ &= \phi(Pref) + \phi(Comp) + \phi(Pos) \end{aligned} \tag{10}$$

where each Shapley value, e.g. $\phi(Pos)$, is the weighted average marginal contribution of this factor across all 2^2 coalitions of the other two factors:

$$\begin{aligned} \phi(Pos) = & \frac{1}{3} \left[\theta_{Right}(Pref_{t'}, Comp_{t'}, Pos_{t'}) - \theta_{Right}(Pref_{t'}, Comp_{t'}, Pos_t) \right] \\ & + \frac{1}{6} \left[\theta_{Right}(Pref_t, Comp_{t'}, Pos_{t'}) - \theta_{Right}(Pref_t, Comp_{t'}, Pos_t) \right] \\ & + \frac{1}{6} \left[\theta_{Right}(Pref_{t'}, Comp_t, Pos_{t'}) - \theta_{Right}(Pref_{t'}, Comp_t, Pos_t) \right] \\ & + \frac{1}{3} \left[\theta_{Right}(Pref_t, Comp_t, Pos_{t'}) - \theta_{Right}(Pref_t, Comp_t, Pos_t) \right], \end{aligned}$$

and $\phi(Comp)$, $\phi(Pref)$ are defined analogously. The weights $\frac{1}{3}, \frac{1}{6}$ follow from $\frac{|S|!(n-|S|-1)!}{n!}$ with $n = 3$ ($|S| = 0, 2$ give $\frac{1}{3}$; $|S| = 1$ gives $\frac{1}{6}$). By construction $\phi(Pref) + \phi(Comp) + \phi(Pos)$ equals the left-hand side of 10 exactly.

To evaluate each counterfactual share, I combine three ingredients. Party positions (Pos_t) are the estimated national median positions in year t ; Voter preferences ($Pref_t$) enter as group-specific ideal-point distributions, estimated as a primitive of the model, $f(x | m, d)$.

Composition ($Comp_t$) comes from the Representative Electoral Statistics. It is a unique data on voting choices per 12 age-gender groups per federal states.⁴¹ These data do not permit disaggregation by education, which I therefore omit.

Table 2 presents the results for the $\phi(Pos)$ across every pair of consecutive years and regions of the country. The last row provides aggregated over the two parts of the country conservative bounds. Table F7 presents also the Shapley values for the other two channels (composition and preferences), but bounds for those channels are wide and uninformative.

The bounds on the Shapley values for the Positions channel suggest that the supply-side channel contributes positively to the increase in the share of Right-motivated voters over 2021–2025, while its contribution for the other pairs of years leans negative. What these results make clear is that the evolution over time in the share of Right-motivated voters documented in this paper is not driven by parties changing their positions.

⁴¹It is calculated based upon calculations from a sample of polling districts in which marked ballots allow votes and turnout to be tabulated by gender and age groups. Hence, this data is based on actual ballots, not surveys.

	2013→2017	2017→2021	2021→2025
<i>West</i>			
Positions	[−0.202, −0.026]	[−0.047, +0.000]	[+0.023, +0.117]
<i>East</i>			
Positions	[−0.103, −0.018]	[+0.000, +0.018]	[−0.041, −0.005]
<i>National (AfD-electorate weighted, conservative)</i>			
Positions	[−0.188, −0.025]	[−0.038, +0.004]	[+0.008, +0.092]

Note: Bounds on the Shapley contribution of positions channel, derived from a single linear program over the pinned feasible set with both position-year catchments held jointly. Positive values raise the Right share. The national row combines the East and West position bounds by each year’s AfD electorate-population weights, taking the worst case over the two year-weights of the pair; it is conservative relative to separate regional bounds.

Table 2. Shapley Decomposition of the Change in the Right Share of AfD Support

In this setup it is hard to pin down demand-side changes and attribute them to either the composition or the preference channel (see Table F7 for the wide bounds of Shapley values on both). I now provide two pieces of evidence that preference evolution within demographic groups is consistent with increasing Right motivation. First, within-person evolution towards Right motivations is consistent with survey-panel evidence showing that repeat AfD voters (i) become both more conservative and more anti-elite over time, and (ii) move faster along the Left–Right dimension.⁴² Second, as shown earlier, there are consistent within-group patterns of preference evolution—for instance, voters without higher education and young voters becoming increasingly Right-motivated.

In turn, the negative demand-side contribution over 2021–2025 could instead be operating through composition. About 19% of AfD voters in 2025 did not vote in the 2021 election (Vehrkamp and Posthofen, 2026), and survey evidence indicates that abstainers tend to have lower trust in elites and in democratic institutions (Grönlund and Setälä, 2007). The entry of these more anti-elite voters shifts aggregate motivations through the composition channel—in line with the fixed-mask findings, where the shares in the Anti-Elite masks rose from 2021 to 2025, and by proportionally more than for the Right group.

7 Discussion

Is populist support rooted in Left-Right ideology or in protest against the elites? This paper develops an approach to recovering voters’ preferences from aggregate electoral data in a multidimensional spatial voting model, and uses it to address exactly this kind of question. Applied to the radical-right Alternative for Germany (AfD), the method reveals that the party’s support became increasingly grounded in its conservative left-right positions rather than its anti-elite stance. Coun-

⁴²The results are based on a 2016–2021 panel of all voters, taking their self-reported issue positions as their ideal points; see Figure G15 and Table G1.

terfactual exercises suggest that establishment parties could recapture a substantial share of AfD voters by modestly hardening their Left–Right positions—particularly on immigration.

These findings speak to a large political economy literature on the rise of populism, which has focused primarily on identifying the causal forces driving voters towards populist parties and candidates. This paper takes a different route: it recovers *the preferences of populist voters themselves*. Taking this route helps inform the design of responses to populism.

That question of response is important. Populism appears to carry real economic and institutional costs (Funke, Schularick and Trebesch, 2023), so how mainstream parties react is consequential. Responses fall broadly into two classes: those addressing voters’ anti-elite grievances, and those addressing their ideological left-right demands. The former may take the form of commitment politics (Bellodi et al., 2025), among other approaches; the latter calls for more economically or culturally conservative platforms. Knowing which margin moves a given electorate therefore determines which response would be most effective. In the German case, the evidence points towards the left-right margin, especially in later years.

More broadly, the paper offers a framework for unpacking the bundle of characteristics embodied in any political option. Any voting option—a candidate, a party, a referendum answer—can be seen as a product with multiple attributes, and recovering preferences over these attributes is far from straightforward. Voting departs from most settings in empirical industrial organization: in a spatial model, individuals hold unobserved ideal points on each dimension, so one cannot impose the linear characteristic–utility relationship standard in that literature. The question “*Do you prefer sweeter or less sweet cereal?*” does not translate into “*Do you prefer a more or less right-wing candidate?*”—voters differ not in how much of an attribute they want, but in where their ideal point lies.

The method developed in this paper is portable to any electoral setting in which variation in candidate choice sets provides identification—across districts, in simultaneous elections, or under ranked-choice voting. Such settings together cover a substantial share of real-world elections. The method can further be extended to incorporate turnout, candidate valence, or additional dimensions of conflict.

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A Identification of dimension weights

This section establishes that the identified set for the relative weight in the two-dimensional weighted spatial voting model of Section 2 is contained in an interval on $\mathbb{R}_+$ that contains the true weight.

The intuition behind the result is as follows. The weight w_2 determines which cells of the MRP are non-empty, and weights are therefore identified only up to the differences in the cells they make available. The argument has two steps. First, a single cell is non-empty only for weights in an interval. Second, any weight consistent with the data must make every cell the data require non-empty; the set of such weights is therefore an intersection of availability intervals, which is again an interval.

A.1 Setting and notation

I use the weighted spatial voting model of Section 2 with the normalization $w_1 = 1$, so the single parameter is the relative weight $w_2 = w > 0$ on the second dimension. Recall that the cell chosen by candidate j under a position vector y is $\mathcal{X}_{j;w}(y) = \{x \in \mathcal{R} : d_W(x, Y_{ij}) \leq d_W(x, Y_{ik}) \ \forall k \neq j\}$, with $d_W(x, Y_{ij})^2 = (x_1 - Y_{ij,1})^2 + w(x_2 - Y_{ij,2})^2$.

Superimposing the partitions induced by every $y \in \mathcal{Y}$ at weight w yields the minimal relevant partition $\mathbb{X}_w$.⁴³ A cell of $\mathbb{X}_w$ is indexed by a *choice profile*

$$\mathbf{c} = (j(y))_{y \in \mathcal{Y}}, \quad j(y) \in \mathcal{J},$$

recording the candidate chosen under each environment, and equals the intersection

$$\mathcal{X}_w(\mathbf{c}) = \bigcap_{y \in \mathcal{Y}} \mathcal{X}_{j(y);w}(y). \quad (\text{A1})$$

The profile $\mathbf{c}$ is *available* at weight w when $\mathcal{X}_w(\mathbf{c}) \neq \emptyset$; only available profiles enter the prediction of observed vote choices (cf. Equation 5). A partition $\mathbb{X}_w$ is thus summarized by its set of available profiles,

$$\mathcal{C}(\mathbb{X}_w) = \{\mathbf{c} : \mathcal{X}_w(\mathbf{c}) \neq \emptyset\},$$

and two weights are observationally equivalent whenever they induce the same $\mathcal{C}(\mathbb{X}_w)$.

The cell as a system of linear constraints. By (A1), $\mathcal{X}_w(\mathbf{c})$ is an intersection of half-planes: one per environment y and rival $k \neq j(y)$, for a total of $M = \sum_{y \in \mathcal{Y}} (|y| - 1)$ constraints. The set of bliss points at least as close to an own position s as to a rival s' is the half-plane $\{x : d_W(x, s)^2 \leq d_W(x, s')^2\}$, which after expansion reads $2(s_1 - s'_1)x_1 + 2w(s_2 - s'_2)x_2 \leq (s_1^2 - s_1'^2) + w(s_2^2 - s_2'^2)$.⁴⁴ Indexing the constraints by $m = 1, \dots, M$, each takes the form

$$\alpha_m x_1 + w \beta_m x_2 \leq \gamma_m + w \delta_m, \quad m = 1, \dots, M, \quad (\text{A2})$$

with fixed constants $\alpha_m = 2(s_1 - s'_1)$, $\beta_m = 2(s_2 - s'_2)$, $\gamma_m = s_1^2 - s_1'^2$, $\delta_m = s_2^2 - s_2'^2$, where for constraint m arising from environment y and rival k the own position is $s = Y_{i,j(y)}$ and the rival is

⁴³This coincides with the coarsest partition of Section 2: superimposing the per-environment partitions merges exactly those bliss points that choose identically under every $y \in \mathcal{Y}$.

⁴⁴Ties (the bisector itself) occur with probability zero under the continuity assumption of Section 2, so the weak/strict inequality convention is immaterial for availability.

$s' = Y_{ik}$. The key structural feature is that w enters (A2) only linearly, both in the coefficient of x_2 and in the threshold, and the same scalar w is shared across all M constraints.

The *availability set* of a profile $\mathbf{c}$ is the set of weights at which its cell is non-empty:

$$\mathcal{W}(\mathbf{c}) = \{ w > 0 : \mathcal{X}_w(\mathbf{c}) \neq \emptyset \} = \{ w > 0 : \exists x \in \mathbb{R}^2 \text{ satisfying (A2) } \forall m \}. \quad (\text{A3})$$

A.2 Set-identification of weights

Lemma 1. *For every choice profile $\mathbf{c}$, the availability set $\mathcal{W}(\mathbf{c})$ is an interval.*

Proof. Fix $w > 0$ and apply the change of variables $(z_1, z_2) = (x_1, wx_2)$, with inverse $x_2 = z_2/w$. For each fixed $w > 0$ this is a bijection on $\mathbb{R}^2$, so $\mathbf{c}$ is available at w if and only if the transformed system

$$\alpha_m z_1 + \beta_m z_2 \leq \gamma_m + w \delta_m, \quad m = 1, \dots, M, \quad (\text{A4})$$

admits a solution $z \in \mathbb{R}^2$. Hence

$$\mathcal{W}(\mathbf{c}) = \{ w > 0 : \exists z \in \mathbb{R}^2 \text{ satisfying (A4)} \}. \quad (\text{A5})$$

The substitution linearizes the bilinear term $w\beta_m x_2$ into $\beta_m z_2$, so w now appears only in the thresholds. Define the augmented set in (z_1, z_2, w) -space

$$\Omega = \left\{ (z_1, z_2, w) \in \mathbb{R}^3 : \alpha_m z_1 + \beta_m z_2 - w \delta_m \leq \gamma_m, \quad m = 1, \dots, M \right\}. \quad (\text{A6})$$

Each defining inequality is linear in the joint variable (z_1, z_2, w) , so Ω is an intersection of M closed half-spaces of $\mathbb{R}^3$, hence a convex polyhedron.

Let $\pi : \mathbb{R}^3 \rightarrow \mathbb{R}$ be the projection $\pi(z_1, z_2, w) = w$. Comparing (A5) and (A6),

$$\mathcal{W}(\mathbf{c}) = \pi(\Omega) \cap (0, \infty).$$

The image of a convex set under a linear map is convex, so $\pi(\Omega)$ is convex; intersecting with the convex set $(0, \infty)$ preserves convexity. Therefore $\mathcal{W}(\mathbf{c})$ is an interval. $\square$

Proposition 3. *Let $\mathcal{C}^+$ be any finite collection of choice profiles whose cells are required to be non-empty. Then the set of weights at which every profile in $\mathcal{C}^+$ is available,*

$$\mathcal{W}(\mathcal{C}^+) \equiv \bigcap_{\mathbf{c} \in \mathcal{C}^+} \mathcal{W}(\mathbf{c}),$$

is an interval. Moreover, if the data are generated at the true weight w_0 , then $w_0 \in \mathcal{W}(\mathcal{C}^+)$, and the identified set of weights satisfies $W^ \subseteq \mathcal{W}(\mathcal{C}^+)$.*

Proof. By Lemma 1, each availability set $\mathcal{W}(\mathbf{c})$ is an interval, and a finite intersection of intervals is again an interval; since $\mathcal{Y}$ is finite, $\mathcal{C}^+$ is finite. If the data are generated at w_0 , then every choice profile observed in the data corresponds to a non-empty cell at w_0 , so any cell that the data require to carry positive mass is available at w_0 ; hence $w_0 \in \mathcal{W}(\mathbf{c})$ for every $\mathbf{c} \in \mathcal{C}^+$, and $w_0 \in \mathcal{W}(\mathcal{C}^+)$. Finally,

any weight $w \in W^*$ must rationalize the data, which requires every cell in $\mathcal{C}^+$ to be non-empty at w ; therefore $W^* \subseteq \mathcal{W}(\mathcal{C}^+)$. $\square$

B Weights estimation: simulation

This section provides a simulation comparing the weight estimator based on 1) the method used in this paper, and 2) an approach that allows for a simultaneous recovery of bliss points and weights from Merlo and De Paula (2017).

I simulate a population of 10,000 voters voting in twenty-four elections with six candidates drawn at random. Bliss points are drawn from a mixture of two dominant clusters and a diffuse background. Voters choose the candidate closest to their bliss point under the weighted Euclidean metric with true weight $w_2 = 1$; the researcher observes only the resulting candidate vote shares in each elections.

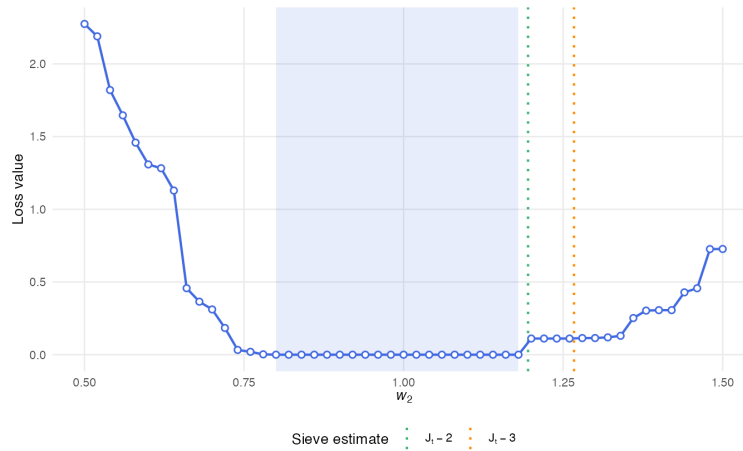
For each candidate value of w_2 on a grid, the implied superimposed Voronoi tessellation determines the set of feasible cells (joint vote patterns). The question then is whether voter mass can be distributed across these cells so as to reproduce the observed shares exactly, subject to masses being non-negative and summing to one. The set of weights for which an exact fit exists is the identified set: since every cell is feasible only on an interval of weights, and concentrated vote shares force particular cells to carry mass, the weights consistent with the data are contained in an interval around the truth; in this simulation, the exact-fit set is itself an interval.

Figure B1 plots the residual fit against w_2 . The loss is exactly zero on the interval $[0.80, 1.18]$, which contains the true value $w_2 = 1$, and rises outside it. The cell geometry alone thus set-identifies the weight without any assumption on the distribution of bliss points.

The dotted lines mark the estimates obtained from the sieve estimator at polynomial orders $J_t = 2, 3$, which impose a flexible parametric form on the bliss-point density. Because a low-order polynomial correction of a Gaussian base cannot reproduce the distribution of voters' bliss points, the resulting weight estimates are far from the truth as well, at 1.20 and 1.27 for $J_t = 2, 3$ respectively.

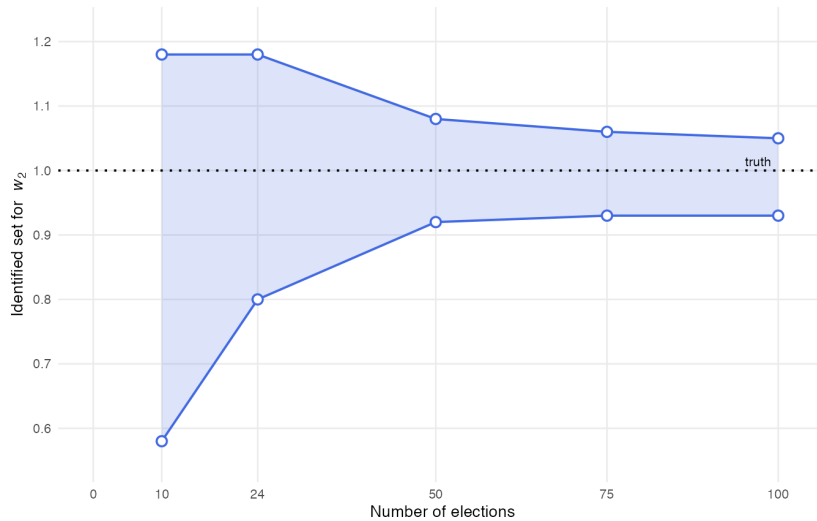
Number of Elections. The identified set shrinks as the number of elections grows. Because each additional election contributes a further set of cells that the observed shares may or may not be able to populate, more elections impose more restrictions on which weights admit an exact fit. Figure B2 traces the identified set for w_2 against the number of elections, holding the data-generating process fixed: the set narrows monotonically around the true value, from $[0.58, 1.18]$ with ten elections to $[0.93, 1.05]$ with one hundred. Note that this simulation illustrates that the size of the identified sets shrinks with the number of elections, the size of the identified sets itself at every elections level depends on the positions of the candidates, and the distribution of preferences.

Noisy perceptions of candidate positions. Realistically, however, there is always noise in the data. I next simulate the behaviour of both estimators under the assumption that each voter observes every candidate's position with an idiosyncratic mislocation error, distributed $\mathcal{N}(0, 0.1^2 I)$, while the estimation procedures remain unchanged. Under noise an exact fit is no longer attainable at any weight. The question, however, is where the minimum is located and how large the basin delivering a near-identical fit is. The region of near-minimal losses becomes *smaller*: the loss now develops a strict interior minimum, and the near-feasible set (weights whose fit is within one percentage point



Notes: Monte Carlo with 10,000 voters, true $w_2 = 1$, twenty-four six-candidate elections; bliss points follow a two-cluster mixture with a diffuse background, and only vote shares are observed. The solid curve is the minimal L^1 distance between observed shares and shares generated by any distribution of voter mass over the cells of the superimposed tessellation at each candidate weight. The shaded region is the identified set $[0.80, 1.18]$, where the fit is exact; it contains the truth. The dotted lines mark the sieve estimates of Merlo and De Paula (2017) at $J_t = 2, 3$ ($\hat{w}_2 = 1.20, 1.27$), both of which lie outside the identified set.

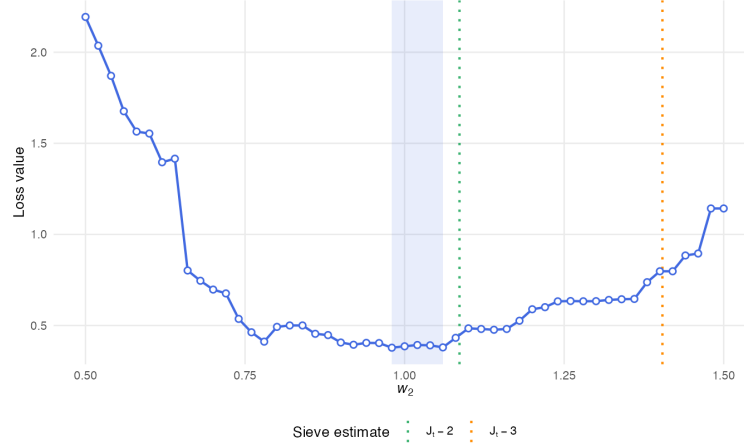
Figure B1. Set identification of the weight via the method used in this paper, and sieve estimates from Merlo and De Paula (2017)



Notes: Each point pair gives the lower and upper bounds of the identified set for w_2 (the weights admitting an exact fit), computed as in Figure B1 but varying the number of elections. The dotted line marks the true value $w_2 = 1$.

Figure B2. Identified set for the weight at different numbers of elections

of share mass of the best attainable) shrinks from the noise-free interval $[0.80, 1.18]$ to $[0.98, 1.06]$, still containing the truth (Figure B3). Intuitively, the smoothed shares are best approximated by the cell structure of the true weight, so mild noise converts the flat exact-fit region into a well-defined basin centred near $w_2 = 1$. The sieve estimator yields $\hat{w}_2 = 1.09$ and 1.40 at $J_t = 2, 3$, both falling outside the near-feasible set. Note that in this simulation the results obtained with $J_t = 2$ are closer to the truth; this suggests that additional flexibility in the estimator does not necessarily translate into more precise estimates.



Notes: Same setup as Figure B1, except that each voter perceives every candidate’s position with an idiosyncratic $\mathcal{N}(0, 0.1^2 I)$ mislocation error. The solid curve is the minimal L^1 distance between observed shares and shares generated by any distribution of voter mass over the cells of the superimposed tessellation at each candidate weight; with noise the loss is strictly positive everywhere. The shaded region is the near-feasible set $[0.98, 1.06]$ (fit within 0.01 of the minimum), which contains the true value $w_2 = 1$. The dotted lines mark the sieve estimates of Merlo and De Paula (2017) at $J_t = 2, 3$ ($\hat{w}_2 = 1.09, 1.40$), both outside the near-feasible set.

Figure B3. Set identification of the weight via the method used in this paper, and sieve estimates from Merlo and De Paula (2017) with voter perception noise

C Exploiting congressional district borders for identification

Longuet-Marx (2025) exploit differences in candidate positions across congressional district borders to identify voters’ preferences, but rely on a different model of voters’ choice. In this section I illustrate how this type of identifying variation could in principle be used to estimate the primitives of vote choice using the model considered in this paper.

For simplicity, I assume a one-dimensional model of choice, with the policy space normalized to the interval $[1, 5]$, and a population that is homogeneous in its observed characteristics across space (i.e. the entire population belongs to a single group d , for example educated females aged 30–35). This assumption simplifies the exposition without affecting the substantive argument.

Suppose that precincts with positions y and y' are paired. Then $f(x | y, m, d) = f(x | y', m, d)$ by the assumption that congressional district borders are exogenous. Likewise, for another pair y'' and y''' we have $f(x | y'', m, d) = f(x | y''', m, d)$. Crucially, this approach does not assume $f(x | y, m, d) = f(x | y'', m, d)$; that is, preferences may differ across distinct pairs of precincts.

Consider a two-party system, which is the most demanding case, as it provides the least variation. Each choice situation is then characterized by the positions of two candidates on a one-dimensional space. From the perspective of a spatial voting model, the object that uniquely defines

this situation is the cut-point between the two candidates. Hence choice situation y yields a cut-point y_c , while y' yields a cut-point y'_c . The researcher observes vote shares for both candidates in both precincts, as illustrated in Figure C1.



Figure C1. Two paired precincts. Candidate positions (A , B), the implied cut-point (thin vertical line), and observed vote shares differ across the pair, while the underlying distribution of bliss points is common by the exogenous-border assumption.

Since the distribution of bliss points is identical within a pair of contiguous precincts, one can overlay the two sets of results to obtain a single, more finely partitioned distribution, as shown in Figure C2.

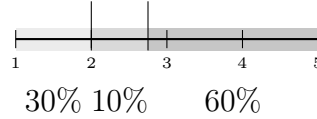


Figure C2. Overlaid distribution for the paired precincts of Figure C1. The two cut-points partition the line into three intervals, yielding refined shares of the common bliss-point distribution.

Repeating this exercise across many pairs of contiguous precincts yields a collection of refined distributions of bliss points. Note, however, that these distributions are assumed to differ across pairs and therefore cannot be overlaid with one another, even when, as in the case considered here, the demographics are identical. What the researcher can do instead is draw inferences from this collection of distributions, using the known population size of each pair of precincts.

For example, suppose the precinct pair above has a population of one thousand. The numbers of individuals lying in each of the three intervals are then 300, 100, and 600.

Now consider a second precinct pair of 1,500 individuals, with cut-points and shares as displayed in Figure C3. The corresponding numbers of individuals in each interval are 450, 300, and 750.

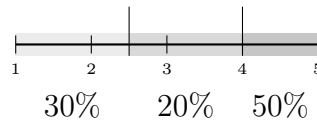


Figure C3. Overlaid distribution for a second precinct pair, with distinct cut-points and shares. This distribution is specific to the pair and cannot be combined with that of Figure C2.

What can be learned from the two distributions jointly? Although they differ and their electoral results cannot be pooled, one can still pose partial-identification questions in the spirit of this paper. For instance, what share of individuals of the given demographic hold left-wing views, defined as a bliss point below 2? The first pair's cut-points lie at 2 and 3.5, and the second pair's at 2.5 and 4. For the first pair, the threshold 2 coincides with the first cut-point, so the 300 individuals lie below

the threshold. For the second pair, the threshold falls strictly inside the first interval $[1, 3)$; since the within-interval distribution is unrestricted, the 450 individuals in that interval may lie anywhere in it, so the pair contributes between 0 and 450. Hence, out of 2,500 individuals in total, the share holding left-wing views is set-identified within $[0.12, 0.30]$.

Note that identifying the dimension weights in this setup would not be possible without further assumptions. As discussed in the main text, the weights are identified through variation in the structure of the partition that candidate positions induce on the policy space, that is, in the composition of the resulting cells. Overlaying only two elections within a pair of contiguous electoral districts generates little to no variation in this cell composition, and is therefore insufficient to pin down the weights.

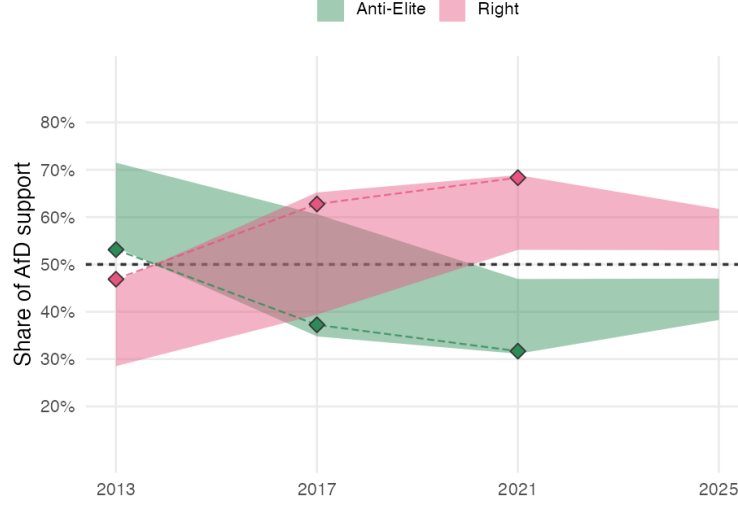
D Stated preferences

This section runs an exercise similar to the one in Section 6.2.2 using a stated preference survey data.

Before turning to the exercise and its results, I want to stress that recovering the weights from these data remains infeasible, so I take the estimated weights from the analysis in this paper. This is so for at least two reasons. First, respondents are surveyed throughout the year, and one cannot expect their perceptions of party positions to remain constant over time; restricting attention to those surveyed no more than three months before the election leaves at most 300 AfD voters in any given year. Second, the problem of voters adjusting their preferences so as to be closer to the party is especially acute for weight estimation, since such closeness would be read as evidence that a dimension is important—which may be precisely the opposite of the truth if individuals are adjusting their stated positions on the issues that matter least to them.

For this exercise I use data from the Politbarometer survey for 2013–2021 (2025 is not yet available). I approximate voters’ Left–Right and Anti-Elite ideal points by their answers to the questions on Left–Right self-placement and satisfaction with the government, as these are the only items available across all years and across all waves within each year. To construct the nationwide AfD position, I take the mean of these answers across AfD voters in a given year. While this need not approximate the nationwide AfD position as accurately as in the main exercise, it is a reasonable approximation and avoids having to place the estimated AfD median values on the same scale as the survey items. I then proceed exactly as in Section 6.2.2.

Figure D1 plots the estimates for the stated preference results over the results from Figure 5b. The results from the surveys replicate the trend well, and remain within the estimated bounds from the revealed preference estimates, while they tend to belong to the upper (lower) bound of the right (anti-elite) shares. This might reflect the specific questions used for the construction of the measure, or the fact that the within-sample means are a poor approximation of the positions used in the revealed preference exercise. Nevertheless, the main takeaway of this exercise is that the identified patterns are consistent with the survey results.



Notes: Survey benchmark constructed from the GESIS Politbarometer surveys (2013, 2017, 2021). Each respondent’s bliss point is built from two items: the left–right self-placement (11-point scale), and satisfaction with the government, rescaled to the same 11-point range. For each year I compute the national survey-weighted mean bliss point among all voters and proceed with the analysis as in Section 6.2.2. The resulting shares of AfD voters are point estimates, not partial-identification bounds. The 2025 survey data are not yet available. The estimates are weighted throughout using the survey weights provided by GESIS.

Figure D1. Comparison to the survey results

E Model Estimation

This section describes how the empirical restrictions implied by the spatial voting model are translated into a finite-dimensional optimization problem, and how weights and bounds for target parameters are computed.

E.1 Weights estimation

This appendix details how the relative weight w_{AE} on the anti-elite dimension is computed. Throughout, the left–right weight is normalised to $w_{LR} = 1$, so the only free parameter is $w_{AE} \in \mathbb{R}_+$.

The exercise is run separately for each region (East, West) and year. As discussed in Section B, weights estimation requires lots of variation in candidate positions, pooling electoral districts within each region helps achieve this. Within a region, the market m is the *state*, the submarket s is the *municipality* (with $s \in m$), and d indexes the 24 age×gender×education demographic groups, with observed demographic shares $q_{s,d}$, $\sum_d q_{s,d} = 1$. An *election* y is the candidate configuration of an electoral district; $\mathcal{Y}_R$ collects the elections of region R , and each municipality s belongs to one election $y(s)$ and one state $m(s)$. The choice set is the six parties.

For a candidate value w_{AE} , the weighting matrix is $W = \text{diag}(1, w_{AE})$. Since w_{AE} enters the distance $d(x, Y_{ij})_W = \sqrt{(x - Y_{ij})^\top W (x - Y_{ij})}$, it rescales the anti-elite axis relative to left–right and thus tilts the bisecting hyperplanes separating any two parties, reshaping each attraction region $\mathcal{X}_{j;w}(y)$ and hence which cells (joint choice sets) appear in the MRP.

Overlaying all elections in $\mathcal{Y}_R$ yields the minimal relevant partition (MRP) $\mathbb{X}_w$, and $\mathbb{X}_{j;w}(y)$ denotes the cells of $\mathbb{X}_w$ on which j wins under y . Because the number of feasible cells governs how well any given partition can fit the data, and this count varies with w_{AE} , I hold it fixed across weights:

I restrict attention to the AfD attraction region, retain the $N = 500$ largest cells, and merge each smaller fragment into the neighbouring cell with which it shares the longest boundary. Let $\mathbb{X}_w$ denote the resulting 500-cell mesh.

The unknowns are the state-and-group mass functions

$$\phi(\mathcal{X} \mid m, d) \geq 0, \quad \sum_{\mathcal{X} \in \mathbb{X}_w} \phi(\mathcal{X} \mid m, d) = 1,$$

one per state m in region R and demographic group d . The weight w_{AE} is held common across all states and groups of the region-year. Given ϕ and w_{AE} , the predicted AfD vote share in municipality s is the finite-dimensional analogue of (9),

$$s_{\text{AfD}}(y(s), s; \phi, w_{AE}) = \sum_d q_{s,d} \sum_{\mathcal{X} \in \mathbb{X}_{\text{AfD};w}(y(s))} \phi(\mathcal{X} \mid m(s), d).$$

In finite samples the exact restriction (9) generically has no solution. For each w_{AE} I choose the mass functions that minimise the mean absolute municipal AfD-share error,

$$\mathcal{E}(w_{AE}) = \min_{\{\phi(\cdot \mid m, d)\} \in \Phi} \frac{1}{|\mathcal{S}_R|} \sum_{s \in \mathcal{S}_R} \left| s_{\text{AfD}}(y(s), s) - s_{\text{AfD}}(y(s), s; \phi, w_{AE}) \right|,$$

where $\mathcal{S}_R$ is the set of municipalities in region R and $s_{\text{AfD}}(y(s), s)$ the observed share. Introducing slacks $u_s \geq 0$, this is the linear program

$$\min_{\{\phi(\cdot \mid m, d)\}, u_s \geq 0} \sum_{s \in \mathcal{S}_R} u_s \quad \text{s.t.} \quad -u_s \leq s_{\text{AfD}}(y(s), s; \phi, w_{AE}) - s_{\text{AfD}}(y(s), s) \leq u_s,$$

solved over $\phi \in \Phi$.

The program is solved for every candidate weight on the grid

$$\mathcal{W} = \{0.2, 0.3, \dots, 0.9, 1.0\} \cup \{1/0.9, \dots, 1/0.2\} \quad (17 \text{ points}),$$

symmetric around 1 in log scale, and the estimate is the fit-minimising weight

$$\hat{w}_{AE} = \arg \min_{w_{AE} \in \mathcal{W}} \mathcal{E}(w_{AE}).$$

Values $\hat{w}_{AE} < 1$ mean voters tolerate larger distances on the anti-elite dimension (the left-right dimension weighs more); $\hat{w}_{AE} > 1$ the reverse. Because the data are noisy the minimizer is generically a single point in most cases. Estimation is carried out independently by region, year. The number of elections per state is capped at 20. The 500-cell mesh is rebuilt at every w_{AE} .

E.2 Target parameters estimation

This appendix describes the bounds used for the main Anti-Elite versus Right exercise. Estimation of bounds for the rest of the exercises follow the same logic. It follows the generic target-parameter LP and adds the two features specific to that exercise: the demographic-group masses are made *state specific* but are pooled onto a single region-wide partition, and they are tied together across states of

a region by a *cross-state coupling* restriction. The exercise is run separately for each region (West, East) and year, with the relative weight fixed at the estimate $\hat{w}_{AE}$ from Appendix E.1.

Under the weight $\hat{w}_{AE}$ I overlay the elections of *all* states of a region to form one minimal relevant partition, and restrict the focus to the AfD attraction region and fix it at the $n_c = 500$ cells, similar to the procedure for weights estimation. Cells are indexed $c \in \{1, \dots, n_c\}$ and demographic groups (age $\times$ gender $\times$ education bins) by $d \in \{1, \dots, D\}$. The masses are indexed by state m : for each state m in region and each group d ,

$$\boldsymbol{\pi}_d^{(m)} = (\pi_{d1}^{(m)}, \dots, \pi_{dn_c}^{(m)})^\top \in \mathbb{R}_+^{n_c}, \quad \mathbf{1}^\top \boldsymbol{\pi}_d^{(m)} = 1,$$

so each state has its own bliss-point distribution over the shared region mesh. Municipalities are indexed $r \in \{1, \dots, R\}$, each belonging to one state $m(r)$; y_r is the observed AfD share and $w_{r,d}$ the turnout-adjusted share of group d among eligible voters in r . With $a_{r,c}^{\text{AfD}} \in \{0, 1\}$ the model-implied AfD choice indicator in r and $\mathbf{a}_r^{\text{AfD}}$ their vector, the predicted AfD share is

$$\hat{y}_r(\boldsymbol{\pi}) = \sum_{d=1}^D w_{r,d} (\mathbf{a}_r^{\text{AfD}})^\top \boldsymbol{\pi}_d^{(m(r))}. \quad (\text{E1})$$

Let $(\bar{X}^{RR}, \bar{X}^{AE})$ be the AfD position (national median value in a year) on the left–right and anti-elite dimensions. I split the AfD attraction region into two target areas by which of two reference points is nearer under the weighted metric $W = \text{diag}(1, \hat{w}_{AE})$:

$$C^{AE} = (\bar{X}^{RR} - 1, \bar{X}^{AE}), \quad C^R = (\bar{X}^{RR}, \bar{X}^{AE} - 1).$$

The *Anti-Elite* region collects the cells nearer C^{AE} ; the *Right* region collects the cells nearer C^R , which pulls on the anti-elite axis and retains the right-wing voters. Writing $b_c^{AE}, b_c^R \in \{0, 1\}$ for the resulting cell indicators, the target parameters are the shares of the region electorate who vote AfD and hold ideal points in each area,

$$\bar{\theta}^\bullet(\boldsymbol{\pi}) = \sum_m \sum_{d=1}^D (\mathbf{b}_d^\bullet)^\top \boldsymbol{\pi}_d^{(m)}, \quad \bullet \in \{AE, R\},$$

where $\mathbf{b}_d^\bullet$ weights the in-area AfD cells by the population mass of group d in state m .⁴⁵

The masses satisfy, besides non-negativity, normalisation, and the fit slacks $-u_r \leq \hat{y}_r(\boldsymbol{\pi}) - y_r \leq u_r$, the second-choice moment restrictions, namely – share of AfD voters for whom the Left is the second best choice. Writing $s_{r,c}^{\text{Left}} \in \{0, 1\}$ for the indicator that in r 's election the Left is the second-nearest party (after the AfD) on cell c , the vector $\mathbf{v}_{m,d}^{\text{Left}}$ has entries $v_{m,d,c}^{\text{Left}} = \sum_{r: m(r)=m} e_r w_{r,d} a_{r,c}^{\text{AfD}} s_{r,c}^{\text{Left}}$, with e_r the number of eligible voters in r , so that $(\mathbf{v}_{m,d}^{\text{Left}})^\top \boldsymbol{\pi}_d^{(m)}$ counts the state- m , group- d AfD voters with the Left as second choice:

$$(\ell_m + \tau_{\text{SC}}) A_m \geq \sum_d (\mathbf{v}_{m,d}^{\text{Left}})^\top \boldsymbol{\pi}_d^{(m)} \geq (\ell_m - \tau_{\text{SC}}) A_m \quad \forall m,$$

with $A_m = \sum_d X_{m,d} T_{m,d}$ the state AfD mass ($T_{m,d}$: eligible voters of group d in state m ; $X_{m,d}$: the

⁴⁵For simplicity, I discuss the case of the target parameter with respect to eligible voters. The parameter with respect to the AfD vote share is analogous, but requires dividing by that share.

group's observed AfD support rate — both data, so the restriction is linear in π) and ℓ_m the observed AfD to Left second-choice rate; $\tau_{\text{SC}} = 0.1$.

Another restriction ties the states of a region together so that pooling is informative without forcing them to be identical. I coarsen the policy space into blocks B on a 0.5 grid and, for each coarse marginal $g \in \{\text{With Higher Education, Without Higher Education, Young, Old, Male, Female}\}$ and block B , require every state's within-marginal mass share on B to stay within a factor $(1 + \tau_{\text{state}})$ of the population-weighted regional mean. With $e_{m,g,B} = \sum_{d \in g} T_{m,d} \sum_{c \in B} \pi_{dc}^{(m)}$ the state's block mass, $T_{m,g} = \sum_{d \in g} T_{m,d}$, and an auxiliary regional mean share $M_{g,B}$,

$$\sum_m e_{m,g,B} = \left(\sum_m T_{m,g} \right) M_{g,B}, \quad \frac{M_{g,B}}{1 + \tau_{\text{state}}} \leq \frac{e_{m,g,B}}{T_{m,g}} \leq (1 + \tau_{\text{state}}) M_{g,B} \quad \forall m.$$

The benchmark configuration uses $\tau_{\text{state}} = 2$ (each state's block share within a factor of three of the regional mean). Note that this is a loose coupling, as it is done only at the level of coarse demographic groups, for large "blocks", and allows for large differences across states within those groups-blocks.

I first solve the best-fit program $\hat{Q} = \min_{\pi, u} \mathbf{1}^\top u$ subject to all of the above. The approximately feasible set caps the total fit slack additively, $\mathbf{1}^\top u \leq \hat{Q} + \epsilon_{\text{abs}}$, with $\epsilon_{\text{abs}} = 10^{-5}$, and the bounds are the two LPs

$$\underline{\theta}^\bullet = \min_{\pi, u} \bar{\theta}^\bullet(\pi), \quad \bar{\theta}^\bullet = \max_{\pi, u} \bar{\theta}^\bullet(\pi),$$

subject to the same constraints and the fit cap. This is the procedure similar to the one in (Tebaldi, Torgovitsky and Yang, 2023).

The reference run pools each region under $\hat{w}_{AE}$, samples at most 100 municipalities per election with population below 10 thousand inhabitants, caps elections per state at 20, fixes the mesh at 500 cells.

E.3 Model Fit

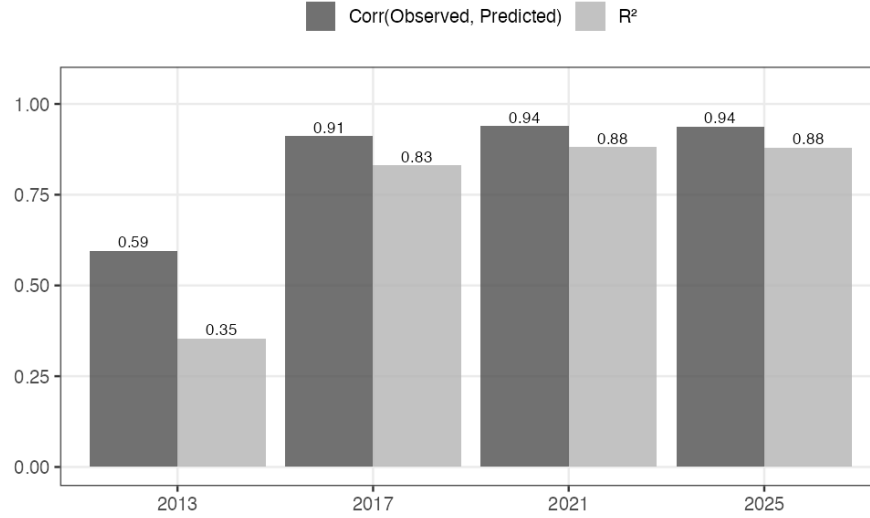
I discuss model fit. Table F3 presents summary statistics on model fit at the municipality level. In every election year, the median residual is exactly 0 by construction. The standard deviation ranges from 0.016 in 2013 to just above 0.03 in 2017 and 2021, and 0.048 in 2025. This increase mirrors an increase in AfD vote shares. The correlation between predicted and observed vote shares is above 0.9 in the later years, and about 0.6 in 2013. Overall, these results suggest a reasonable model fit, especially in the later years.

The correlations of predicted and observed municipal AfD shares with municipalities' demographic characteristics are almost identical (Figure E2).

To sum up, both the model fit and the observed heterogeneity in the data suggest that the model performs well in explaining observed AfD voting patterns.

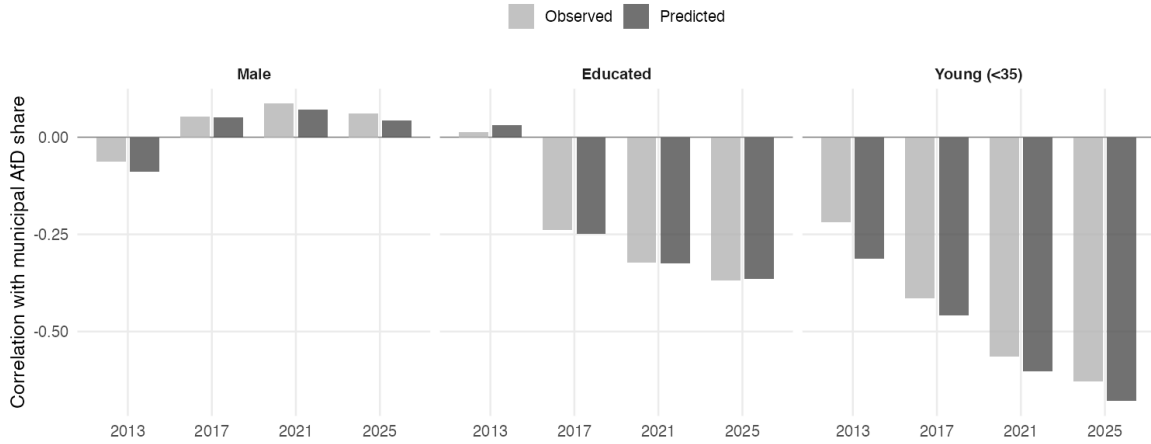
E.4 Statistical inference

The target is the *Right share* of AfD support in the benchmark specification (Section 6.2.2). In this section I account for the sampling uncertainty in the sample of municipalities. The analysis was run only for 2021 to ease the computation.



Notes. Bars plot the values for correlation between predicted and observed vote shares, and R^2 from a regression of one on another. Predicted AfD vote shares are chosen at a random seed.

Figure E1. Correspondence between predicted and observed AfD vote shares across municipalities.



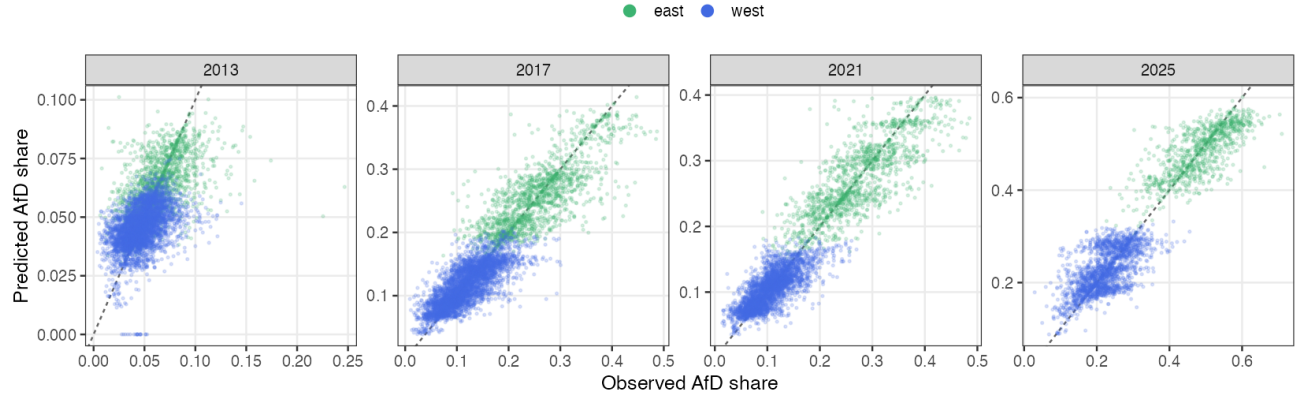
Notes. Each bar is an estimate of a correlation between a predicted (observed) AfD vote share and a given municipality demographic characteristic.

Figure E2. Correlation between predicted and observed AfD vote shares with demographic characteristics of municipalities.

Sampling uncertainty is quantified with a nonparametric bootstrap. Each replication $b = 1, \dots, B$ ($B = 100$):

1. resamples municipality observations with replacement within each (state $\times$ election) group, preserving the rest of the design;
2. re-solves the full estimator on the resampled data, yielding endpoints $(\hat{\theta}_{(b)}^{lb}, \hat{\theta}_{(b)}^{ub})$.

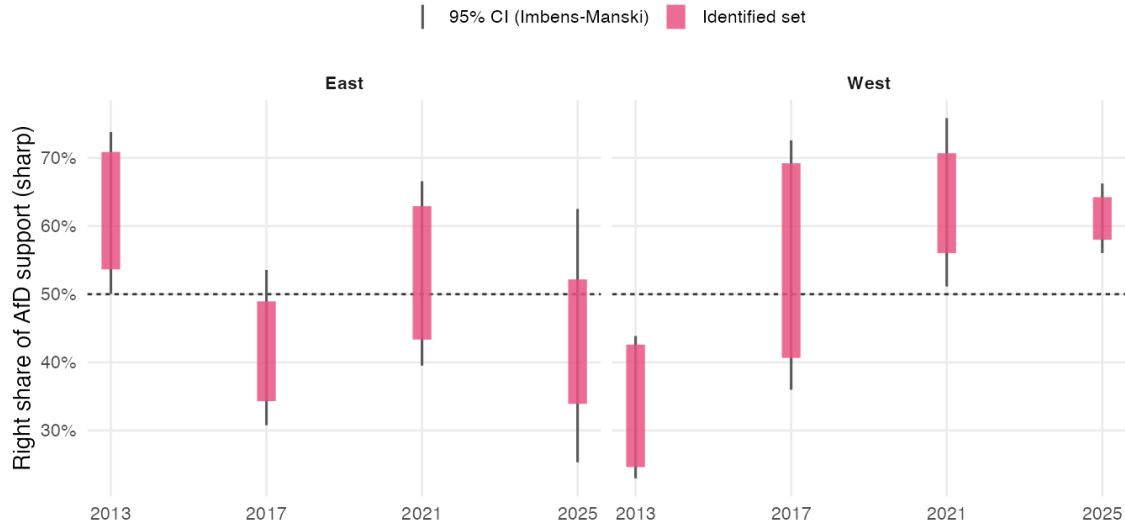
From the bootstrap distribution of the endpoints I report an interval for the parameter following [Imbens and Manski \(2004\)](#), $[\hat{\theta}^{lb} - c \hat{se}_{lb}, \hat{\theta}^{ub} + c \hat{se}_{ub}]$ with c solving $\Phi(c + \hat{\Delta} / \max(\hat{se})) - \Phi(-c) = 0.95$ and $\hat{\Delta}$ the estimated width of the identified set.



Notes. Each dot is a municipality, with the respective values for predicted and observed vote shares. Predicted AfD vote shares are chosen at a random seed.

Figure E3. Correspondence between predicted and observed AfD vote shares in Eastern and Western Germany.

Such 95% CI are presented on the Figure E4; the results remain quantitatively similar to the main results in the paper.



Bars show the sharp identified set for the Right share of AfD support, θ_R , under the benchmark specification. Whiskers are 95% confidence intervals for the parameter following (Imbens and Manski, 2004). Endpoint standard errors are obtained from a nonparametric bootstrap ($B = 100$) that resamples municipalities with replacement within state-election strata and re-solves the full linear program on each draw.

Figure E4. Bootstrapped CI, Imbens-Manski

F Tables & Figures

	<i>Dependent variable:</i>			
	<i>Perceived Party Position, L-R</i>			<i>Own L-R Position</i>
	All Parties	Chosen Parties	Chosen AfD	Chosen Parties
	(1)	(2)	(3)	(4)
Candidate, L-R	-0.042 (0.068)	0.392** (0.168)	1.843*** (0.582)	0.114 (0.164)
State F.E.	Yes	Yes	Yes	Yes
Party F.E.	Yes	Yes	Yes	Yes
Observations	49,176	7,023	269	6,878
Adjusted R ²	0.622	0.484	0.091	0.400

*p<0.1; **p<0.05; ***p<0.01

Standard errors clustered by respondent

Table F1. Correlation of LR Positions between Candidates (estimated) and Parties (reported)

	Candidate LR Position		
	All Parties	Chosen Parties	Chosen AfD
Self, LR	0.001 (0.001)	0.002 (0.002)	0.003 (0.010)
State F.E.	Yes	Yes	Yes
Party F.E.	Yes	Yes	Yes
Observations	22,715	3,236	124
R ²	0.883	0.859	0.213
Adjusted R ²	0.883	0.858	0.144

Note: *p<0.1; **p<0.05; ***p<0.01. Estimation is based on German Electoral Survey 2021 Data, which records each individual's electoral district. Self-reported views take values from 0 (very left) to 11 (very right).

Table F2. Candidates' LR positions are uncorrelated with voters' self-reported views

	2013	2017	2021	2025
Mean	0.001	0.001	0.001	-0.001
SD	0.016	0.033	0.032	0.048
P10	-0.017	-0.036	-0.034	-0.062
Median	0.000	0.000	0.000	0.000
P90	0.020	0.041	0.040	0.058

Note: Unit of observation is municipality. Residual = observed AfD vote share – model-predicted share.

Table F3. Distribution of Residuals Across Municipalities

Year	Region	Total SD		Between-party		Within-party		AfD only	
		econ	anti-el.	econ	anti-el.	econ	anti-el.	econ	anti-el.
2013	East	0.63	0.60	0.66	0.58	0.21	0.29	0.22	0.11
2013	West	0.64	0.64	0.65	0.61	0.24	0.31	0.25	0.38
2017	East	0.66	0.58	0.67	0.56	0.24	0.28	0.29	0.26
2017	West	0.72	0.64	0.75	0.62	0.22	0.30	0.26	0.28
2021	East	0.72	0.63	0.71	0.56	0.32	0.36	0.48	0.43
2021	West	0.71	0.61	0.74	0.58	0.24	0.30	0.29	0.29
2025	East	0.69	0.42	0.72	0.33	0.22	0.29	0.23	0.32
2025	West	0.75	0.48	0.80	0.41	0.19	0.30	0.19	0.35

Notes: Standard deviations of district-level (Wahlkreis) candidate positions; between-party = SD of party means, within-party = SD of positions around the own-party mean. The between-party dispersion on the anti-elite dimension compresses sharply in 2025, reflecting party convergence and the thinner 2025 item battery.

Table F4. Dispersion of estimated candidate positions by year and region

Survey item (question)	Theme	GESIS	Scale	σ (1 pt)
<i>Anti-elite dimension</i>				
Satisfaction with how democracy works in Germany	Current elites	c20	1–4	0.175
Parties are citizens’ central intermediaries	Current elites	d6c	1–5	0.144
Referendums lead to badly designed laws	Direct democracy	d6g	1–5	0.138
Legislation reflects citizens’ interests	Current elites	d6b	1–5	0.137
Citizens have lost trust in politics	Current elites	d6h	1–5	0.137
There are enough opportunities to participate	Direct democracy	d6a	1–5	0.117
There should be federal-level referendums	Direct democracy	d6f	1–5	0.098
<i>Left–right dimension</i>				
Same-sex marriage should be banned	Culture	c2d	1–5	0.096
Immigrants are good for the German economy	Immigration	c2i	1–5	0.093
Women should decide themselves about abortion	Culture	c2j	1–5	0.085
Providing social security is a core government task	Econ. policy	c2g	1–5	0.081
Government spending on development aid	Econ. policy	c18h	1–5	0.080
Immigrants should adapt to German culture	Culture	c2a	1–5	0.073
The government should stay out of the economy	Econ. policy	c2b	1–5	0.071
The government should reduce income differences	Econ. policy	c2h	1–5	0.070
Government spending on defense	Econ. policy	c18c	1–5	0.070
Own position on immigration (self-placement)	Immigration	c6	1–11	0.032

Notes. σ is the change in the standardized dimension score (Left–right or anti-elite) induced by moving a single item one scale point; it is identical across parties because the score is linear in the answers. To move a dimension score by a 0.25/0.5/1 step using a single item requires about $0.25/\sigma$, $0.5/\sigma$, $1/\sigma$ scale points: e.g. $\approx 1.4/2.9/5.7$ points on the strongest anti-elite item (satisfaction with democracy) and far more than the scale allows for the weak eleven-point immigration self-placement, so sizeable steps must come from moving many items at once. Units and item set are held fixed across years, 2017 survey items are provided as an example.

Table F5. Item-level potency (2017): shift in the dimension score from a one-point move on each survey item

Survey item (question)	Theme	GESIS	Scale	σ (1 pt)
<i>Anti-elite dimension</i>				
Satisfaction with how democracy works in Germany	Current elites	c25	1–4	0.245
Parties are citizens' central intermediaries	Current elites	d4c	1–5	0.201
Legislation reflects citizens' interests	Current elites	d4b	1–5	0.192
Citizens have lost trust in politics	Current elites	d4g	1–5	0.192
There should be federal-level referendums	Direct democracy	d4f	1–5	0.138
<i>Left–right (economic) dimension</i>				
Marriage should be only for heterosexual couples	Culture	c2d	1–5	0.106
Immigrants are good for the German economy	Immigration	c2i	1–5	0.102
Women should decide themselves about abortion	Culture	c2j	1–5	0.093
Providing social security is a core government task	Econ. policy	c2g	1–5	0.089
Immigrants should adapt to German culture	Culture	c2a	1–5	0.080
The government should stay out of the economy	Econ. policy	c2b	1–5	0.078
The government should reduce income differences	Econ. policy	c2h	1–5	0.076
Own position on immigration (self-placement)	Immigration	c6	1–11	0.035
Own position on taxes vs. social spending	Econ. policy	c9	1–11	0.030

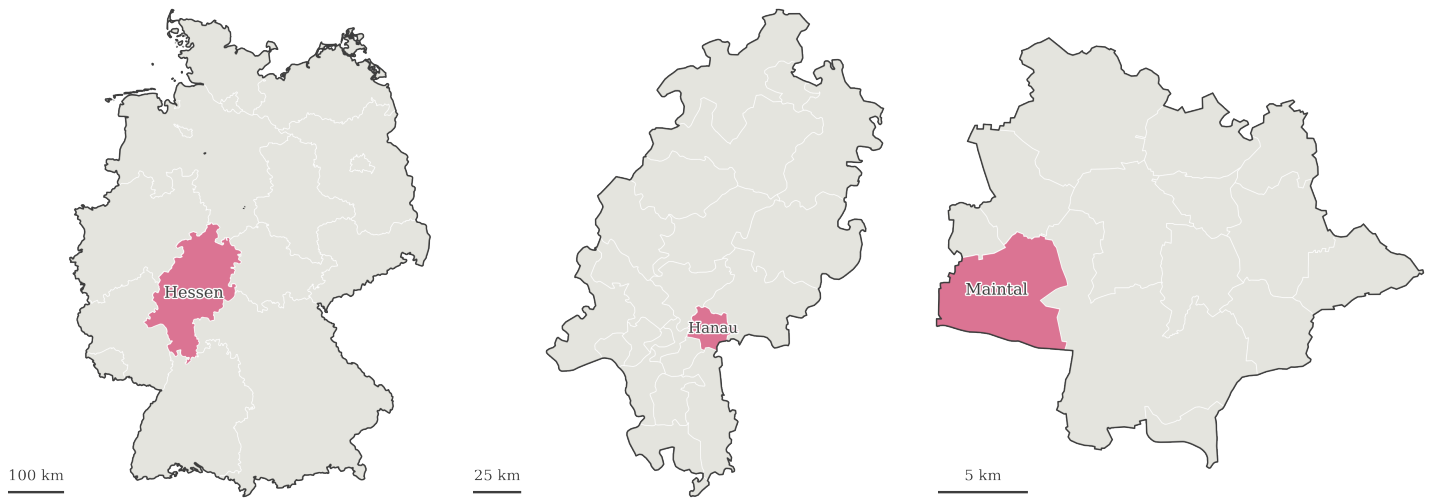
Notes. σ is the change in the standardized dimension score (Left–right or anti-elite) induced by moving a single item one scale point; it is identical across parties because the score is linear in the answers. To move a dimension score by a 0.25/0.5/1 step using a single item requires about $0.25/\sigma$, $0.5/\sigma$, $1/\sigma$ scale points: e.g. $\approx 1.4/2.9/5.7$ points on the strongest anti-elite item (satisfaction with democracy) and far more than the scale allows for the weak eleven-point immigration self-placement, so sizeable steps must come from moving many items at once. Units and item set are held fixed across years, 2025 survey items are provided as an example.

Table F6. Item-level potency (2025): shift in the dimension score from a one-point move on each survey item

	2013→2017	2017→2021	2021→2025
<i>West</i>			
Composition	[−0.102, +0.141]	[−0.138, +0.119]	[−0.122, +0.106]
Positions	[−0.202, −0.026]	[−0.047, +0.000]	[+0.023, +0.117]
Preferences	[−0.287, +0.506]	[−0.432, +0.466]	[−0.345, +0.295]
<i>East</i>			
Composition	[−0.090, +0.048]	[−0.066, +0.042]	[−0.031, +0.045]
Positions	[−0.103, −0.018]	[+0.000, +0.018]	[−0.041, −0.005]
Preferences	[−0.384, +0.234]	[−0.233, +0.312]	[−0.302, +0.157]

Notes: Bounds on the Shapley contribution of Composition, Positions and Preferences channels, derived from a single linear program over the pinned feasible set with both position-year catchments held jointly. Positive values raise the Right share (θ_R from the Subsection 6.2.2).

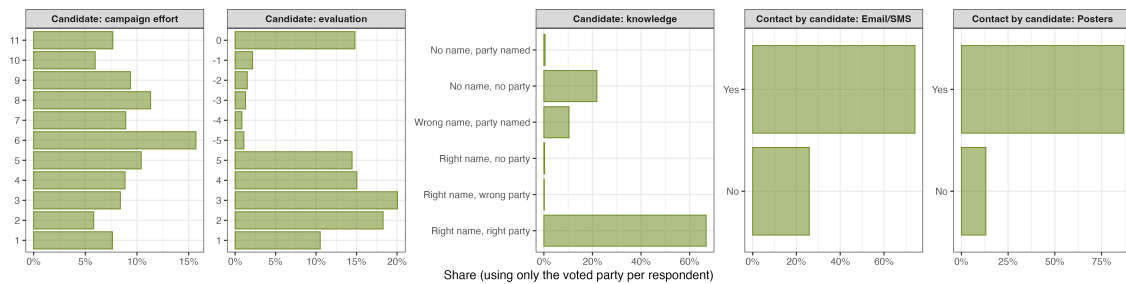
Table F7. Shapley Decomposition of the Change in the Right Share of AfD Support



(a) **Federal states.** Hessen highlighted in pink. (b) **Electoral districts** within Hessen. Hanau highlighted in pink. (c) **Municipalities** within Hanau. Maintal highlighted in pink.

Figure F1. Administrative and electoral hierarchy in Germany.

Municipalities are nested within electoral districts, which are in turn nested within federal states. The figure illustrates these levels for one example state and one example electoral district.



Campaign effort (1–11): 1 = mostly about the candidate; 11 = mostly about the party; **Candidate knowledge:** respondents' ability to name the constituency candidate and their party; **Contact:** whether the respondent was contacted via the listed means

Figure F2. Voters knowledge and interaction with SMD candidates

Partei	Insgesamt		davon Wählerinnen und Wähler in der Geburtsjahresgruppe											
			1997 – 2003		1987 – 1996		1977 – 1986		1962 – 1976		1952 – 1961		1951 und früher	
			entspricht Altersgruppe etwa von ... bis ... Jahren											
			18 – 24		25 – 34		35 – 44		45 – 59		60 – 69		70 und mehr	
	1.000	%	1.000	%	1.000	%	1.000	%	1.000	%	1.000	%	1.000	%
Mecklenburg-Vorpommern														
Insgesamt														
Insgesamt	935,0	100	52,1	100	91,9	100	135,3	100	241,1	100	205,3	100	209,4	100
ungültig	17,0	1,8	0,9	1,7	0,9	1,0	2,2	1,6	3,2	1,3	4,1	2,0	5,8	2,8
gültig	918,1	98,2	51,2	98,3	91,0	99,0	133,1	98,4	237,9	98,7	201,2	98,0	203,7	97,2
CDU	180,7	19,7	5,3	10,4	11,2	12,3	22,2	16,7	48,9	20,6	42,9	21,3	50,0	24,6
SPD	261,4	28,5	9,7	18,9	18,8	20,7	29,0	21,8	59,6	25,1	64,4	32,0	80,0	39,3
AfD	170,0	18,5	7,4	14,4	19,2	21,1	33,5	25,2	52,7	22,1	35,7	17,7	21,5	10,6
FDP	66,3	7,2	8,4	16,4	9,1	10,0	11,5	8,6	18,8	7,9	11,5	5,7	7,0	3,4
DIE LINKE	116,0	12,6	6,7	13,1	9,7	10,7	12,5	9,4	25,3	10,7	28,3	14,1	33,5	16,4
GRÜNE	62,7	6,8	8,4	16,5	12,7	14,0	11,4	8,5	15,0	6,3	8,9	4,4	6,3	3,1
CSU	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Sonstige	61,0	6,6	5,3	10,3	10,3	11,3	13,1	9,9	17,5	7,3	9,5	4,7	5,4	2,6
darunter:														
FREIE WÄHLER ¹	18,5	2,0	0,9	1,7	2,9	3,2	3,2	2,4	5,5	2,3	4,1	2,0	1,9	0,9
dieBasis ¹	15,4	1,7	0,8	1,5	1,9	2,1	3,8	2,8	5,3	2,2	2,2	1,1	1,4	0,7

Figure F3. Representative Electoral Statistics: Example

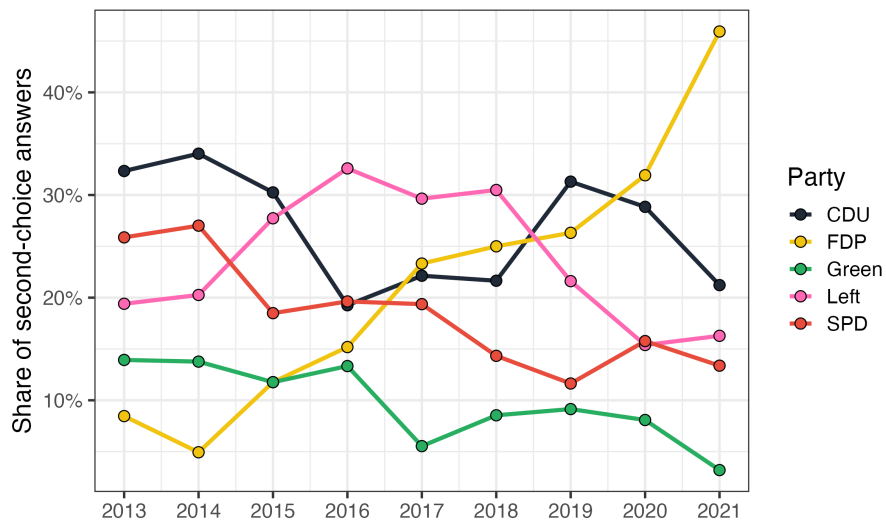
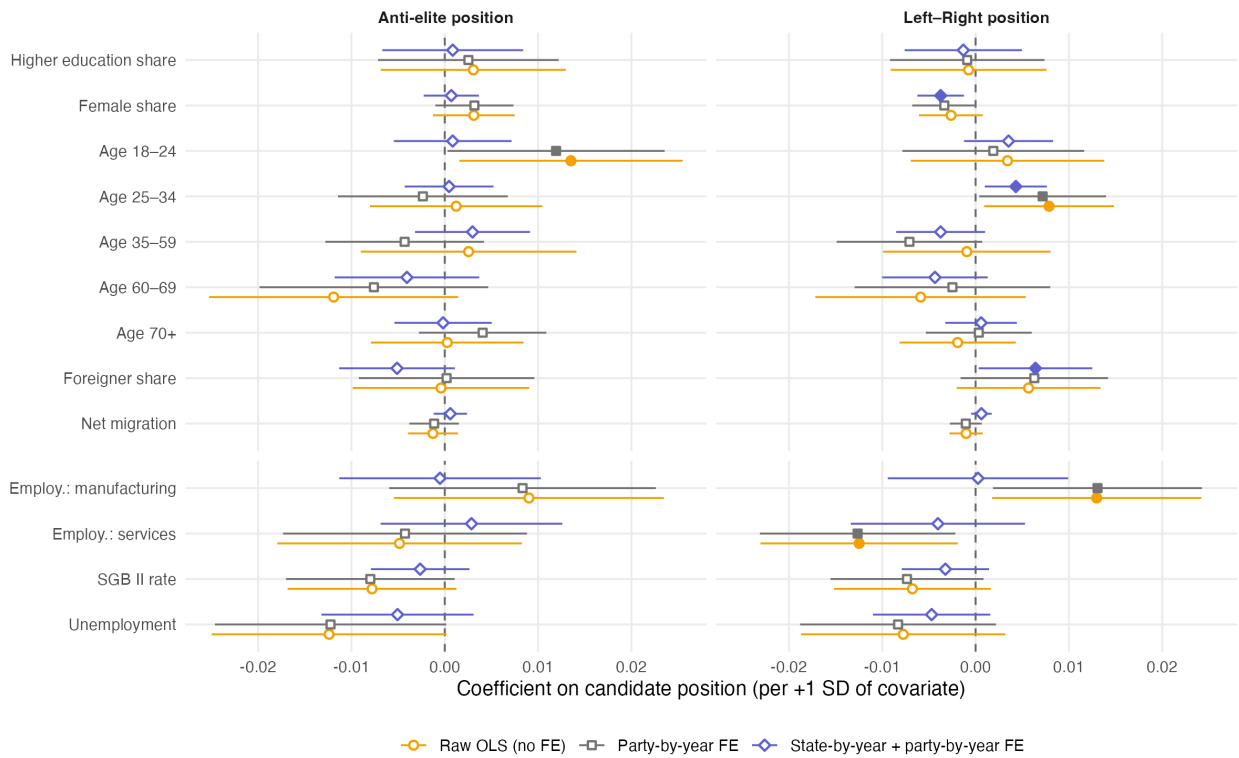
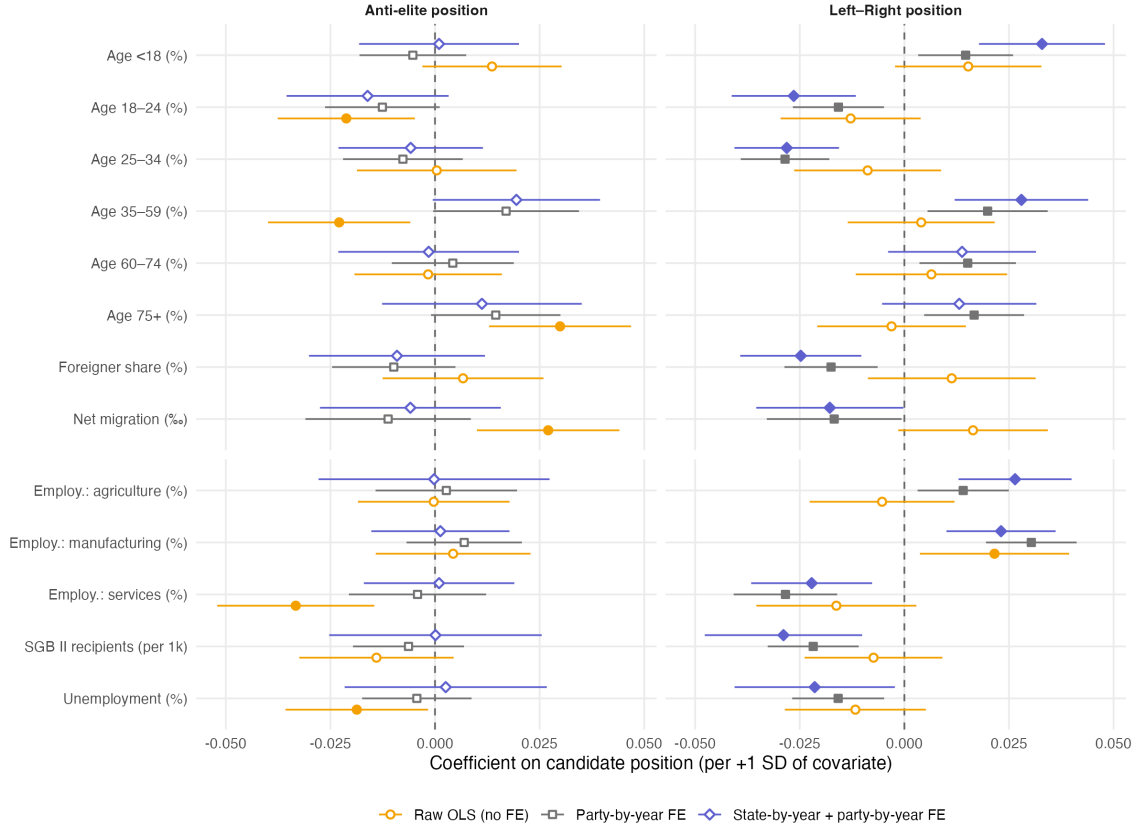


Figure F4. Second best party choices of AfD voters, federal level



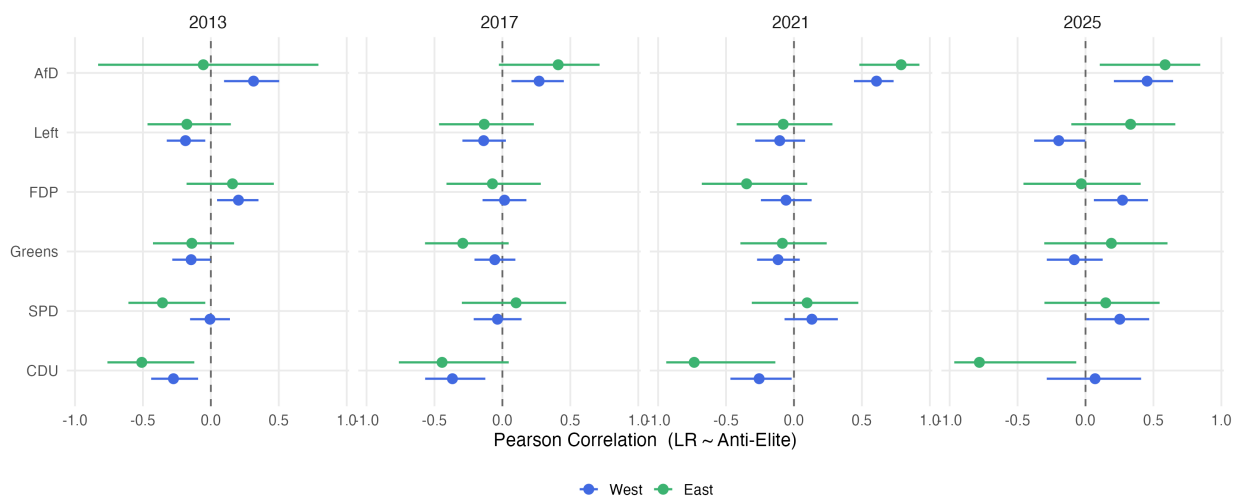
Notes. Each marker is the coefficient from a separate regression of a candidate position on one characteristic. The unit of observation is a municipality-by-party cell in elections 2013-2025. The sample consists of municipalities used in the estimation of the structural model (see Section 4.2, 5 for details). All covariates are standardized to mean zero and unit standard deviation. The specifications are (i) Raw OLS with no fixed effects, (ii) party-by-year fixed effects, and (iii) state-by-year plus party-by-year fixed effects. Standard errors are clustered by district-by-year. Horizontal bars are 95% confidence intervals. Filled markers denote coefficients significant at the 5% level ($|t| > 1.96$).

Figure F5. Association between candidates' positions on each dimension and municipalities' demographic and socio-economic characteristics.

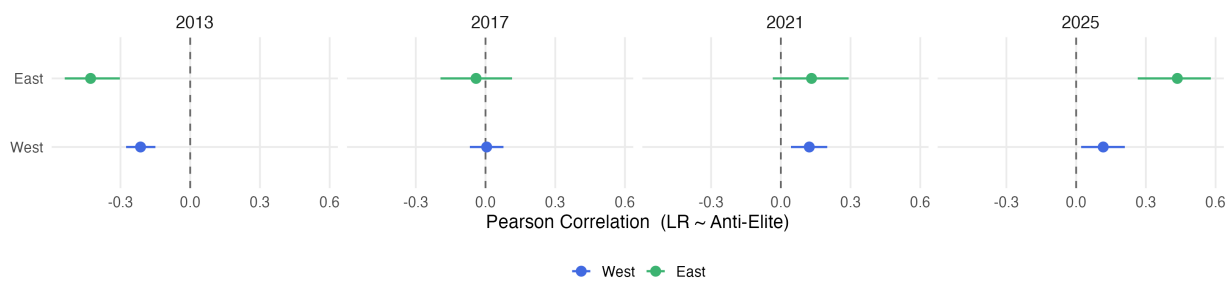


Notes. Each marker is the coefficient from a separate regression of a candidate position on one characteristic. The unit of observation is an electoral district-by-party cell in elections 2013-2025. The sample consists of electoral districts used in the estimation of the structural model (see Section 4.2, 5 for details). All covariates are standardized to mean zero and unit standard deviation. The specifications are (i) Raw OLS with no fixed effects, (ii) party-by-year fixed effects, and (iii) state-by-year plus party-by-year fixed effects. Standard errors are clustered by district-by-year. Horizontal bars are 95% confidence intervals. Filled markers denote coefficients significant at the 5% level ($|t| > 1.96$).

Figure F6. Association between candidates' positions on each dimension and their electoral districts' demographic and socio-economic characteristics.



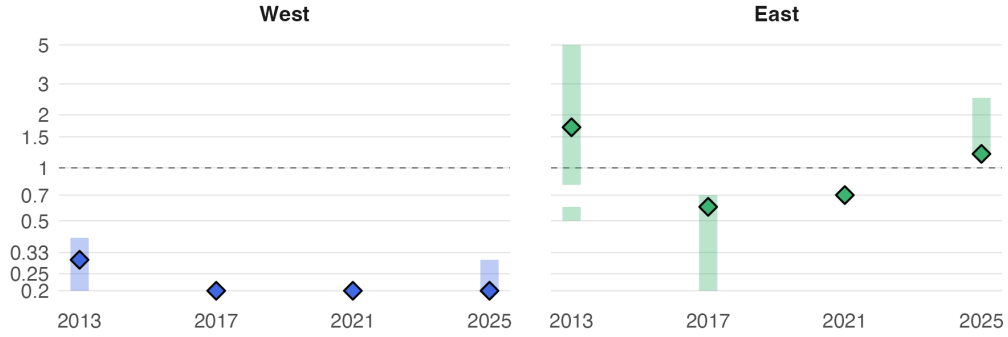
(a) Within parties



(b) Across parties

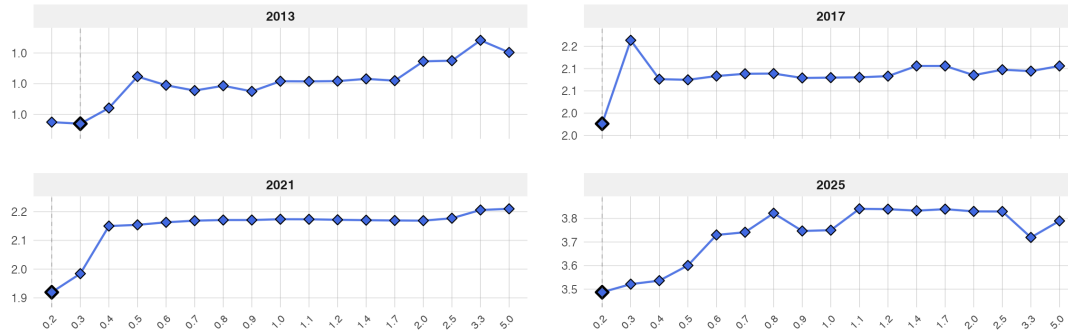
Figure F7. Correlation between candidate positions on Left-Right and Anti-Elite positions

G Additional Results

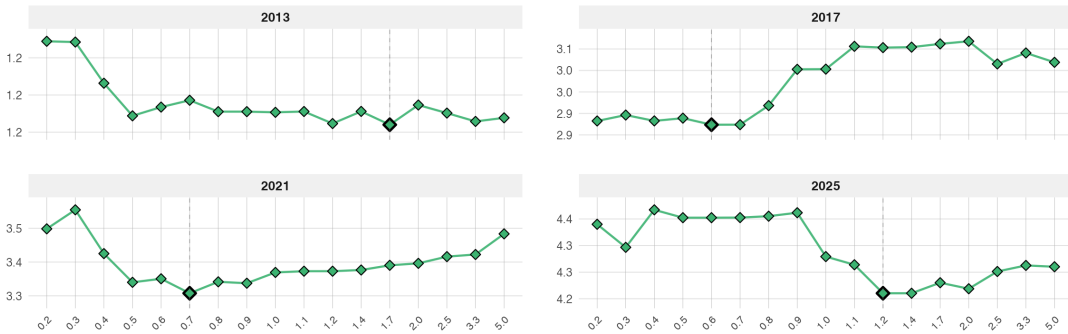


Notes. $w_{LR} = 1$ is fixed throughout, w_{AE} should be interpreted as the relative weight placed on the Anti-Elite dimension. Points denote the point estimates corresponding to the best-fitting weight, shaded intervals denote weights within 1% of min objective function value. Separate exercises are run for East and West Germany. I allow ideal-point distributions to vary across federal states – that is, each demographic group d in each market m (federal state) has its own bliss point distribution $f(x | m, d)$; while imposing the same weight w_{AE} for all groups within a given year and region. To run this exercise, the grid of possible weights is constructed. The grid is symmetric around 1 and imply relative weights across the two dimensions of $(0.2, 0.3, \dots, 0.9, 1)$ and the corresponding inverse ratios. See G2 for full results over the grid.

Figure G1. Set Best Weights for Anti-Elite dimension w_{AE} relative to Left-Right $w_{LR} = 1$ across times and parts of the country



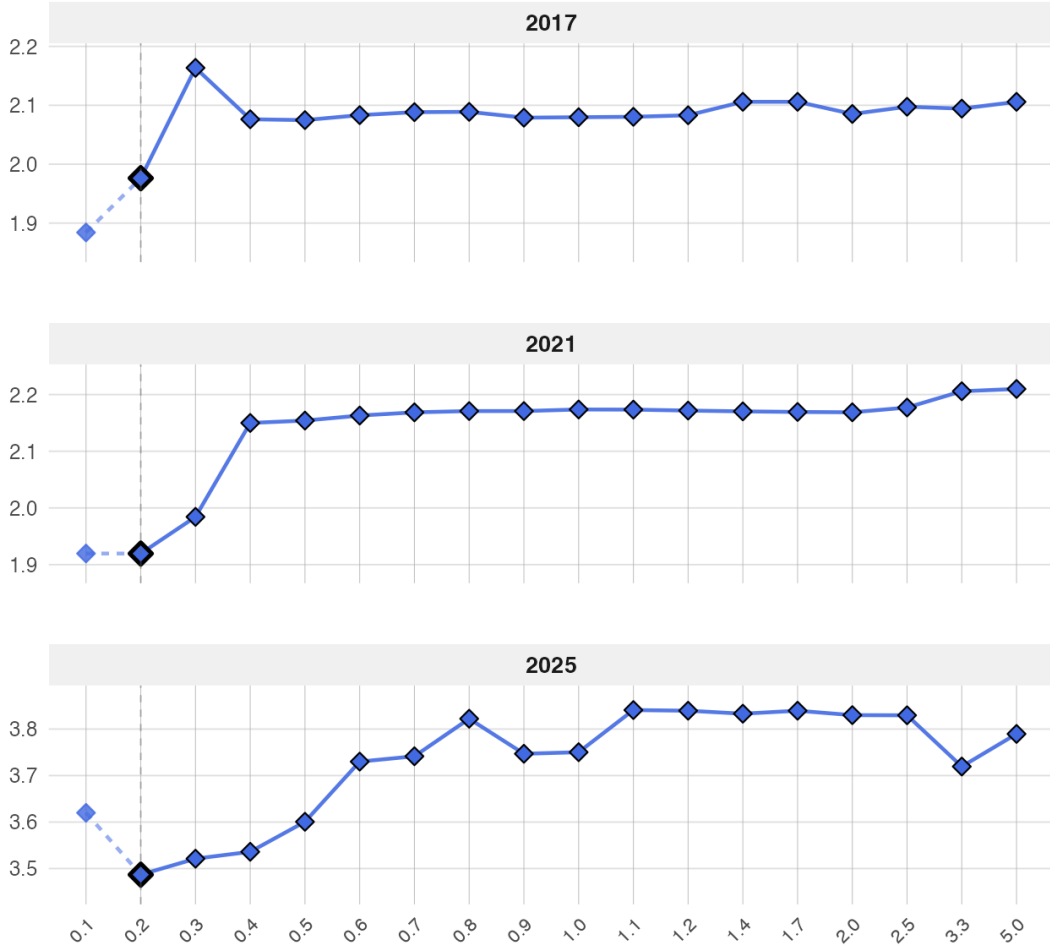
(a) West



(b) East

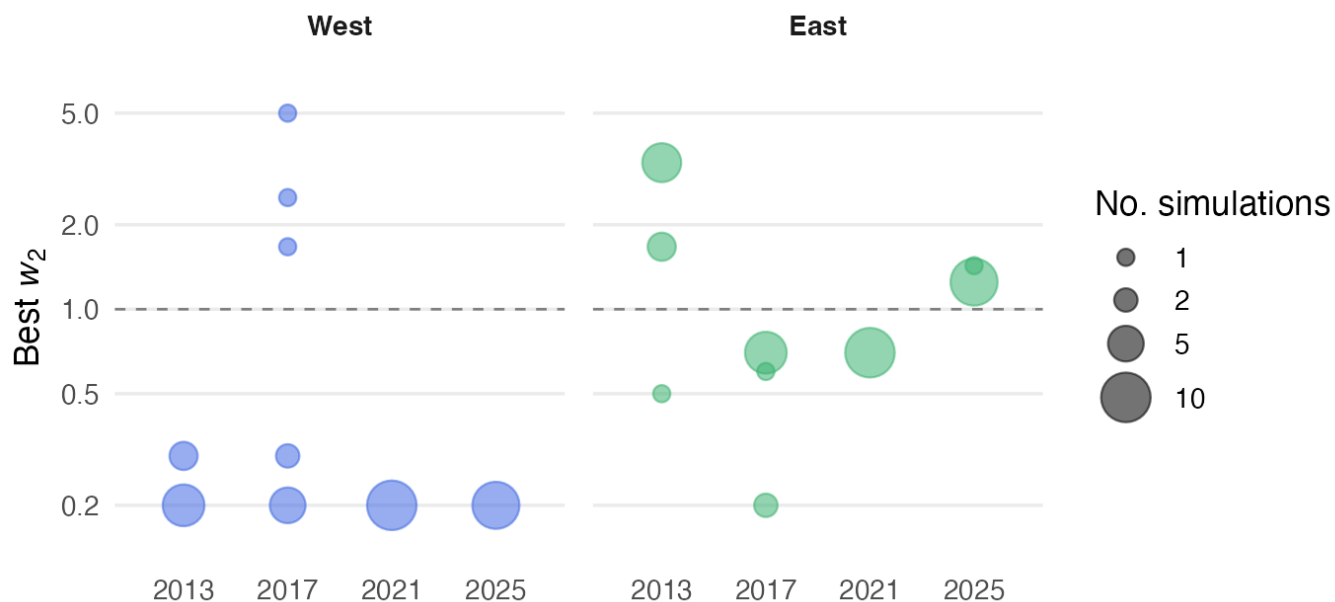
Notes. $w_{LR} = 1$ is fixed throughout, w_{AE} should be interpreted as the relative weight placed on the Anti-Elite dimension. Separate exercises are run for East and West Germany. I allow ideal-point distributions to vary across federal states – that is, each demographic group d in each market m (federal state) has its own bliss point distribution $f(x | m, d)$; while imposing the same weight w_{AE} for all groups within a given year and region. Note that the weights are symmetric around 1 and imply relative weights across the two dimensions of (0.2, 0.3, ..., 0.9, 1) and the corresponding inverse ratios.

Figure G2. Model Error Value by different relative weight values, years and parts of the country



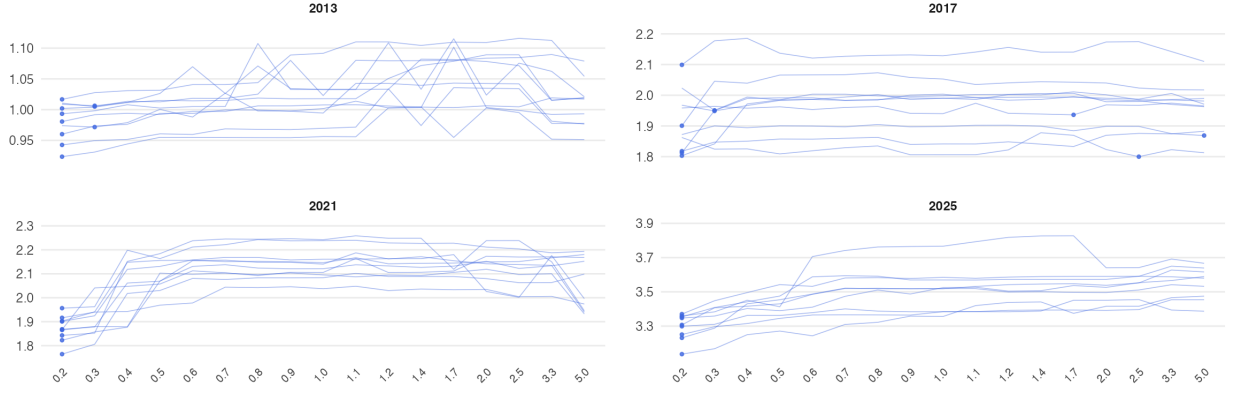
Notes. $w_{LR} = 1$ is fixed throughout, w_{AE} should be interpreted as the relative weight placed on the Anti-Elite dimension. Separate exercises are run for East and West Germany. I allow ideal-point distributions to vary across federal states – that is, each demographic group d in each market m (federal state) has its own bliss point distribution $f(x | m, d)$; while imposing the same weight w_{AE} for all groups within a given year and region.

Figure G3. Model Error Value by different relative weight values for the West 2017-2025 with a new point $w = 0.1$ (where corner $w = 0.2$ is the best fitting weight).

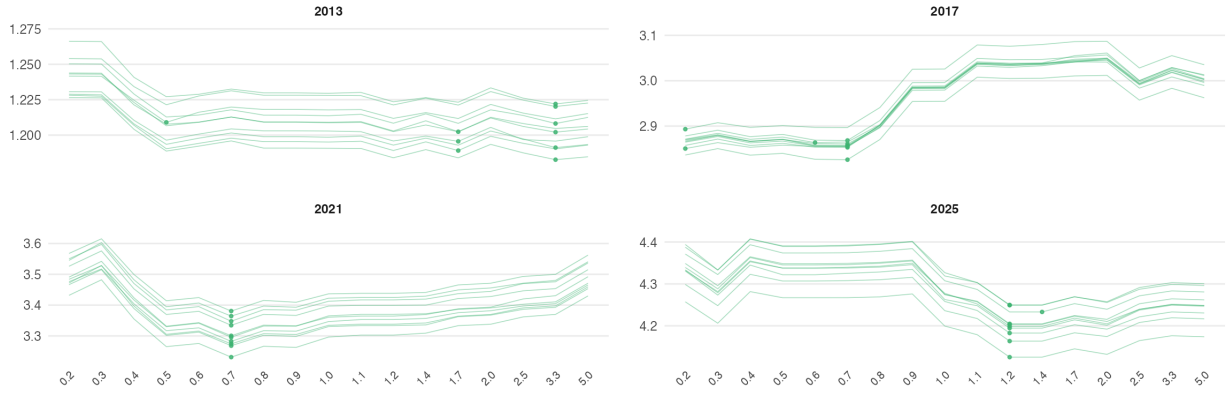


Notes. $w_{LR} = 1$ is fixed throughout, w_{AE} should be interpreted as the relative weight placed on the Anti-Elite dimension. Separate exercises are run for East and West Germany. I allow ideal-point distributions to vary across federal states – that is, each demographic group d in each market m (federal state) has its own bliss point distribution $f(x | m, d)$; while imposing the same weight w_{AE} for all groups within a given year and region. Note that the weights are symmetric around 1 and imply relative weights across the two dimensions of $(0.2, 0.3, \dots, 0.9, 1)$ and the corresponding inverse ratios. The exercise was repeated 10 times for each part of the country and each year by subsampling at most 10 electoral districts from each state, with at most 50 municipalities in each district.

Figure G4. Optimal relative weight values by years and parts of the country



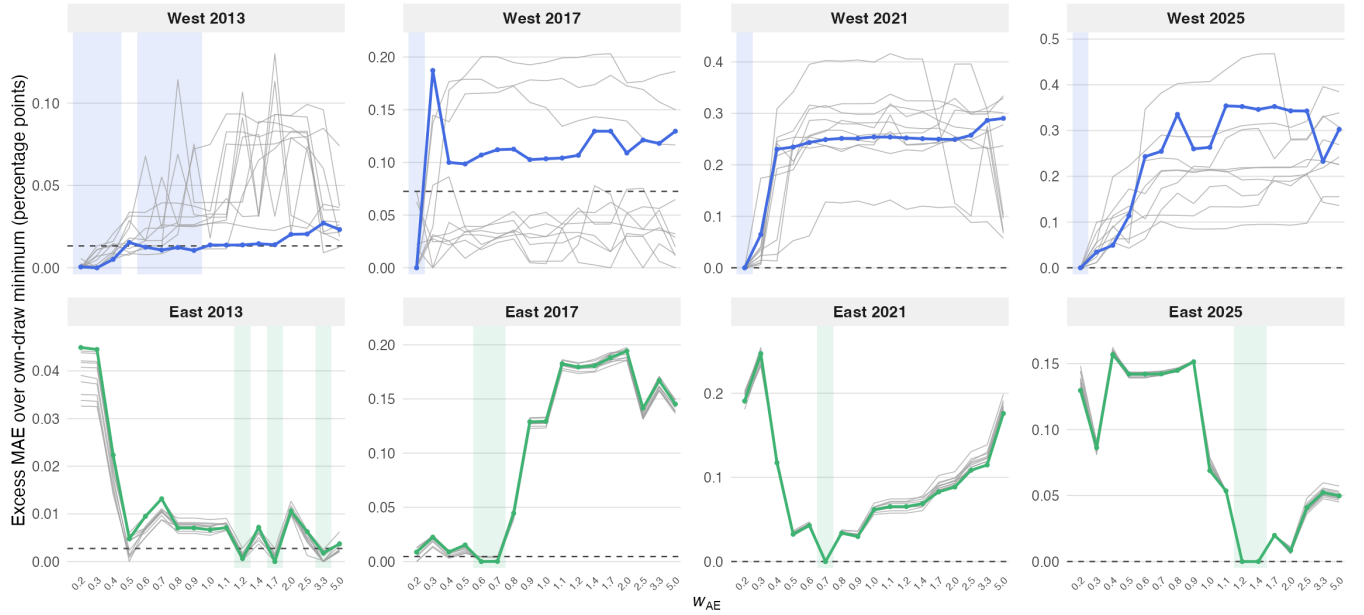
(a) West



(b) East

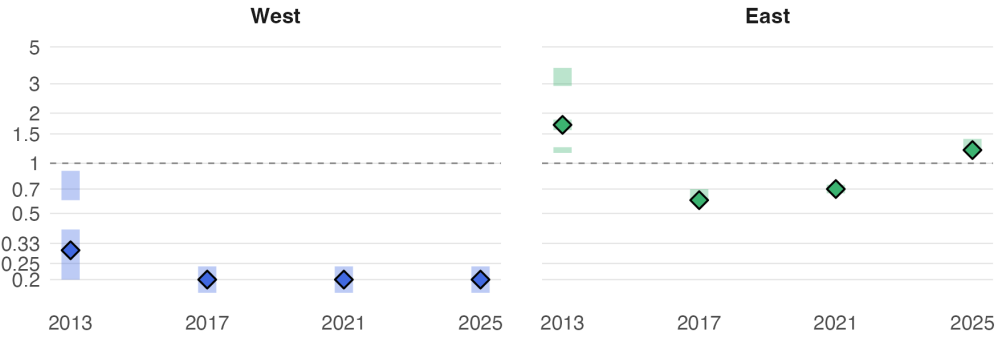
Notes. $w_{LR} = 1$ is fixed throughout, w_{AE} should be interpreted as the relative weight placed on the Anti-Elite dimension. Separate exercises are run for East and West Germany. I allow ideal-point distributions to vary across federal states – that is, each demographic group d in each market m (federal state) has its own bliss point distribution $f(x | m, d)$; while imposing the same weight w_{AE} for all groups within a given year and region. Note that the weights are symmetric around 1 and imply relative weights across the two dimensions of $(0.2, 0.3, \dots, 0.9, 1)$ and the corresponding inverse ratios. The exercise was repeated 10 times for each part of the country and each year by subsampling at most 10 electoral districts from each state, with at most 50 municipalities in each district.

Figure G5. Model Error Value by different relative weight values, years and parts of the country by subsamples



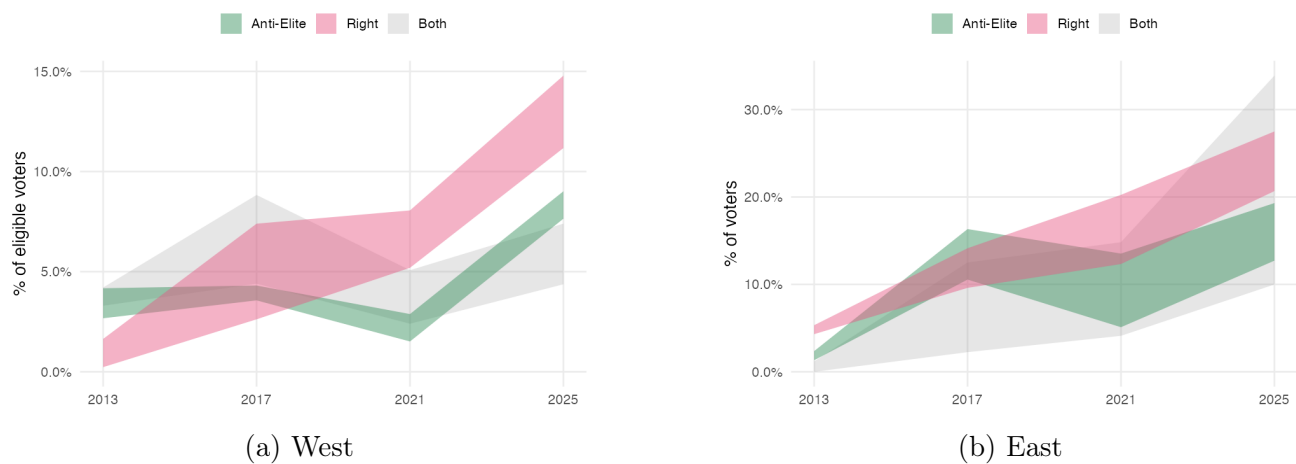
Notes. One panel per region-year. Grey lines are the $B = 10$ subsample MAE curves of Figure G5, each recentred on its own minimum ($D_{bj} = E_{bj} - \min_j E_{bj}$); recentring removes the vertical level shift between draws, which is what makes the raw curves look near-parallel, and leaves only the shape. The coloured line is the full-sample curve of Figure G2, recentred the same way ($\hat{d}_j$). The dashed horizontal line is the cutoff: the $(1 - \alpha)$ inverse-ECDF quantile of D_{bj^*} across draws at the full-sample minimiser j^* , with $\alpha = 0.05$.

Figure G6. Calibration of the weight confidence sets from the subsample re-runs



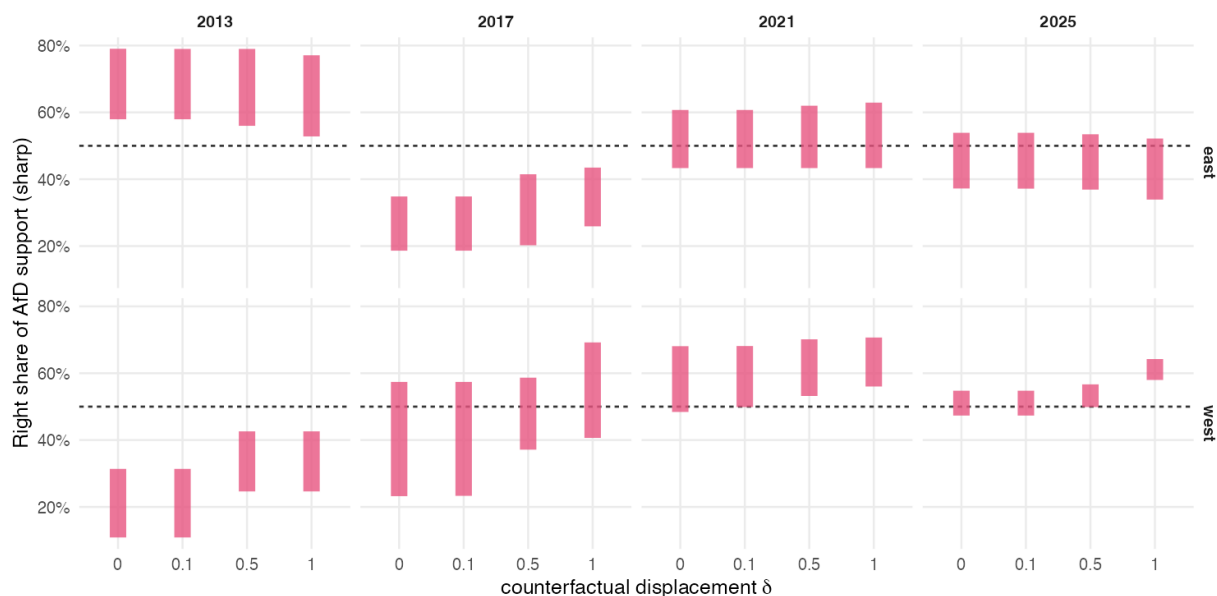
Notes. $w_{LR} = 1$ is fixed throughout, w_{AE} should be interpreted as the relative weight placed on the Anti-Elite dimension. Diamonds denote the point estimates corresponding to the best-fitting weight, identical to those in Figure G1. Shaded bars denote the confidence set CS , calibrated from the $B = 10$ subsample re-runs of Figures G4 and G5 rather than from the fixed 1% tolerance used in Figure G1. Each draw's MAE curve is recentred on its own minimum, $D_{bj} = E_{bj} - \min_j E_{bj}$, which removes the level shift between draws and retains only the curvature; the cutoff is the $(1 - \alpha)$ empirical quantile (inverse-ECDF, $\alpha = 0.05$) of D_{bj^*} . Separate exercises are run for East and West Germany. See Figure G6 for the calibration itself.

Figure G7. Confidence sets for the Anti-Elite weight w_{AE} , calibrated from the subsample re-runs



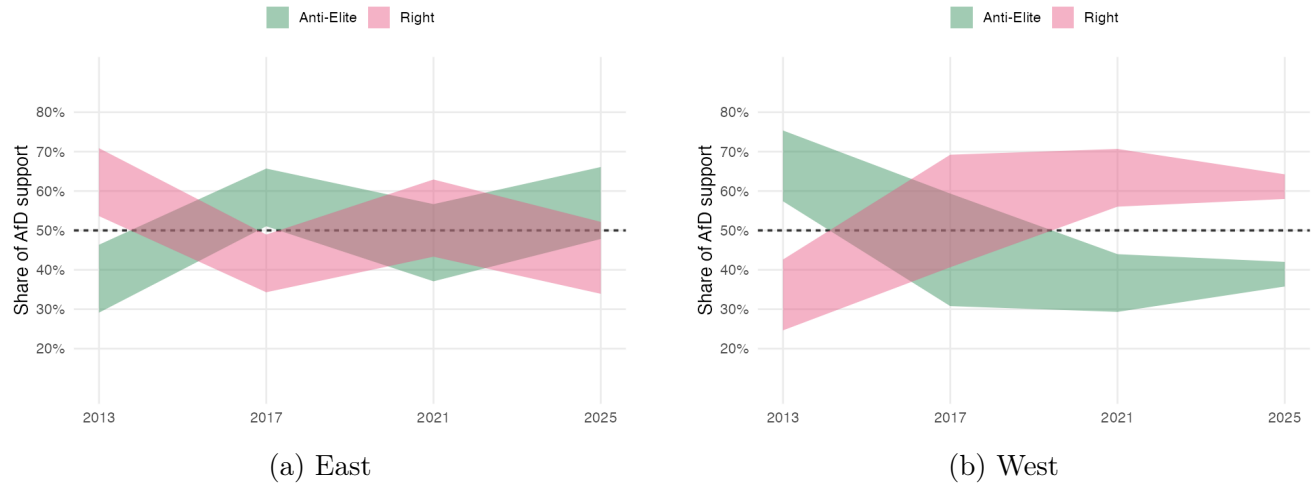
Notes: Panels plot bounds on the shares of voters whose bliss points lie in the fixed masks – green, pink and gray regions of the Figure 4 across years.

Figure G8. Estimated shares of voters in fixed Anti-Elite, Right and Both areas in Eastern and Western Germany



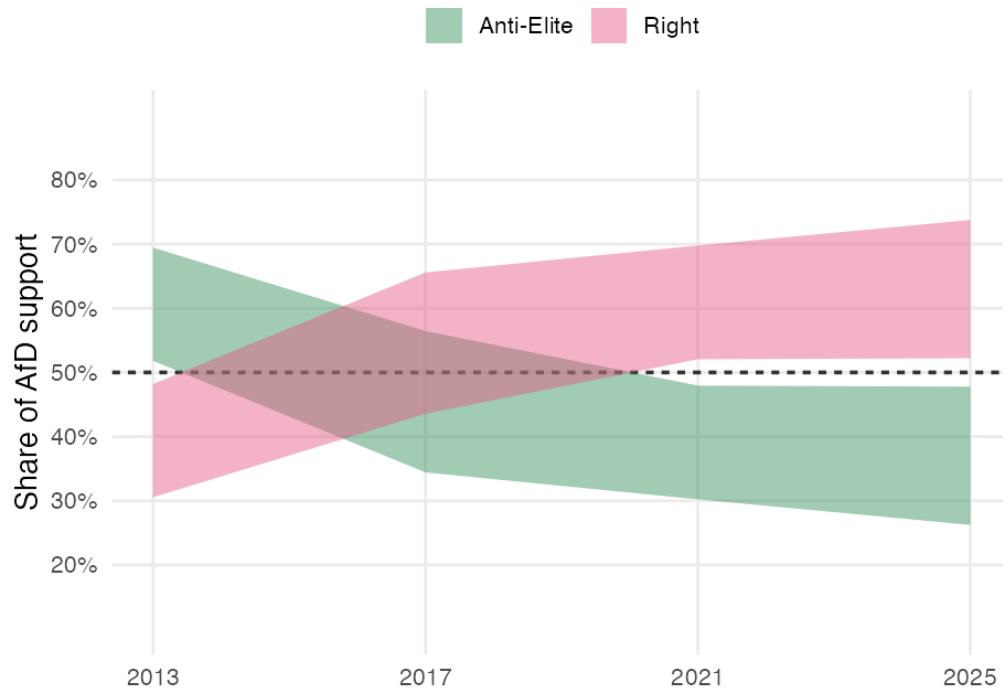
Notes: The smallest used $\delta = 10^{-6}$ (not identical 0).

Figure G9. Bounds on the shares of preferring the hypothetical Right party among AfD voters for different steps δ



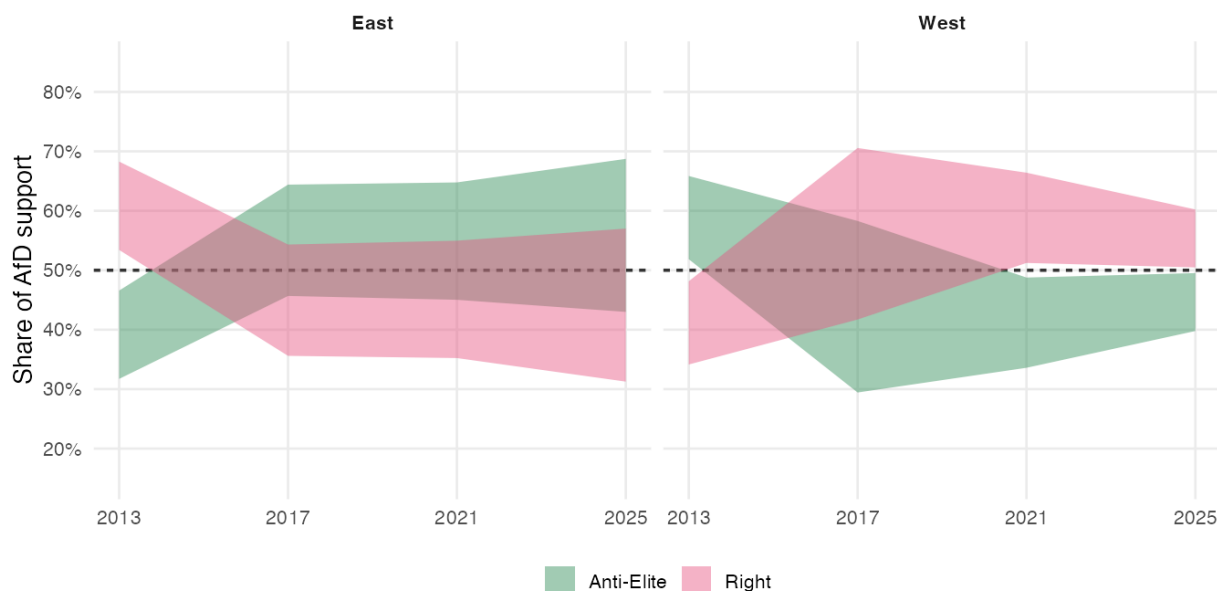
Notes: Panel plots bounds on the shares of voters whose bliss points lie in the attraction areas of two hypothetical parties that retain the AfD's median position within the state on one dimension while adopting a more moderate position on the other: the *Anti-Elite* party is more left-wing (green), whereas the *Right* party is more pro-elite (pink). Together, these areas recover the full AfD support. See Figure 5 for an illustration.

Figure G10. Regional Heterogeneity: Results separately for West and East Germany.



Notes. Bounds on the shares of voters whose bliss points lie in the green and pink regions on the Figure 5a across years are plotted. Second choice moments are not used in the estimation.

Figure G11. Estimated shares of AfD voters in both subspaces, estimation without second choice moments.



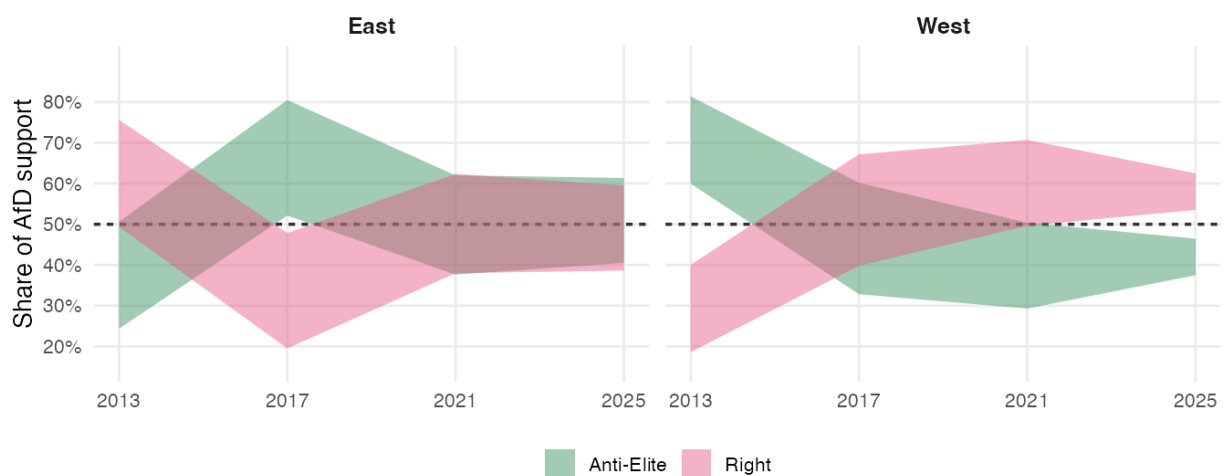
Notes. Bounds on the shares of voters whose bliss points lie in the green and pink regions on the Figure 5a across years are plotted.

Figure G12. Estimated shares of AfD voters in both subspaces on the sample of municipalities with population < 5 thous. inhabitants.



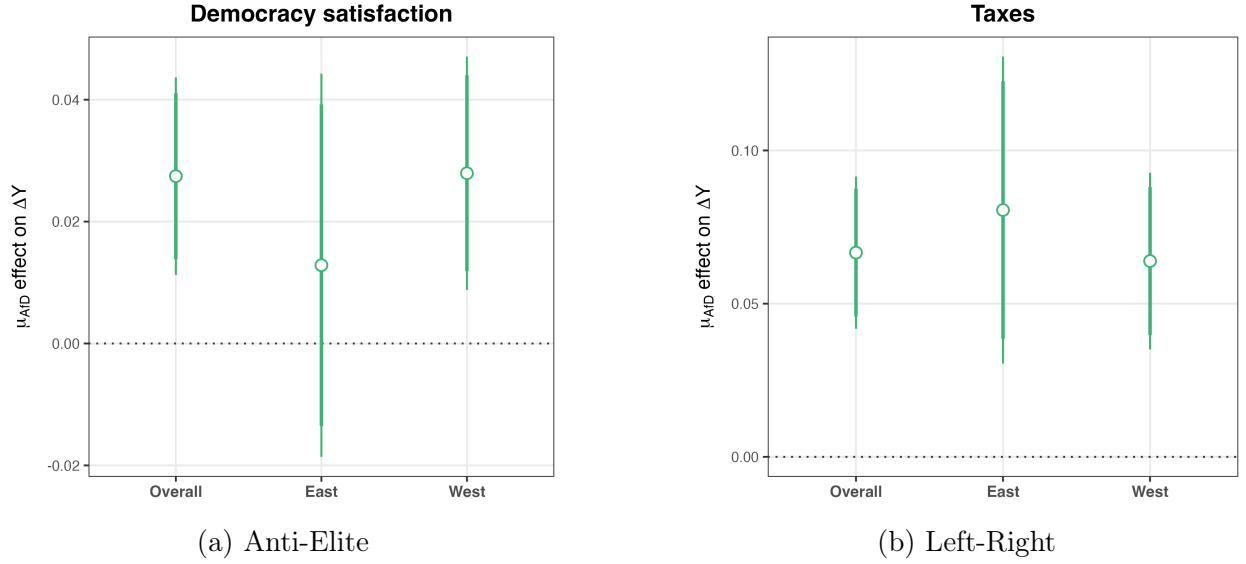
Notes: The figure plots bounds, across years, on the shares of voters whose bliss points lie in the green and pink regions of Figure 5a. Only districts in which the AfD candidate's vote share falls short of the winner's by more than 5% (10%) in the East (West) are retained, for the years 2017 and 2021 (2025). These restrictions drop 12 districts in the East in 2017 and 28 in 2021, and 22 districts in the West in 2025. The East in 2013 and the West in 2013–2021 are omitted because there were no serious contenders, so the results from the main text stand; the East in 2025 is omitted because the AfD won in almost every district, making the analysis infeasible.

Figure G13. Estimated shares of AfD voters in both subspaces on the sample of noncontender electoral districts.



Notes: Bounds on the shares of voters whose bliss points lie in the green and pink regions on the Figure 5a across years are plotted. The analysis allows the estimated ideal-point mass of each demographic group, in each partition cell, to differ by up to a factor of three between conservative and moderate electoral districts of the same state — i.e. the conservative-district mass may range from one-third to thrice the moderate-district mass in every cell. A district is classified as conservative if its AfD candidate is positioned above the within-state median AfD position on both the Left–Right and the anti-elite dimension, and moderate otherwise. This analysis is similar to an instrument in [Tebaldi, Torgovitsky and Yang \(2023\)](#).

Figure G14. Estimated shares of AfD voters in both subspaces allowing for the differences in ideal points distributions between *conservative* and *moderate* electoral districts.



Notes: The panels plot an average yearly within-person evolution of preferences of repeat AfD voters relative to voters of other parties. Based on self-reported positions $Y_{i,t}$ I construct $\Delta Y_{i,t} = Y_{i,t} - Y_{i,t-1}$ for consecutive years $t-1, t$, and drop any respondent-year pair for which AfD status changes across the two waves. That is, keep only pairs where either $\text{AfD}_{i,t-1} = \text{AfD}_{i,t} = 1$ (consistent AfD voters) or $\text{AfD}_{i,t-1} = \text{AfD}_{i,t} = 0$ (consistent non-AfD voters). Plotted values are μ_{AFD} from: $\Delta Y_{i,t} = \mu + \mu_{\text{AFD}} \cdot \mathbf{1}\{\text{AfD}_{i,t-1} = \text{AfD}_{i,t} = 1\} + \delta_t + \varepsilon_{i,t}$. Higher values correspond to positions closer to the AfD. The point estimate of μ_{AFD} together with 90% (thick) and 95% (thin) confidence intervals in the Overall, East, and West samples are reported. Standard errors clustered at the respondent level.

Figure G15. Results for within-person preference evolution for repeat AfD voters relative to the voters from other parties

	Overall		East		West	
	No FE	Year FE	No FE	Year FE	No FE	Year FE
μ	0.0229*** (0.0047)	0.0260*** (0.0048)	0.0165* (0.0099)	0.0183* (0.0102)	0.0249*** (0.0053)	0.0282*** (0.0054)
Year FE	No	Yes	No	Yes	No	Yes
Obs. (ΔY)	4806	4806	1155	1155	3651	3651
Individuals	2182	2182	506	506	1688	1688

Notes: Sample: AfD voters only (from their first AfD vote onward). For each year t we compute the *median* AfD voter's position in the normalized (taxes, democracy-satisfaction) space, where both outcomes are oriented so that higher values are closer to the AfD, and draw a 45 degree line through the point, and define a dummy $\text{Below45}_{i,t}$ for being 'below' the 45 degree line. Estimate, on consecutive-year pairs in which i is an AfD voter in t , $\text{Below45}_{i,t+1} - \text{Below45}_{i,t} = \mu + \delta_t + \varepsilon_{i,t}$. Standard errors are clustered at the respondent level. Columns are estimated separately on the Overall, East, and West subsamples.

Table G1. Annual drift in AfD voters "below 45°" dummy

H Candidate data

H.1 Candidate nominations across state lists and single member districts

The data on all the nominated politicians with their names and party affiliations was obtained by scraping the non-partisan platform *Wen Wählen?* (*Whom To Vote For?*). This platform sets as its goal providing comprehensive and easy-access information about federal elections in Germany. In particular, it compiles a comprehensive set of candidates participating in elections (which would not be available otherwise), as well as reaches to the candidates to ask them to participate in an electoral survey about their positions. While the portal provides information on the universe of nominated candidates, I focus just on the candidates from the main parties considered in the text of the paper (AfD, CDU/CSU, FDP, Left, Green, SPD).

The nominations done by parties are, given mixed electoral voting system, separated into two parts. In Bundestag elections, the nomination of direct candidates is formally decentralized to constituency-level members' or delegates' assemblies. State/Land party actors do not normally nominate direct candidates themselves, though they possess limited gatekeeping powers. Their strategic influence is stronger at Land level because Land-list rankings determine many actual Bundestag careers, and local selectorates often choose district candidates with Land-list prospects in mind. In practice, around 64% of candidates who appear on the state lists, also run in single member districts. This implies that in many ways the state lists, chosen by state leadership, is at least constraining the choice of the candidates to the electoral districts by local electoral units. These values differ across parties and years – Figure H2 illustrates that.

Importantly for the purpose of this study, the share of AfD candidates who are nominated both ways is very high, especially in 2021 (78%). Suggesting that most of the candidates were nominated, or at least approved, by the state level leadership. Such patterns could also indicate lack of candidates who would like to be balloted. It is also clear from the data that successful candidates in the SMD are more likely to be added to the state lists – Figure H3. Overall, politicians who are on the state lists are more successful, and heterogeneously so – those on higher positions are more successful, (Figure H4).

The presence of a candidate on both state and single member districts lists lower motives

for these politicians to target their constituency, as they could reasonably expect to be elected to Bundestag no matter the electoral outcome in the SMD. One example could be the leader of AfD, Alice Weidel. She is being nominated from the Electoral district 293, Bodensee in Baden-Württemberg. She does not have realistic chances of winning the district, but she is being placed first on the state list, and, hence, gets reelected.

I use this data to explore the patterns of candidate nominations to single member districts. In particular, I ask how likely a candidate is to be 1) renominated, 2) renominated to the same district, and how it depends on his electoral performance.

Across 2013, 2017, 2021 there were 5215 unique nominations⁴⁶ (at the level of elections-candidates). Among them, 985 were repeated nominations from one electoral cycle to another – e.g. the same candidate could be nominated in 2013 and 2017, or in 2017 and 2021. Out of all the nominations in 2017 and 2021 repeated nominations make up 28.7% of all the nominations. Among the renominations only 58 (5.8%) happen to be in another electoral district, and only 4 (< 0.5%) in another federal state. This suggests that when a candidate is renominated, it is usually within the same federal state, which is consistent with strong influence of state-level party organizations over nominations.

Renomination patterns are different by electoral success of the candidate. Among those candidates who win their electoral district, 58% get renominated; among those who lose – 21.7%. Figure H1 demonstrates this pattern in more details – for each of the observed largest parties the probability of renomination is positively correlated with first-vote share.

Upon a victory and renomination, a candidate gets renominated to the same district in 100% of cases⁴⁷. Among those who lose and get renominated 87.4% get nominations in the same electoral district.

The strong attachment of candidate to their electoral districts (and federal states) is in part due to where these candidates reside at the time of the elections. To take a closer look at this question, I exploit the question of the GLES Candidate Survey which asks Bundestag Candidates how long they have been residing in this electoral district. Importantly, participation in the survey is voluntary, so not all candidates respond, and the information is self-reported. That said, the survey is conducted for research purposes and is not released before the election, which should limit any “demand-side” effects.

The analysis of this data reveals that share of candidates born in the electoral district in which they run was 28%, 16%, 31% in years 2013, 2017, 2021 respectively. Mean number of years of residence in a given electoral district are 25, 25.3, 25.9 across the same years. The share of candidates who lived in these districts for less than 10 years is about 20% across all the years.

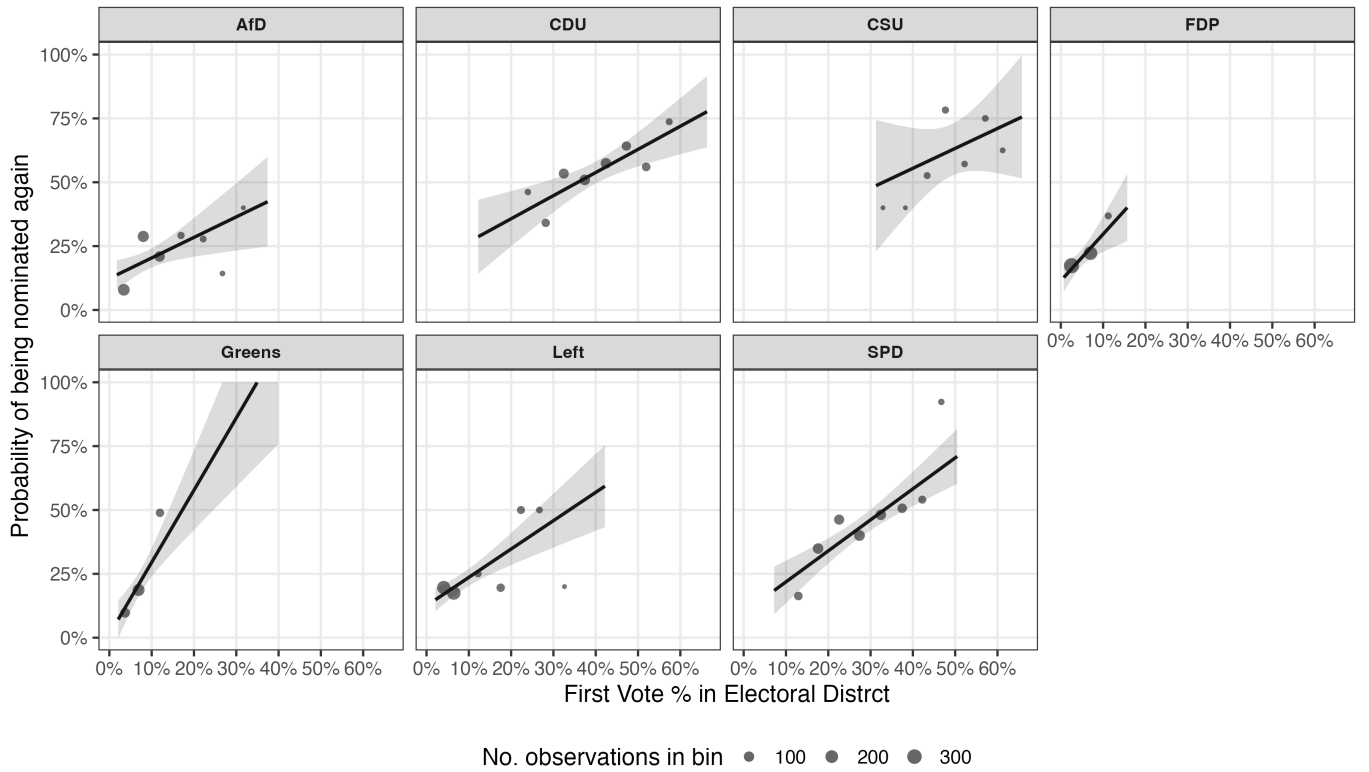
This highlights that while almost one third of candidates are local, there also is a non-negligible share of those who moved not that long before the elections. This could happen for strategic reasons related to the career of a politician, as well as for completely unrelated issues.

⁴⁶Note that the max number is 5382. It is slightly because of two parties – AfD and Greens – not nominating candidates to all the districts in some years.

⁴⁷There is 1 case in which a candidate was nominated to *another* district due to the change in districts’ boundaries. This was Albert Helmut Weiler who won the district of Gera – Jena – Saale-Holzland-Kreis in 2013 which was dissolved by 2017 elections. He was nominated and won 2017 elections in a district Saalfeld-Rudolstadt – Saale-Holzland-Kreis – Saale-Orla-Kreis, which was partly composed from the district in which he ran in 2013.

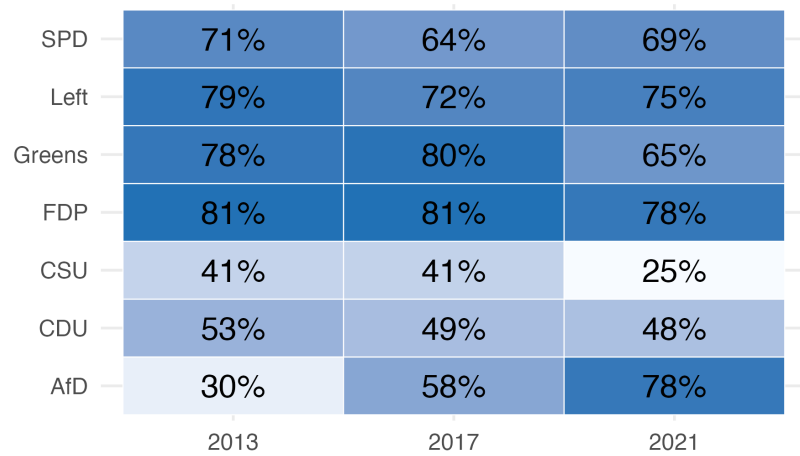
In light of the main analysis in the paper, I interpret this as following. When nominating candidates at the federal level, parties take into account candidates' personal histories – this could be for the logistics reason, as well as for the reasons related to the fact that such candidates could be more recognizable and likable by voters. In practice, candidates who are elected via a direct vote are expected to spend some time in their electoral district, and communicate with their constituency. This would often require easier presence and connection with the district. My identification claims rests on the assumption that this mechanism of candidate selection could be happening even at the expense of ideological congruence.

The picture of renomination I highlighted before draws a similar picture – many candidates get renominated to the same districts, even if they lose their elections. I claim that this happens mostly because it is logistically easier for the parties to keep going with the same candidate, instead of trying to find a new, and better one.



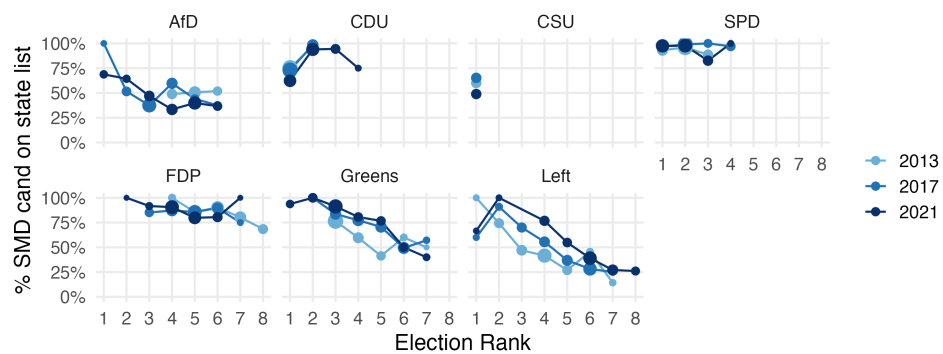
Notes. The data for this figure are drawn from the non-partisan candidate platform Wen Wählen?, which provides information on all candidate lists for federal elections. Vote share refers to the candidate's first-vote (Erststimme) share in their electoral district. The spline plots fitted values from an OLS regression of the form $Renominated_{i,t+1} = \beta_0 + \beta_1 Erststimme_{i,t} + \epsilon_{i,t}$.

Figure H1. Probability of renomination by candidate vote shares across main parties.



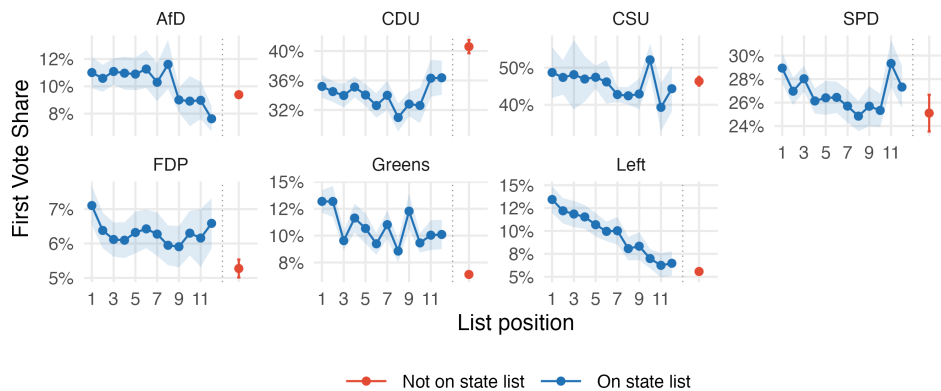
Notes. The data for this figure are drawn from the non-partisan candidate platform Wen Wählen?, which provides information on the candidate lists and state lists for federal elections. This data provides the universe of all party nominations.

Figure H2. Share of candidates from the state lists nominated in single member districts



Notes. The data for this figure are drawn from the non-partisan candidate platform Wen Wählen?, which provides information on the candidate lists and state lists for federal elections. This data provides the universe of all party nominations. Only dots representing more than 1 candidate are plotted.

Figure H3. Share of SMD candidates nominated in state lists by their electoral success



Notes. The data for this figure are drawn from the non-partisan candidate platform *Wen Wählen?*, which provides information on the candidate lists and state lists for federal elections. This data provides the universe of all party nominations.

Figure H4. First Vote Share of SMD candidates by their list position

H.2 Estimation of candidates' positions

1. Estimation of candidates' positions using GLES Candidate Survey Data Candidate positions on two dimensions — left-right and anti-elite — are estimated from the GLES Candidate Studies for 2013, 2017, 2021 and 2025. For each dimension, a set of survey items is first harmonized (see Table H1 for the full list of questions). Each item is then z-scored using a pooled mean and standard deviation computed across all waves in which it appears. This strategy places all items on a common metric and preserves comparability across years.

Each candidate's score on a dimension is the unweighted mean of all available z-scored items within that dimension. The anti-elite dimension comprises seven questions on satisfaction with democracy and current institutions; the left-right dimension uses questions covering redistribution, state intervention, immigration, and social and cultural issues, such as abortion and same-sex marriages. It is also supplemented by a left-right self-placement scale.

Because of the low response rate to this survey (around 35% across all years and parties), the survey was complemented with a non-partisan candidate survey, which also asks candidates about their positions on different issues in each electoral year.

2. Complementing GLES Candidate Survey Data with non-partisan candidate survey *Wen Wählen*. For each election year, candidates from the GLES Candidate Survey are merged with candidate data from the non-partisan platform *Wen Wählen*. Because the two surveys cannot be linked at the level of the individual candidate, the merge is at the district-party level: each *Wen Wählen* candidate is matched to the mean GLES-derived score of their party's candidates in their district. The GLES scores themselves come from the pooled cross-year normalisation, so the ideological scale is comparable across waves. The elastic net procedure is then used to impute ideological scores for those candidates missing in the GLES Survey but present in *Wen Wählen*. Since the *Wen Wählen* questionnaire changes from election to election, the models are estimated separately by wave (2013, 2017, 2021, 2025).

Two separate elastic net models are estimated per wave, one for each ideological dimension. The outcome y_i is the GLES-derived district-party score attached to candidate i . Only candidates

whose district-party cell contains at least one GLES-surveyed candidate enter the training set.

The feature matrix $\mathbf{X}$ contains, for candidate i , the full battery of that wave’s policy questions (65 in 2013, 76 in 2017, 69 in 2021, 50 in 2025; the same set enters both dimension models). Each question is encoded as one-hot indicators for the five response categories: *Yes*, *Rather Yes*, *Undecided*, *Rather No*, *No*, plus an explicit indicator for a missing answer. being pulled towards party means; no year indicators are needed since each wave is fit separately.

The elastic net minimises a penalised least-squares criterion that interpolates between Ridge (ℓ_2) and Lasso (ℓ_1) regularisation:

$$\hat{\boldsymbol{\beta}}(\lambda, \alpha) = \arg \min_{\boldsymbol{\beta}} \frac{1}{n} \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2 + \lambda \left[\alpha \sum_j |\beta_j| + \frac{1-\alpha}{2} \sum_j \beta_j^2 \right]$$

where n is the number of candidates in the training set, $\lambda \geq 0$ controls the overall amount of regularisation, and $\alpha \in [0, 1]$ determines the mix between the two penalty types. Setting $\alpha = 1$ recovers the Lasso, which drives many coefficients to exactly zero (implicit variable selection); $\alpha = 0$ recovers Ridge, which shrinks all coefficients towards zero while retaining all predictors. Intermediate values of α combine both effects.

Both λ and α are selected jointly by 10-fold cross-validation on the labelled candidates (those with a GLES-based score). The procedure is:

1. The labelled training set is randomly partitioned into 10 roughly equal folds.
2. A full Cartesian grid of 550 hyperparameter combinations is constructed:
 - $\lambda \in \{10^{-4}, \dots, 10^2\}$ (50 values, log-uniform spacing);
 - $\alpha \in \{0.0, 0.1, 0.2, \dots, 1.0\}$ (11 values).
3. For each of the $550 \times 10 = 5\,500$ fold-hyperparameter combinations the fit is run exclusively on the nine training folds and then applied to the held-out validation fold.
4. The root mean squared error (RMSE) between predicted and observed scores is recorded for each held-out fold.
5. The pair $(\hat{\lambda}, \hat{\alpha})$ that minimizes the mean RMSE across the 10 folds is selected.

The winning model is then refit on the wave’s full labelled training set using $(\hat{\lambda}, \hat{\alpha})$, and this final fit is used to predict scores for the wave’s unlabelled candidates. The selected mixtures are Ridge-leaning ($\hat{\alpha}$ between 0.1 and 0.5 across waves and dimensions), retaining 20–119 non-zero coefficients per model. The most potent questions per each year and dimension are presented in the Table [H2](#). The most potent questions and directions are as expected given the underlying dimensions.

The model performs well, explaining a large share of variation in observed GLES-based Left–Right position scores (cross-validated $R^2 = 0.8$), with a root mean squared error of 0.336. This indicates that the elastic net predictions are able to capture meaningful within-party variation in candidate positions across districts.

Performance is somewhat lower for the anti-elite score with cross-validated $R^2 = 0.6$. The anti-

elite score is harder to map from the policy-stance questions available in *Wen Wählen*. Nonetheless, the model explains more than a half the variance in observed anti-elite scores.

To mitigate overfitting, all performance metrics are computed from out-of-fold cross-validation predictions. Across the four waves, 1,147 *Wen Wählen* candidates carry a GLES-based training score and 1,348 receive an elastic-net imputed score; imputed scores are clamped to $[-1.5, 1.5]$ before entering the estimation data. Combining the two candidate surveys yields a locally estimated position for about 60% of all district $\times$ party cells. Missing positions are imputed with the within-state-wave median; if that median is also unavailable, the national within-wave median serves as a fallback. ⁴⁸

Dimension	Question	Values
Anti-Elite	Satisfaction with democracy in Germany	1–4
Anti-Elite	Democracy offers enough opportunities for citizens to participate	1–5
Anti-Elite	Laws reflect citizens’ interests	1–5
Anti-Elite	Political parties are the central intermediary between citizens and government	1–5
Anti-Elite	There should be nationwide referendums (direct votes) at the federal level	1–5
Anti-Elite	Referendums tend to lead to bad laws	1–5
Anti-Elite	Citizens have lost trust in politicians	1–5
Left–Right	In a crisis, the government should spend to stimulate economic growth	1–5
Left–Right	In a crisis, Germany should provide financial support to other countries/EU partners	1–5
Left–Right	Immigrants should adapt to German culture	1–5
Left–Right	Government should stay out of the economy	1–5
Left–Right	Same-sex marriage should be banned	1–5
Left–Right	Providing social security should be a key goal of government	1–5
Left–Right	Government should reduce income differences	1–5
Left–Right	Immigration is good for the German economy	1–5
Left–Right	Women should be free to decide about abortion	1–5
Left–Right	Self-placement on the left–right scale	1–11
Left–Right	Immigration policy: make immigration easier vs. restrict it	1–11

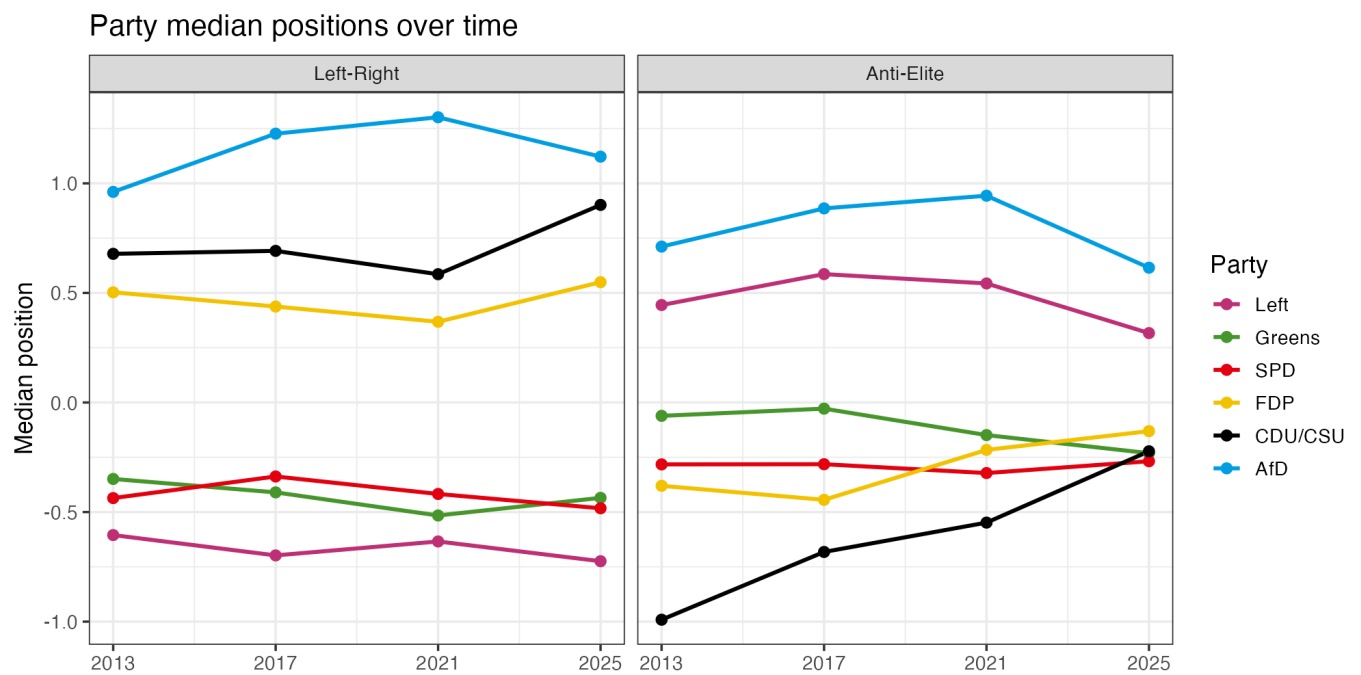
Table H1. Survey questions used to estimate candidate positions (GESIS Candidate Survey Data)

⁴⁸The global median fallback is applied in 5% of the used electoral districts, and only in Eastern Germany in 2013 and 2017.

Year	Item (translated)	Answer	Coef.
<i>Economic Left-Right (positive = more right)</i>			
2013	Reintroduce the Deutsche Mark!	Yes!	+0.17
	Reintroduce the Deutsche Mark!	Rather yes	+0.14
2017	Repeal “marriage for all”!	Yes!	+0.12
	Push forward European integration!	Rather not	+0.12
2021	Ban gender-inclusive language!	Yes!	+0.20
	Legalise soft drugs!	No!	+0.19
2025	Raise the minimum wage substantially!	Yes!	−0.39
	Ban gender-inclusive language!	Yes!	+0.32
<i>Anti-Elite (positive = more anti-elite)</i>			
2013	Referendums at the federal level!	No!	−0.30
	Referendums at the federal level!	Rather not	−0.23
2017	Referendums at the federal level!	No!	−0.26
	Referendums at the federal level!	Yes!	+0.26
2021	Leave the EU!	Rather not	+0.20
	Bring qualified immigrants to Germany!	Undecided	+0.16
2025	Repeal the Self-Determination Act!	Rather yes	+0.29
	Jobs are more important than climate protection!	Yes!	+0.22

Notes. Coefficients on one-hot answer-category indicators from the wave- and dimension-specific elastic nets used to impute candidate positions, at the cross-validated $(\hat{\lambda}, \hat{\alpha})$. For each wave the two substantive answer categories with the largest absolute coefficients are shown; an item can appear twice when two of its answer categories both rank in the top two. Pooled rare answers, explicit refusals, and missing responses are excluded. Answer categories (translated from German): *Yes!* (*Ja!*), *Rather yes (Eher ja)*, *Undecided (Unentschieden)*, *Rather not (Eher nicht)*, *No! (Nein!)*.

Table H2. Most Predictive *Wen Wählen* Items in the Elastic-Net Imputation



Notes. The data is based on the estimated single-member candidate positions using GLES Candidate Survey and non-partisan platform Wen Wählen.

Figure H5. Medians of left-right and anti-elite positions across parties over years